

Antihypertensive Therapy

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CHAPTER OUTLINE

PHARMACOLOGY OF THE NONDIURETIC ANTIHYPERTENSIVE DRUGS, 1654
SELECTION OF ANTIHYPERTENSIVE DRUG THERAPY, 1690

DRUG TREATMENT OF HYPERTENSIVE URGENCIES AND EMERGENCIES, 1702

This chapter is divided into three major sections. The first section reviews the pharmacology of nondiuretic antihypertensive drugs to provide clinicians with a complete overview of how to use these therapies safely in practice (Table 49.1; see Chapter 50 for a review of diuretic drugs). The first section also discusses individual drug classes and highlights the class mechanisms of action, renal effects, and efficacy and safety. Individual similarities and differences within and between classes are then addressed. The second section reviews clinical decision making with regard to blood pressure (BP) goals, selection of antihypertensive therapy, and methods of treating resistant hypertension. The third section reviews the pharmacology of the parenteral and oral drugs available for the management of hypertensive urgencies and emergencies and discusses clinical considerations in seeking to achieve rapid reduction of BP. The history of the development of modern antihypertensive drug therapy is shown in Fig. 49.1.

PHARMACOLOGY OF THE NONDIURETIC ANTIHYPERTENSIVE DRUGS

ANGIOTENSIN-CONVERTING ENZYME INHIBITORS

CLASS MECHANISMS OF ACTION

Angiotensin-converting enzyme (ACE) inhibitors inhibit the activity of ACE, which converts the inactive decapeptide angiotensin I (Ang I) into the potent hormone angiotensin II (Ang II) (Fig. 49.2). Because Ang II plays a crucial role in maintaining and regulating BP levels by promoting vasoconstriction and renal sodium and water retention, ACE inhibitors are powerful tools for targeting multiple pathways that contribute to hypertension. ACE inhibitors directly reduce the circulating and tissue levels of Ang II, thus blocking the potent vasoconstriction induced by the hormone (Table 49.2).¹ The resulting decrease in peripheral vascular resistance is not accompanied by changes in cardiac output or glomerular filtration rate (GFR) changes; the heart rate is

unchanged or may be reduced in patients with baseline heart rates higher than 85 beats per minute.²

A reduction in systemic and local levels of Ang II leads to effects beyond vasodilation that contribute to the antihypertensive efficacy of ACE inhibitors (Table 49.2).³ Additional mechanisms include the following: (1) inhibition of the breakdown of vasodilatory bradykinins catalyzed by ACE or kininase II (the hypotensive action of ACE inhibitors is blocked, in part, by bradykinin antagonists)⁴; (2) enhancement of vasodilatory prostaglandin synthesis; (3) improvement of nitric oxide-mediated endothelial function^{5,6} and upregulation of endothelial progenitor cells⁷; (4) reversal of vascular hypertrophy⁸; (5) decrease in aldosterone secretion; (6) augmentation of renal blood flow to induce natriuresis⁹; (7) blunting of sympathetic nervous system (SNS) activity^{10–12} through presynaptic modulation of norepinephrine release; (8) inhibition of postjunctional pressor responses to norepinephrine or Ang II^{11,13}; (9) inhibition of central Ang II-mediated sympathoexcitation, norepinephrine synthesis, and arginine vasopressin release; (10) inhibition of centrally controlled baroreceptor reflexes, which results in increased baroreceptor sensitivity¹³; (11) decrease in vasoconstrictor endothelin-1 levels¹⁴; (12) inhibition of thirst; (13) inhibition of cholesterol oxidation¹⁵; and (14) inhibition of collagen deposition in target organs.¹⁶

CLASS MEMBERS

Currently, there are more than 15 ACE inhibitors in clinical use. Each drug has a unique structure that determines its potency, tissue receptor binding affinity, metabolism, and prodrug compound, but they have remarkably similar clinical effects (Tables 49.3 and 49.4).³ The drugs are classified into sulfhydryl, carboxyl, or phosphinyl categories on the basis of the ligand that binds to the ACE–zinc moiety.

Sulfhydryl Angiotensin-Converting Enzyme Inhibitors

Captopril is a sulfhydryl-containing ACE inhibitor that is available in tablets of 12.5, 25, and 50 mg (see Tables 49.3

Table 49.1 Pharmacologic Classification of Nondietetic Antihypertensive Drugs

Angiotensin-converting enzyme inhibitors
Sulfhydryl
Carboxyl
Phosphinyl
Angiotensin II type 1 receptor antagonists
Biphenyl tetrazoles
Nonbiphenyl tetrazoles
Nonheterocyclics
β -Adrenergic antagonists and α 1- and β -adrenergic antagonists
Nonselective β -adrenergic antagonists
Nonselective β -adrenergic antagonists with partial agonist activity
β 1-Selective adrenergic antagonists
β 1-Selective adrenergic antagonists with partial agonist activity
Nonselective β -adrenergic and α 1-adrenergic antagonists
Calcium antagonists
Benzothiazepines
Dihydropyridines
Diphenylalkylamines
Tetralines
Central α 2-adrenergic agonists
Central and peripheral adrenergic-neuronal blocking agents
Direct-acting vasodilators
Moderately selective peripheral α 1-adrenergic antagonists
Peripheral α 1-adrenergic antagonists
Peripheral adrenergic-neuronal blocking agents
Renin inhibitors
Selective aldosterone receptor antagonists
Tyrosine hydroxylase inhibitors
Vasopeptidase inhibitors ^a

^aNot approved for the treatment of hypertension.**Table 49.2 Antihypertensive Mechanisms of Action of Angiotensin-Converting Enzyme Inhibitors**

Lower peripheral vascular resistance.
Inhibit the breakdown of vasodilatory bradykinins.
Enhance vasodilatory prostaglandin synthesis.
Improve nitric oxide-mediated endothelial function.
Reverse vascular hypertrophy.
Decrease aldosterone secretion.
Induce natriuresis.
Augment renal blood flow.
Blunt sympathetic nervous system activity and pressor responses.
Inhibit norepinephrine and arginine vasopressin release.
Inhibit baroreceptor reflexes.
Reduce endothelin-1 levels.
Inhibit thirst.
Inhibit oxidation of cholesterol.
Inhibit collagen deposition in target organs.

and 49.4).^{12,17,18} The usual starting dosage for hypertension treatment is 25 mg two or three times daily (Table 49.3), and the dosage can be titrated at 1- to 2-week intervals.¹² Captopril has 75% bioavailability, with peak onset within 1 hour. The half-life is 2 hours; with long-term administration, the hemodynamic effects are maintained for 3 to 8 hours.¹² Food may decrease captopril absorption by up to 54%, but this decrease is clinically insignificant.¹⁹ Captopril is partially metabolized in the liver into an inactive compound; 95% of the parent compound and metabolites are eliminated in the urine within 24 hours. The elimination half-life increases markedly in patients with creatinine clearances of less than 20 mL/min/1.73 m². In such patients, the initial dosages should be reduced, and smaller increments should be used for titration. Hemodialysis removes approximately 35% of the dose.^{12,20}

Carboxyl Angiotensin-Converting Enzyme Inhibitors

Benazepril hydrochloride is a long-acting, nonsulfhydryl-containing, carboxyl ACE inhibitor that is available as 10- or 20-mg tablets alone or in combination with amlodipine.¹⁸ The usual initial dosage is 10 mg daily, with maintenance dosages of 20 to 40 mg daily. Some patients respond better to twice-daily dosing (Tables 49.3 and 49.4).²¹ The onset of action occurs in 2 to 6 hours; maximal antihypertensive responsiveness occurs in 2 weeks. Benazepril is a prodrug that is rapidly bioactivated in the liver into the active benazeprilat compound, which is 200 times more potent than benazepril. The elimination half-life of benazeprilat is 22 hours. Benazeprilat is excreted primarily in the urine. Dialysis does not remove benazepril, but the initial dose should be no more than 10 mg in patients with a creatinine clearance of less than 60 mL/min/1.73 m², and the dose should be reduced to 5 mg in patients with a creatinine clearance of less than 30 mL/min/1.73 m².²²

Cilazapril is a nonsulfhydryl prodrug of the long-acting ACE inhibitor cilazaprilat.^{12,18} The usual dosage is 2.5 to 10 mg daily or in divided doses. After absorption, cilazapril is rapidly deesterified in the liver to its active metabolite, cilazaprilat. The initial antihypertensive response occurs in 1 to 2 hours, peaks at 6 hours, and lasts for 8 to 12 hours.²³ Dosages should be reduced by 25% in patients with end-stage kidney disease (ESKD).^{22,24}

Enalapril maleate is a nonsulfhydryl prodrug of the long-acting ACE inhibitor enalaprilat.^{18,25} The oral preparations are available in tablets of 2.5, 5, 10, and 20 mg. The initial dosage of enalapril is 5 mg once daily (Table 49.3). The usual daily dose is 10 to 40 mg, singly or in divided doses. Initial responses occur in 1 hour, and peak serum levels of enalaprilat are achieved in 3 to 4 hours. Enalapril undergoes biotransformation in the liver into the active compound enalaprilat (Table 49.4). Enalapril is excreted primarily in the urine. Dosages should be reduced by 25% to 50% in patients with ESKD.²²

Imidapril is the nonsulfhydryl prodrug of the long-acting ACE inhibitor imidaprilat.^{18,26} The usual daily dose is 10 to 40 mg (Table 49.3). The peak response occurs in 5 to 6 hours and lasts for 24 hours. Imidapril is metabolized in the liver (Table 49.4). The elimination half-life of the metabolites is 10 to 19 hours. No dosage adjustments are necessary in patients with impaired kidney function. Imidapril has a unique

THE HISTORY OF ANTIHYPERTENSIVES

Reserpine, Pentolinium, Guanethidine, Methyldopa (1950–1960), Clonidine (1980)

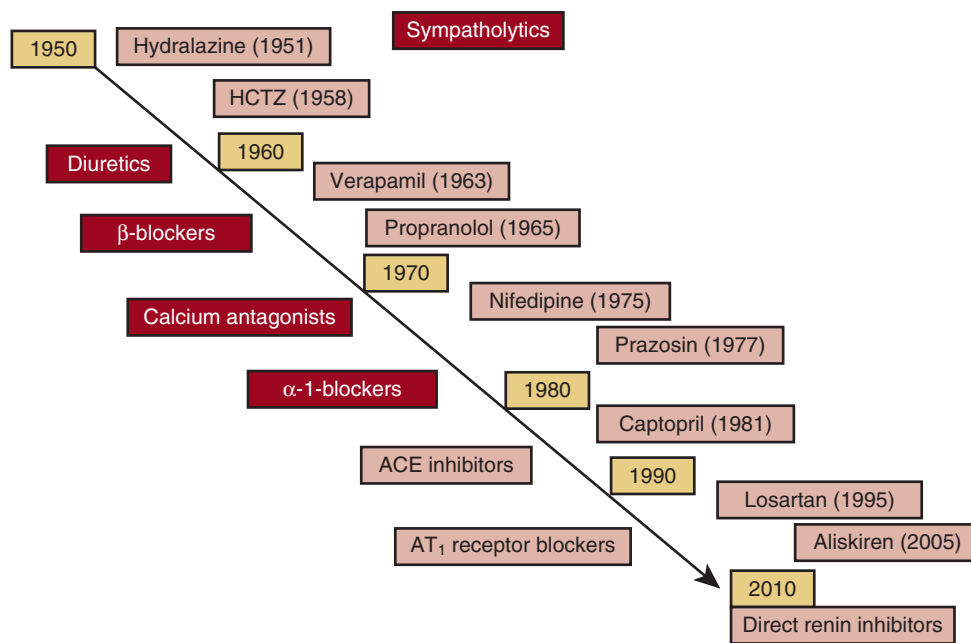


Fig. 49.1 History of modern antihypertensive drug therapy. The first antihypertensive drug, hydralazine, was a nonspecific vasodilator discovered in the 1950s. This was followed by blockade of calcium channels on vascular smooth muscle cells, the calcium channel blockers in the 1960s, and blockade of postsynaptic α -adrenoceptors on peripheral sympathetic neurons, the α -blockers in the late 1970s. Blockade by the renin-angiotensin-aldosterone system by angiotensin-converting enzyme inhibitors was discovered in the 1980s, angiotensin receptor blockers in the 1990s, and direct renin inhibitors just 10 years ago. ACE, Angiotensin-converting enzyme; AT_1 , angiotensin II type 1; HCTZ, hydrochlorothiazide. (From Sever PS, Messerli FH. Hypertension management 2011: optimal combination therapy. *Eur Heart J*. 2011;32:2499–2506.)

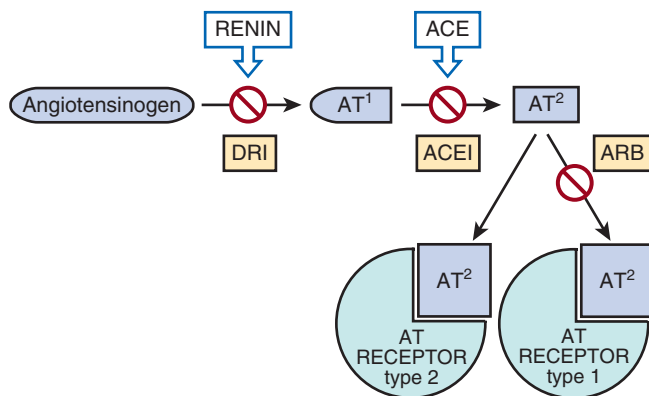


Fig. 49.2 Renin-angiotensin-aldosterone system (RAAS)—effect site for each type of RAAS blocking drugs. The two major classes of drugs that target the RAAS are the angiotensin-converting enzyme (ACE) inhibitors and the selective AT_1 receptor blockers (ARBs). Both these drug classes target angiotensin II, but the differences in their mechanisms of action have implications for their effects on other pathways and receptors. Both ACE inhibitors and ARBs are effective antihypertensive agents that have been shown to reduce the risk of cardiovascular and renal events. Direct inhibition of renin—the most proximal aspect of the RAAS—became clinically feasible in 2007 with the introduction of aliskiren. AT, Angiotensin; DRI, direct renin inhibitor. (From Robles NR, Cerezo I, Hernandez-Gallego R. Renin-angiotensin system blocking drugs. *J Cardiovasc Pharmacol Ther*. 2014;19:14–33.)

advantage over other ACE inhibitors in that it is less likely to cause cough.²⁷

Lisinopril is a nonsulfhydryl analog of enalaprilat.^{18,28} The initial dosage is 10 mg/day, and the usual daily dose is 20 to 40 mg (Table 49.3). The initial antihypertensive response occurs in 1 hour, peaks at 6 hours, and lasts for 24 hours (Table 49.4). The maximal effect may not be observed for 24 hours. The elimination half-life is 12 hours. Lisinopril is not metabolized, and it is exclusively eliminated in the urine unchanged. Lisinopril is dialyzable, and patients undergoing dialysis may require supplemental doses. The initial dosage should be reduced to 2.5 to 7.5 mg/day in patients with moderate to advanced chronic kidney disease (CKD).²²

Moexipril hydrochloride is the nonsulfhydryl prodrug of the ACE inhibitor moexiprilat.^{18,29} The usual daily dose is 7.5 to 30 mg in single or divided doses (Table 49.3).¹² The oral bioavailability of moexipril is approximately 20%, and its absorption is impaired by high-fat meals. The peak response occurs at 3 to 6 hours and lasts for 24 hours (Table 49.4). Moexipril is rapidly converted in the liver to moexiprilat, which is 1000 times more potent than the parent compound. The dosage should be reduced by 50% in patients with ESKD.^{22,30}

Perindopril is a nonsulfhydryl prodrug of the long-acting ACE inhibitor perindoprilat.^{12,30} The usual daily dose is 4 to 8 mg (Table 49.3). The response peaks at 3 to 7 hours. A single dose has a duration of action of 24 hours. Perindopril undergoes extensive first-pass hepatic metabolism into the

Table 49.3 Pharmacodynamic Properties of Angiotensin-Converting Enzyme Inhibitors

Generic Name (Trade Name)	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Alacepril (Cetapril)	12.5	12.5–100	100	qd	3	24
Captopril (Capoten)	25	12.5–50	150	tid	1–2	3–8
Benazepril (Lotensin)	10	20–40	40	qd	2–6	24
Enalapril (Vasotec)	5	10–40	40	qd, bid	3–4	12–24
Moexipril (Univasc)	7.5	7.5–30	30	qd, bid	3–6	24
Quinapril (Accupril)	10	20–80	80	qd	2	24
Ramipril (Altace)	2.5	2.5–20	40	qd, bid	2	24
Trandolapril (Mavik)	1	2–4	8	qd	2–12	24
Fosinopril (Monopril)	5	5–40	40	qd, bid	2–7	24
Cilazapril (Dynorm)	2.5	2.5–10	10	qd, bid	6	8–12
Perindopril (Aceon)	4	4–8	8	qd	3–7	24
Spirapril (SCH 33844)	6	6	6	qd	3–6	24
Zofenopril (SQ 26991)	15	30–60	60	qd	—	—
Lisinopril (Zestril, Prinivil)	10	20–40	40	qd	6	24
Imidapril (TA 6366)	5	10–40	40	qd	5–6	24

Table 49.4 Pharmacokinetic Properties of Angiotensin-Converting Enzyme Inhibitors

Drug	Absorption (%)	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Alacepril	—	70	—	1	1.9	L	U (70%)	Captopril
Captopril	60–75	75	Yes	1	2	L	U	Inactive
Benazepril	35	>37	No	2–6	22	L, K	F, U	Benazeprilat
Enalapril	55–75	73	—	3–4	11–35	L	F, U	Enalaprilat
Lisinopril	25	6–60	—	1	12	K	U	Enalaprilat
Moexipril	>20	13–22	Yes	1.5	2–10	L	F (50%), U	Moexiprilat
Quinapril	60	50	Yes	1	25	L	U (50%)	Quinaprilat
Ramipril	50–60	60	—	1–2	13–17	L	F, U	Ramiprilat
Trandolapril	70	10	No	2–12	16–24	L	F (66%), U	Trandolaprilat
Fosinopril	36	36	—	1	12	L, K, I	F, U	Fosinoprilat
Cilazapril	—	57–76	No	1–2	30–50	L	U (52%)	Cilazaprilat
Spirapril	—	50	Yes	1	33–41	L	F (60%), U (40%)	Spiraprilat
Perindopril	—	75	Yes	1.5	3–10	L	F, U (75%)	Perindoprilat
Imidapril	—	40	—	3–10	10–19	L	U	Imidaprilat
Zofenopril	>80	96	Yes	5	5	K	F (26%), U (69%)	Zofenoprilat

F, Feces; I, intestine; K, kidney; L, liver; U, urine.

active metabolite perindoprilat (Table 49.4). Renal excretion accounts for 75% of the clearance. The dosage should be reduced by 75% and 50% in patients with creatinine clearances of less than 50 and less than 10 mL/min/1.73 m², respectively.^{22,30}

Quinapril hydrochloride is a nonsulphydryl prodrug of the ACE inhibitor quinaprilat.^{18,30,31} The initial dose is 10 mg, and the usual daily dose is 20 to 80 mg, which should be adjusted at 2-week intervals (Table 49.3). Twice-daily therapy may provide a more sustained BP reduction. The onset of action occurs in 1 hour, and the peak response occurs in 2 hours and lasts for 24 hours. Quinapril is extensively metabolized in the liver into the active metabolite, quinaprilat (Table 49.4). Renal excretion by way of filtration and active tubular secretion accounts for 50% of the clearance. Quinapril is not dialyzable. The dosage should be reduced by 25% to 50% in patients with ESKD.²²

Ramipril is a potent, nonsulphydryl prodrug of the ACE inhibitor ramiprilat.^{12,18} Ramipril capsules are available in 1.25, 2.5, or 5 mg. The initial daily dose is 2.5 mg (Table 49.3). The usual daily dose is 2.5 to 20 mg, and it can be titrated by doubling the current dose at 2- to 4-week intervals. Ramipril is well absorbed from the gastrointestinal tract; peak concentrations are achieved in 1 to 2 hours (Table 49.4). The peak response occurs in 2 hours and lasts for 24 hours. Ramipril is extensively metabolized in the liver into the active metabolite ramiprilat. The elimination half-life of the active compound is 13 to 17 hours, and it is prolonged in renal failure patients to approximately 50 hours. The dosage should be reduced by 50% to 75% in patients with a creatinine clearance of less than 50 mL/min/1.73 m².^{18,22}

Trandolapril is a nonsulphydryl ethyl ester prodrug of the ACE inhibitor trandolaprilat.^{12,18} It is available in tablets of 1, 2, and 4 mg or in combination with verapamil. The usual

starting dosage is 1 mg/day (Table 49.3). Trandolapril is only 10% bioavailable, and its absorption is not affected by food (Table 49.4).³² Trandolapril undergoes extensive first-pass hepatic metabolism into trandolaprilat. The peak serum concentrations of trandolaprilat occur within 2 to 12 hours; the duration of action is 24 hours, but it may be as long as 6 weeks. The recommended starting dose in patients with a creatinine clearance of less than 30 mL/min/1.73 m² is 0.5 mg.

Phosphinyl Angiotensin-Converting Enzyme Inhibitor

Fosinopril sodium is a nonsulfhydryl prodrug of fosinoprilat, a long-acting ACE inhibitor.^{18,33} The usual daily dose is 5 to 40 mg (Table 49.3). Its maximal effects may not occur until 4 weeks. The initial response occurs in 1 hour, the peak response occurs in 2 to 7 hours, and the duration of response is 24 hours, which is prolonged in patients with ESKD (Tables 49.3 and 49.4). The elimination half-life of fosinoprilat is 12 hours. All metabolites are excreted in the urine and feces. Hepatic biliary clearance increases significantly as kidney function declines. Thus the dosage must be reduced by 25% in patients with ESKD.²²

CLASS RENAL EFFECTS

There has been considerable interest in the ability of ACE inhibitors to protect the kidney from the unrelenting deterioration that occurs with hypertension and many etiologies of CKD. ACE inhibitors have extensive hemodynamic and nonhemodynamic effects that afford such protection (Table 49.5). In hypertensive patients, ACE inhibitors can restore the pressure-natriuresis relationship to normal, thereby maintaining sodium balance at a lower arterial BP.³⁴ The response is exaggerated in the setting of dietary sodium restriction. The mechanism responsible for this effect is the direct inhibition of proximal, and possibly distal, tubule sodium reabsorption.³⁵ The increased renal excretory capacity plays a major role in the long-term antihypertensive activity of the drugs. Clinically, the increase in sodium excretion is transitory because the reduced arterial pressure allows the sodium excretion to return to normal. However, the maintenance of normal sodium excretion at lower arterial pressures correlates with increased excretion in the setting of hypertension.³⁴ After several days, inhibition of Ang II and aldosterone contributes

to the natriuresis.³⁶ The long-term effects on water excretion are less certain. ACE inhibitors initially induce an increase in free water clearance, but there are no long-term changes in total body weight, plasma, or extracellular fluid volume. The decrease in aldosterone caused by ACE inhibition also correlates with decreased potassium excretion,³⁶ particularly in patients with impaired kidney function. The antidiuretic effect appears to be transient, but it can be exacerbated by concomitant administration of potassium-sparing diuretics, oral potassium supplements, and nonsteroidal antiinflammatory drugs (NSAIDs); it should be monitored rigorously.

The effects of ACE inhibitors on angiotensin peptide levels depend on the responsiveness of renin secretion.^{37,38} All ACE inhibitors decrease circulating concentrations of Ang II and increase circulating concentrations of angiotensin-(1–7) (a potential vasodilator) and plasma renin. When renin levels show little increase in response to ACE inhibition, the levels of Ang II and its metabolites decrease markedly, with little change in the levels of Ang I. Large increases in renin levels in response to ACE inhibition increase the levels of Ang I and its metabolites. The increased levels of Ang I can produce higher levels of Ang II through uninhibited ACE and other pathways, thereby blunting the effect of reduced Ang II. This phenomenon is termed “ACE escape” and may contribute to reduced ACE inhibitor efficacy when used over a long term.³⁹

Recent experimental reports have highlighted the potential importance of the tissue specificity of ACE inhibitors,⁴⁰ as well as disagreement about the role played by systemic generation of Ang II versus the local generation of Ang II within the kidneys and its relative importance in terms of hypertension. An important role for intrarenal ACE has been demonstrated in murine hypertension models, as mice lacking only renal ACE are protected against hypertension.⁴⁰ The intrarenal Ang II production and local regulation of renal sodium transport represents a new mechanistic understanding of the underlying inappropriate regulation of salt and water balance. However, what remains unclear is the role that intrarenal renin–angiotensin–aldosterone system (RAAS) plays in human disease. It is clear that local inhibition of ACE in the vascular wall and renal vessels contributes to the antihypertensive activity of hypertension drugs. ACE inhibitor–induced changes in BP correlate with the degree of inhibition of RAAS in plasma and tissues.⁴¹ The ACE inhibitors with the greatest tissue specificity, however, are associated with prolonged activity at the tissue level, even after serum ACE levels have returned to normal.⁴¹ Consequently, they are more efficacious with once-daily dosing.⁴¹ Other potential renoprotective effects that have been noted in experimental models include attenuation of oxidative stress,⁴² scavenging of free radicals, and attenuation of lipid peroxidation.⁴³ The clinical importance of these effects is under investigation.

Because the degree of proteinuria correlates (inversely) with the rate of decline in kidney function, and a decrease in proteinuria correlates better with renal protection than with decreased BP, the reduction of proteinuria can have a substantial impact.⁴⁴ All ACE inhibitors decrease urinary protein excretion^{45–47} in normotensive and hypertensive patients with kidney disease of various etiologies. Individual response rates vary from an increase of 31% to a decrease of 100%, and they are strongly influenced by drug dosage and dietary sodium intake.^{46,48} There is a clear dose–response

Table 49.5 Renal Protective Mechanisms of Angiotensin-Converting Enzyme Inhibitors

- Restore pressure-natriuresis relationship to normal.
- Inhibit tubule sodium reabsorption.
- Decrease arterial pressure.
- Decrease aldosterone production.
- Decrease proteinuria.
- Improve altered lipid profiles.
- Decrease renal blood flow.
- Decrease filtration fraction.
- Decrease renal vascular resistance.
- Reduce scarring and fibrosis.
- Attenuate oxidative stress and reduce free radicals.

relation between increased doses and reduced proteinuria, and this relationship is not dependent on changes in BP, renal plasma flow, or GFR. Furthermore, the effect of ACE inhibitors on the reduction of proteinuria is eliminated with high salt intake.

Studies have demonstrated that in normotensive patients with diabetes mellitus, ACE inhibitors can normalize GFR, markedly reduce the progression of kidney disease, and normalize microalbuminuria.^{49,50} These findings are discussed in depth in Chapter 39. The effect is noted in the first month of therapy and is maximal at 14 months. Several mechanisms account for the reduction in urinary protein excretion, including the following: a decrease in glomerular capillary hydrostatic pressure; a decrease in mesangial uptake and clearance of macromolecules; and improved glomerular basement membrane perm-selectivity.^{3,51} ACE inhibitors have better antiproteinuric efficacy than other classes of antihypertensive agents, with the exception of angiotensin receptor blockers (ARBs). In the treatment of proteinuria, the noninferiority of ARBs to ACE inhibition was confirmed in a meta-analysis of 24 studies.⁴⁷ Furthermore, the antiproteinuric effect is additive with the ARBs, but because of the risk of hyperkalemia and higher risk of acute kidney injury (AKI), combination therapy is not recommended. The beneficial effects of ACE inhibition may not be enhanced by combined nondihydropyridine calcium channel antagonist therapy beyond BP lowering.^{52,53} A suggested goal of ACE inhibition or ARB therapy is to reduce proteinuria to less than 1000 mg/day or reduce proteinuria to more than 50% from baseline, in addition to protein excretion at less than 3.5 g/day.⁵⁴ BP should be reduced to less than 130/80 mm Hg as recommended by the recent American College of Cardiology/American Heart Association Task Force (ACC/AHA) guidelines.⁵⁵

Most of the vasoconstrictive action of Ang II is confined to the efferent arteriole. ACE inhibitors preferentially dilate the efferent arteriole by reducing the systemic and intrarenal levels of Ang II. The effect is a reduction in intraglomerular capillary pressure. In patients with hypertension, ACE inhibitors uniformly increase renal blood flow, decrease filtration fraction, decrease renal vascular resistance, reduce urinary protein excretion, impair microvascular autoregulation, and restore normal circadian BP patterns,^{56–58} with variable to no effect on GFR. In patients with nondiabetic glomerular disease, short-term ACE inhibitor administration causes a decrease in renal perfusion, glomerular filtration and pressure, and an increase in afferent resistances.⁵⁹ Long-term administration is associated with a decrease in renal perfusion, as well as a tendency toward a higher filtration fraction and lower afferent resistances. Marked improvement in GFR occurs and is sustained for up to 3 years.^{60,61} However, many patients with impaired renal function exhibit a reversible fall in GFR with ACE inhibitor therapy that is not detrimental. The GFR declines initially because of hemodynamic changes, but the long-term reduction in perfusion pressure is renoprotective. Even in patients receiving hemodialysis, ACE inhibitor therapy significantly preserves residual kidney function and helps maintain urine output.⁶² Patients with type 1 diabetes who have the largest initial decline in GFR have the slowest rate of loss of kidney function over time.⁶³ It should be emphasized that ACE inhibitors need not be withdrawn immediately in response to an increase in serum creatinine; a 20% to 30%

decline in GFR can be expected, and close monitoring is warranted.

An inherited trait of disordered regulation of the RAAS contributes to the pathogenesis of hypertension in approximately 45% of patients.⁶⁴ Such patients have sodium-sensitive hypertension, abnormalities in renal vascular responses to changes in sodium intake and Ang II, blunted decrements of renin release in response to saline or Ang II, and accentuated vasodilator responses to ACE inhibition; they have been termed “nonmodulators.”⁶⁴ In these patients, ACE inhibition not only increases renal blood flow substantially more than it does in normal patients, but also restores the renal vascular and adrenal responses to Ang II, renin release, renal sodium handling, and BP.^{64,65}

Treatment with ACE inhibitors is frequently associated with an initial decrease in kidney function, as reflected by an increase in the serum creatinine concentration.⁶⁶ The significance of an increase in serum creatinine after the initiation of RAAS inhibitor therapy is uncertain because it is due in part to a reduction in intraglomerular pressure, which might be expected to contribute to the slowing of disease progression in the long term. An initial elevation in the serum creatinine level of up to 30% above baseline, which stabilizes within the first 2 months of therapy, has been considered acceptable, and it is no reason to discontinue therapy in the absence of hypotension or hyperkalemia.^{67,68} Suggesting otherwise, a review of 12 randomized trials has shown that in patients receiving ACE inhibitors, a stable rise in the serum creatinine of less than 30% was associated with long-term preservation of kidney function.^{67,68}

In patients with an activated RAAS, however, ACE inhibitors can decrease GFR and may precipitate AKI. Patients with severe bilateral renal artery stenosis,⁶⁹ unilateral renal artery stenosis of a solitary kidney, severe hypertensive nephrosclerosis, volume depletion, congestive heart failure, cirrhosis, or a transplanted kidney⁷⁰ are at higher risk for deterioration in kidney function with ACE inhibitor therapy.^{71,72} These patients typically have a precipitous drop in BP and deterioration of kidney function when treated with ACE inhibitors. In these states of reduced renal perfusion related to low effective arterial circulating volume or reduced perfusion because of narrowing of arterial inflow, the maintenance of renal blood flow and GFR is highly dependent on increased efferent arteriolar vasoconstriction mediated by Ang II. Interruption of the increased tone causes a critical reduction in perfusion pressure and can lead to dramatic reductions in GFR and urinary flow, worsening of renal ischemia and, in rare cases, anuria. The hemodynamic effect described here is typically reversible with cessation of therapy.⁷³ Even paradoxical severe hypertension has been described in rare patients with renal artery stenosis treated with ACE inhibitors.⁷⁴ Among 26 older patients with hemodynamically significant renal artery stenosis who received renin–angiotensin blockers and experienced a decline in kidney function of 25% or more, 19% (5 of 26) developed ESKD, and 73% (19 of 26) showed improvement in estimated GFR (eGFR).⁷⁵

Although instances of reversible azotemia have provoked disproportionate fear among prescribers, multiple randomized controlled trials (RCTs) have shown that RAAS blockade effectively decreases the progression of kidney disease,^{76–78} and observational studies have demonstrated the benefit of RAAS blockade on BP control, adverse cardiovascular

outcomes, and all-cause mortality. Recent studies of atherosclerotic renovascular disease demonstrating the efficacy of medical therapy, including use of ACE inhibitors and ARBs over renal artery stenting, have indicated that the risks of RAAS blockade (e.g., azotemia, hyperkalemia, angioedema) can be balanced by their benefits when patients are carefully monitored.^{79–81}

It should be noted that in the studies of renoprotection by ACE inhibitors, those with advanced stage 4 or nondialysis-dependent stage 5 CKD were excluded. Some have raised the possibility that withdrawal of ACE inhibitors/ARB therapy in advanced CKD may preserve kidney function. In an observational study, withdrawal of ACE inhibitor/ARB therapy in 52 older patients with advanced CKD led to a mean increase in eGFR of 10 mL/min/1.73 m² over 12 months and an increase or stabilization in eGFR in all but four patients.⁸² However, RCTs are needed to establish the safety and efficacy of this approach.

CLASS EFFICACY AND SAFETY

ACE inhibitors are recommended for initial monotherapy in patients with mild, moderate, and severe hypertension, regardless of age, sex, or ancestry.⁸³ They are effective in patients with diabetes, obesity, and/or following kidney transplantation,⁸⁴ and are safe to use in patients with mild, moderate, and severe CKD. In general, patients with high-renin hypertension and chronic renal parenchymal disease respond particularly well, presumably because they have inappropriately high intrarenal renin and Ang II levels. African Americans with hypertension have been found to respond less well (*vis-à-vis* BP reduction) to lower dosages than whites, but higher dosages are equally effective.^{85–87} In most studies, ACE inhibitors elicit an adequate response in 40% to 60% of patients.⁸⁸ An immediate fall in BP occurs in 70% of patients. The enhanced efficacy of ACE inhibitors in the presence of salt restriction is paralleled by the additive effects of diuretic therapy.⁸⁹ The addition of low-dose hydrochlorothiazide (HCTZ) enhances the efficacy more than 80%, normalizing BP in another 20% to 25% of patients.⁹⁰ Adding a diuretic is more effective than increasing the dosage of the ACE inhibitor.^{90,91} Recent clinical trials have examined the use of the combination of an ACE inhibitor and ARB to prevent target-organ damage.⁹² The Renal Outcomes with Telmisartan, Ramipril, or Both, in People at High Vascular Risk (ONTARGET) trial compared the ACE inhibitor ramipril with the ARB telmisartan, alone and in combination, in patients at high risk for vascular disease.^{93,94} Although the achieved mean BP was lower in the patients who received telmisartan or both agents than in those who received ramipril alone, there was no difference in the primary outcomes between any of the groups, and more adverse outcomes were noted in the combination group. Importantly, this trial did not evaluate ARB and ACE inhibitor therapy in patients with advanced proteinuric renal disease. The VA NEPHRON-D study, a trial using combination therapy (ACE inhibitor and ARB therapy vs. ARB monotherapy) in patients with proteinuric diabetic nephropathy, was stopped because of increased adverse events of hyperkalemia and AKI.⁹⁵ The Aliskiren Trial in Type 2 Diabetes Using Cardiorenal Endpoints (ALTITUDE) randomly assigned 8561 patients to aliskiren (300 mg daily), a direct renin inhibitor, or placebo as an adjunct to ACE or ARB monotherapy. The trial was stopped prematurely because

of adverse events (hyperkalemia and hypotension).⁹⁶ Therefore, among patients with type 2 diabetes and kidney disease, ACE inhibitors should not be used concomitantly with ARBs and renin inhibitors because of an increased risk of hypotension, hyperkalemia, and aggravated renal dysfunction.⁹⁷

Many studies have attempted to achieve further benefit from ACE inhibitors and other RAAS blocking agents by increasing the doses of these drugs. This reasoning is based on the original observation that the optimal antiproteinuric dose does not necessarily equal the optimal antihypertensive dose. Many of these study results have shown further proteinuria reduction,^{98–102} whereas others have not.^{103–105} However, similar to the initial studies of the combination therapy with RAAS blocking agents, many of these high-dosage studies were also short-term studies that used BP and albuminuria as outcome variables. These studies have not been sufficiently powered, or they lacked duration to detect safety signals and side effect rates that may emerge with endpoint trials.⁹⁷ Therefore before ultrahigh-RAAS blocking agent dosing as renoprotective therapy can be recommended, more studies with hard kidney and cardiovascular event points are warranted.⁹⁷

Neither the duration nor the degree of BP lowering is predicted by the effect on blood ACE or Ang II levels, and all ACE inhibitors appear to have comparable efficacy. The response may be due in part to interindividual variability of the ACE genotype.^{87,106} The activity of ACE is partially dependent on the presence or absence of a 287–base pair element in intron 16, and this insertion–deletion (I/D) polymorphism of a human Al repetitive DNA element accounts for 47% of the total phenotypic variation in plasma ACE. Deletion–deletion cases have the highest ACE concentrations, I/D cases have intermediate ACE concentrations, and insertion–insertion cases have the lowest. Genotype also influences tissue ACE activity, but the clinical implications remain incompletely understood.

ACE inhibitors are indicated as first-line therapy in hypertensive patients with heart failure and systolic dysfunction,¹⁰⁷ type 1 diabetes and proteinuria,⁶³ reduced systolic function after myocardial infarction (MI),¹⁰⁸ coronary artery disease or new atrial fibrillation,^{109,110} and left ventricular dysfunction,^{83,110,111} as well as patients undergoing dialysis.¹¹² ACE inhibitors reduce ventricular hypertrophy independent of the BP decrease.¹¹³ All patients with diabetes, even those without evidence of nephropathy, should be given ACE inhibitors for cardiovascular risk reduction.^{114–118} Primary and secondary prevention trials^{114,119,120} have shown that ACE inhibitors improve endothelial dysfunction and cardiac and vascular remodeling, retard the progression of atherosclerosis, improve arterial distensibility, and reduce the risk of myocardial ischemia and infarction, stroke, and cardiovascular death. Some but not all studies have shown a small decrease in the risk of new and recurrent atrial fibrillation because of the beneficial structural and electrical effects on the atria.^{121,122} The use of ACE inhibitors is also associated with improved exercise performance in patients with hypertension and intermittent claudication,¹²³ reduced pain perception,¹²⁴ and reduced perioperative myocardial ischemia,¹⁰⁸ as well as the retardation of the progression of aortic stenosis,¹²⁵ protection against cognitive decline and dementia,^{126,127} promotion of atrial structural remodeling,¹²⁸ prolongation of the survival of arteriovenous polytetrafluoroethylene grafts,¹²⁹ and possible

reduced risk of pneumonia¹³⁰ and prevention of osteoporosis.¹³¹ Long-term use of these agents has been associated with a lower risk for breast cancer, but the importance of this finding has yet to be substantiated.¹³²

ACE inhibitors may cause fetal or neonatal injury or death when used during the second and third trimesters of pregnancy¹³³ because angiotensin appears to be required for normal fetal growth and development.¹³⁴ Concern has been expressed regarding first-trimester use of ACE inhibitors because their use during this period might be associated with congenital malformations, but these malformations may also be related to maternal factors.¹³⁵ A recent study has found that maternal use of ACE inhibitors in the first trimester shows a risk profile similar to that of other antihypertensive agents.¹³⁶ Recognizing that the benefits of ACE inhibitors are largely long term, we advise women attempting conception to transition to an alternative antihypertensive agent with a proven safety profile in pregnancy (e.g., labetalol).

A systemic review of published case reports and case series dealing with intrauterine exposure to ACE inhibitors or ARBs has shown 118 cases of ACE inhibitor exposure and 68 cases of ARB exposure. Among 118 cases of ACE inhibitor exposure, 27% were taken only in the first trimester; in 68 cases of ARB exposure, 38% were taken throughout the entire pregnancy. Neonatal complications were more frequent following exposure to ARBs; 52% of the newborns exposed to ACE inhibitors did not exhibit any complications, whereas only 13% of the newborns exposed to ARBs did not ($P < .0001$); however, many of the patients exposed to ARBs received therapy throughout the pregnancy.¹³⁷ In 26 children, 23% developed kidney failure, 8% required dialysis, 15% demonstrated hypertension, 8% developed acidosis, 12% developed polyuria, 15% showed small or echogenic kidneys on ultrasound, 8% had polycythemia, 15% had growth retardation, and 12% experienced neurodevelopmental delay. The outcome was normal in 50% of cases, mild in 42%, and bad in 8%. Complications were similar whether the children were exposed to ACE inhibitors or ARBs. The authors commented that the congenital malformations that occurred following in utero RAAS exposure might have resulted from the drug as well as the underlying maternal illness, usually hypertension or diabetes.

A prospective, observational, controlled cohort study of ACE inhibitors and ARB exposure during the first trimester was reported in 2012.¹³⁸ Participants were enrolled from an unselected sample of women contacting a teratogen information service. There were two comparison groups, women with hypertension treated with other antihypertensives (including methyldopa or calcium channel blockers [CCBs]) and healthy controls. In the ACE-ARB and disease-matched groups, the offspring exhibited significantly lower birth weights and gestational ages than those of the healthy controls ($P < .001$ for both variables). A significantly higher rate of miscarriage was noted in the ACE-ARB group ($P < .001$). These results suggest that ACE inhibitors and ARBs are not major human teratogens during the first trimester. There was, however, a higher rate of spontaneous abortions in the ACE-ARB group.

When a patient becomes pregnant during treatment with an ACE inhibitor or ARB, the drug should be discontinued immediately, and alternative antihypertensive therapy should be started. Pregnancy termination is at the discretion of the patient and treatment team, but physicians and patients

can derive some reassurance from the small studies quoted earlier if the drug is discontinued during the first trimester. Recognizing that the benefits of ACE inhibitors are largely long term, we advise women attempting conception to transition to an alternative antihypertensive agent with a proven safety profile in pregnancy (e.g., labetalol) (Table 49.6; see also Chapter 48).

ACE inhibitors are transferred into breast milk, but the drug levels in milk are low. Captopril and enalapril have been reviewed by the American Academy of Pediatrics¹³⁹ and the National Institute for Health and Care Excellence (NICE) guidelines for the management of hypertension in pregnancy from the United Kingdom,¹⁴⁰ and they are compatible with lactation. However, newborns may be more susceptible to the hemodynamic effects of these drugs (e.g., hypotension) and sequelae (e.g., oliguria, seizures).

Overall, ACE inhibitors are well tolerated and have relatively neutral or beneficial metabolic effects. ACE inhibitors are associated with 8% to 11% reductions in levels of low-density lipoprotein (LDL) cholesterol and triglycerides and 5% increases in levels of high-density lipoprotein (HDL) cholesterol.¹⁴¹ They do not cause perturbations of serum sodium or uric acid levels. ACE inhibitors reduce the levels of plasminogen activator inhibitor-1 and may improve fibrinolysis.¹⁴²

The effects of ACE inhibitors on glucose metabolism are favorable.¹⁴³ They may improve glucose tolerance by augmenting the insulin secretory response to glucose¹⁴³ and may also help ameliorate obesity and hyperinsulinemia.¹⁴⁴ The use of ACE inhibitors has been clinically associated with a 25% to 30% reduction in the risk of developing diabetes.¹⁴⁵ Several large clinical trials have evaluated the clinical relevance of this finding.¹⁴⁶ Many of the ACE inhibitors require dosage adjustment in the presence of impaired kidney function, as described earlier²² (Table 49.7).

There are relatively few adverse effects (AEs) of this class of drugs, and AEs may occur with all ACE inhibitors. The newer agents appear to have a lower incidence of AEs, possibly because of lack of the sulfhydryl moiety found in captopril. The most common AE of ACE inhibitors is a dry, hacking, nonproductive cough, which has been reported in up to 20% of patients.¹⁴⁷ In the ONTARGET trial, cough sufficiently severe to permanently discontinue the drug was described in 4.2% of patients treated with ramipril¹⁴⁸; cough is much less common with ARBs. The ACE inhibitor-induced cough is thought to be secondary to hypersensitivity to bradykinins, which are increased by ACE,¹⁴⁹ increased levels of prostaglandins,¹⁵⁰ accumulation of substance P,¹⁴⁷ a potent bronchoconstrictor,¹⁵¹ or polymorphisms in the neurokinin-2 receptor gene.¹⁵² The cough can begin initially or many months after the start of therapy.¹⁵³ It is more common in women, African Americans,¹⁵⁴ and Asians in Hong Kong¹⁵⁵ and Korea,¹⁵⁶ and it may spontaneously disappear.¹⁵³ It may be more common in patients with bronchial hyperreactivity, but ACE inhibitors are safe to use in asthmatic patients.¹⁵⁷ NSAIDs, oral iron supplements, and sodium cromoglycate have been reported to improve the cough,¹⁵⁸ but cessation of ACE therapy is the only absolute cure. In a dosage of 600 mg twice daily, picotamide (a thromboxane antagonist) was effective in treating ACE inhibitor-induced cough.¹⁵⁹ Patients may be effectively switched to ARBs if an antihypertensive effect is observed with ACE inhibitors.

Table 49.6 Antihypertensive Therapies Commonly Used in Pregnancy

Agent	Indication	Dose Range	FDA Classification	Potential Side Effects	Comments
α 2-Methylidopa	Often used as first line	250 mg to 1.5 g orally twice a day	B	Lethargy	Data on offspring up to 7.5 years of age demonstrating long-term safety
Labetalol	Often used as first line	100–1200 mg	C	Exacerbation of asthma	Widely used in pregnancy
Calcium channel blockers	Second line of alternative first line (nifedipine)	Varies according to drug used	C	Concern for synergy with magnesium sulfate for neuromuscular depression	
β -Blockers	Second line	Varies according to drug used	C	Exacerbation of asthma	Some recommend avoiding atenolol during pregnancy and lactation
Thiazide diuretics	Second line	12.5–50 mg orally once a day	C	Volume depletion and hypokalemia	

Angiotensin-converting enzyme inhibitors and angiotensin receptor blockers are contraindicated in the second and third trimesters of pregnancy and are class D. Safety in the first trimester is controversial: in this trimester, they are class C.
 FDA, U.S. Food and Drug Administration.

FDA Classification of Drugs in Pregnancy

Category A: Controlled studies in pregnant women have demonstrated no risk of fetal abnormalities

Category B: Reproduction studies in animals indicate no fetal risk, but there are no adequate and well-controlled studies in pregnant women; or animal reproduction studies have shown an adverse effect of the drug but not in well-controlled studies in pregnant women.

Category C: Animal reproduction studies have shown an adverse effect on the fetus but no adequate human or animal studies; or animal reproduction studies have been conducted and it is not known whether the drug can cause fetal harm when administered to a pregnant woman.

Category D: Evidence of human fetal risk, but benefits outweigh risks. Women taking this drug during pregnancy should be informed of potential fetal risk.

Category X: Evidence of human fetal risk. Drug is contraindicated in women who are pregnant or who may become pregnant. If this drug is used during pregnancy or if the patient becomes pregnant while taking this drug, the patient should be apprised of the potential hazard to a fetus.

Table 49.7 Dose Modifications of Antihypertensive Drugs Required for Renal Insufficiency

	Estimated Glomerular Filtration Rate (Creatinine Clearance; mL/min/1.73 m ²)			
Drug	>50	10–15	<10	Dialysis ^a
Angiotensin-Converting Enzyme Inhibitors				
Benazepril	No change	50%	25%	Negligible
Captopril	No change	50%	25%	H: 50%
Cilazapril	No change	50%	25%	H: 50%
Enalapril	No change	50%	25%	H: 50%
Fosinopril	No change	No change	75%	—
Imidapril	No change	No change	—	—
Lisinopril	No change	50%	25%	H: 50%
Moexipril	No change	50%	25%	—
Perindopril	No change	75%	50%	—
Quinapril	No change	50%	25%	—
Ramipril	No change	50%	25%	—
Trandolapril	No change	50%	25%	—
Zofenopril	No change	—	—	—
Angiotensin Receptor Blockers				
Candesartan	No change	No change	No change	Negligible
Eprosartan	No change	No change	50%	Negligible
Irbesartan	No change	No change	—	Negligible

Table 49.7 Dose Modifications of Antihypertensive Drugs Required for Renal Insufficiency (Cont'd)

	Estimated Glomerular Filtration Rate (Creatinine Clearance; mL/min/1.73 m²)			
Drug	>50	10–15	<10	Dialysis ^a
Losartan	No change	No change	No change	Negligible
Olmesartan	No change	—	—	—
Telmisartan	No change	No change	No change	Negligible
Valsartan	No change	No change	No change	—
Adrenergic Antagonists				
Nadolol	No change	50%	25%	H: 50%
Carteolol	No change	50%	25%	—
Penbutolol	No change	No change	50%	Negligible
Pindolol	No change	No change	50%	Negligible
Atenolol	No change	50%	25%	H: 50%
Betaxolol	No change	No change	50%	H: 50%
Bisoprolol	No change	50%	25%	Negligible
Acebutolol	No change	50%	30%–50%	H: 50%
Celiprolol	No change	50%	Avoid	—
Nebivolol	No change	50%	—	—
Calcium Channel Blockers				
Diltiazem	No change	No change	No change	Negligible
Verapamil	No change	No change	No change	Negligible
Nifedipine	No change	No change	No change	Negligible
Amlodipine	No change	No change	No change	Negligible
Felodipine	No change	No change	No change	Negligible
Isradipine	No change	No change	No change	Negligible
Manidipine	No change	No change	No change	Negligible
Nicardipine	No change	No change	No change	Negligible
Nisoldipine	No change	No change	No change	Negligible
Lacidipine	No change	No change	No change	Negligible
Lercanidipine	Dosage adjustment in renal failure unknown			
Central α ₂ -Adrenergic or I ₁ Imidazole Receptor Agonists				
Methyldopa	No change	No change	50%	H: 50%
Clonidine	No change	50%	25%	Negligible
Moxonidine	No change	50%	—	—
Rilmenidine	No change	50%	25%	—
Direct-Acting Vasodilators				
Hydralazine	No change	No change	75% ^b	Negligible
Minoxidil	No change	50%	50%	H and P: 50%
Peripheral Adrenergic Neuronal Blocking Agents				
Guanethidine	No change	No change	—	—
Guanadrel	No change	50%	25% (avoid)	—
Renin Inhibitor				
Aliskiren	No change	No change	Not studied	Not studied
Tyrosine Hydroxylase Inhibitor				
Metyrosine	No change	50%	25%	—
Selective Aldosterone Receptor Antagonist				
Eplerenone	Dosage adjustment in renal failure unknown Caution in regard to hyperkalemia			

Percentage of usual dose given.

^aReplacement dose at end of dialysis as percentage of dose prescribed for patient with glomerular filtration rate <10 mL/min.

^bSlow acetylators.

H, Hemodialysis; P, peritoneal dialysis; —, not applicable.

Angioedema is a rare but potentially life-threatening complication of ACE inhibitor therapy. It occurs in 0.1% to 0.7% of patients within hours of the first dose of ACE inhibitor or after prolonged use.^{160–162} In the ONTARGET Trial, the occurrence of angioedema, although potentially life-threatening, was reported and was listed as a reason to discontinue from the study permanently in just 0.3% of more than 8500 individuals given ramipril.¹⁴⁸

This absolute risk of ACE inhibitor–induced angioedema is low, but with large numbers of prescriptions written annually, many patients are at risk for this disorder.¹⁶³ ACE inhibitor–induced angioedema accounts for one-third of all cases of angioedema seen in emergency departments.

Our understanding of the mechanism of ACE inhibitor–induced angioedema is evolving. The side effect occurs five times more frequently in individuals of African ancestry. ACE inhibitors act by inhibiting bradykinin breakdown in addition to blocking the conversion of Ang I to Ang II. ACE inhibitor–induced angioedema involves several components, including tissue accumulation of bradykinin and inhibition of C1 esterase activity.^{160,164} Susceptible individuals typically have defects in non-ACE, nonkininase I vasoactive pathways of bradykinin degradation, possess the XPNPEP2 gene variant,^{165–172} have elevated des-Arg9-BK,¹⁶⁷ or are taking dipeptidyl peptidase inhibitors (e.g., sitagliptin, saxagliptin, linagliptin) to treat diabetes.^{160,168–170} Some patients also have defective degradation of substance P, thereby increasing vascular permeability.¹⁷¹

Clinical features, including asymmetric swelling confined to the face, subcutaneous or submucous membranes, and lips, usually resolve with discontinuation of the therapy, but obstructive sleep apnea may be exacerbated.¹⁷² If the ACE inhibitor is not discontinued, the episode usually resolves, but the frequency and severity of future episodes will escalate.^{173,174} Angioedema of the small intestine and acute appendicitis have also been reported.^{175–178}

Involvement of the glottis and larynx requiring airway management occurs in 10% of all cases and may result in laryngeal obstruction and death.¹⁷⁹ Administration of epinephrine, histamine-2 blockers, glucocorticoids, and/or fresh-frozen plasma is indicated.¹⁸⁰ Recently, icatibant, a selective bradykinin B2 receptor antagonist approved for the treatment of hereditary angioedema, has also been shown to be effective for the treatment of ACE inhibitor–induced angioedema in a multicenter, double-blind, phase 2 study. In 27 patients, the median time to resolution was 8 hours with icatibant (interquartile range, 3.0 to 16.0 hours) compared with 27.1 hours with standard therapy (interquartile range, 20.3 to 48.0 hours) with a glucocorticoid and an antihistamine. Patients experiencing an episode of angioedema associated with ACE inhibitor usage should be switched to ARBs or other agents.¹⁸¹ ACE inhibitors are contraindicated in patients with a known hypersensitivity to ACE inhibitors.

First-dose hypotension, with a reduction in BP of up to 30%, has been reported with all ACE inhibitors in up to 2.5% of patients. In the Studies of Left Ventricular Dysfunction (SOLVD trial), hypotension was observed in 14.8% of patients versus 7.1% of individuals who received a placebo ($P < .0001$).¹⁸² In the ONTARGET Trial, of the 8579 patients who received ramipril, only 1.7% and 0.2% permanently stopped therapy because of hypotension and syncope, respectively.¹⁶⁰

Hypotension occurs more commonly in patients with effective arterial volume depletion, patients with high-renin hypertension, and those with systolic heart failure.^{182,183} Hypotension is usually well tolerated, although occasionally it is associated with syncope. In older patients, ACE inhibitor therapy more frequently causes nocturnal hypotension.¹⁸⁴ The accompanying increase in the plasma norepinephrine concentration may explain the low incidence of orthostatic symptoms.¹⁸⁴ In patients at high risk of orthostatic symptoms, therapy should be initiated at lower dosages, preferably after discontinuing diuretics. Rebound hypertension has not been reported with discontinuation of ACE inhibitors.

AEs related to the chemical structure are more frequently seen with the sulfhydryl-containing captopril than with the other agents. Dysgeusia appears to be related to the binding of zinc by the ACE inhibitors.¹⁸⁵ Approximately 2% to 4% of patients experience a diminution or loss of taste sensation that is associated with a metallic taste. It is usually self-limited and resolves in 2 to 3 months, even with continued therapy. However, it may be severe enough to interfere with nutrition and cause weight loss.¹⁸⁶

Cutaneous reactions manifest as a nonallergic, pruritic, maculopapular eruption that appears during the first few weeks of therapy. These reactions may be associated with a fever or arthralgias and may disappear, even with continuation of the ACE inhibitor.¹⁸⁷

Leukopenia and anemia have been reported with ACE inhibitor therapy. Among patients with normal kidney function, ACE inhibitors may reduce hemoglobin concentrations in a dose-dependent manner.¹⁸⁸ ACE inhibitors have been demonstrated to interfere with the response to erythropoietin. Patients receiving hemodialysis and kidney transplant recipients receiving erythropoietin frequently require higher dosages to maintain hemoglobin concentrations.¹⁸⁹ Consequently, ACE inhibitors can be used effectively to reduce posttransplantation erythrocytosis,¹⁹⁰ but appear to have little effect on erythropoiesis in patients receiving hemodialysis.¹⁹¹ Neutropenia (<1000 neutrophils/mm³) with myeloid hypoplasia occurs almost exclusively in patients with impaired kidney function, immunosuppression, collagen vascular disease, or autoimmune disease.¹² Neutropenia occurs within 3 months of initiation of therapy and generally resolves 2 weeks after therapy is discontinued.^{12,187}

Anaphylactoid reactions ranging from mild pruritus to bronchospasm and cardiopulmonary collapse have been reported in patients treated with ACE inhibitors who undergo dialysis with equipment that uses high-flux polyacrylonitrile, cellulose acetate, or cuprophane membranes or who undergo apheresis with equipment that uses dextran sulfate membranes.¹⁹² The frequency of reactions is unknown, but they occur within the first few minutes of treatment. Such membranes should be avoided when patients are receiving ACE inhibitors. The use of ACE inhibitors in patients on plasmapheresis or an exchange protocol is safe as long as effective arterial volume is managed appropriately.

Few significant drug interactions occur with ACE inhibitors. Studies have shown that aspirin dosages of 100 mg/day or less do not negate the effects of ACE inhibitors.¹⁹³ Ang II stimulates the production of vasodilatory prostaglandins. Aspirin inhibits the production of vasodilator and antithrombotic prostaglandins. Theoretically, either agent may antagonize the effectiveness of the other. Concomitant use of ACE

inhibitors and cyclosporine may exacerbate renal hypoperfusion.¹² Use of the mammalian target of rapamycin inhibitors sirolimus or everolimus in transplant recipients decreases the metabolism of bradykinins and predisposes them to angioedema with ACE inhibitors.¹⁹⁴

Hyperkalemia of more than 5.5 mmol/L was observed in 3.3% of patients taking ramipril in the ONTARGET study.¹⁴⁸ Production of Ang II systemically and locally in the zona glomerulosa of the adrenal gland, which is blocked by ACE inhibitors, will reduce subsequent aldosterone synthesis and urinary potassium excretion. In a Veterans Administration Medical Center case-control study of 1818 patients using ACE inhibitors, 194 (11%) developed hyperkalemia.¹⁹⁵ The results of laboratory studies indicated that a serum urea nitrogen concentration higher than 6.4 mmol/L and a serum creatinine concentration higher than 136 μ mol/L, as well as congestive heart failure and long-acting ACE inhibitors, were independently associated with hyperkalemia; concurrent use of a loop or thiazide diuretic agent was associated with reduced risk. After 1 year of follow-up, 15 (10%) of 146 patients who remained on a regimen of an ACE inhibitor developed severe hyperkalemia (serum potassium >6.0 mmol/L). A serum urea nitrogen level higher than 8.9 mmol/L and age older than 70 years were independently associated with subsequent severe hyperkalemia.

Hyperkalemia has been effectively and safely treated with patiromer and sodium zirconium cyclosilicate recently in an outpatient setting.^{196,197} These two new drugs add to the outpatient pharmacopoeia that until now has been limited to sodium and calcium polystyrene sulfonate. Correction of metabolic acidosis, if present, may also reduce the incidence and severity of hyperkalemia.

ANGIOTENSIN II TYPE 1 RECEPTOR ANTAGONISTS

CLASS MECHANISMS OF ACTION

The Ang II receptor blockers (ARBs) allow more specific and complete blockade of the RAAS than the ACE inhibitors because they circumvent all pathways that lead to the formation of Ang II (Fig. 49.2). For example, Ang I is metabolized not only by ACE to form Ang II but also by chymase, cathepsin G, tissue plasminogen activator, and other enzymes.¹⁹⁸ Ang II can be formed at sites other than those in the systemic circulation, such as the brain, kidney, and heart. Furthermore, long-term ACE inhibitor therapy is associated with a return of Ang II levels to baseline, which possibly contributes to reduced efficacy. The ARBs selectively antagonize Ang II directly at the Ang II type 1 (AT₁) receptor, regardless of the source of production. Because Ang II plays a crucial multifactorial role in maintaining and regulating BP, blockade of the AT₁ receptor with ARBs is a powerful tool for targeting multiple pathways that contribute to hypertension.

Like ACE inhibitors, ARBs directly block the vasoconstrictive action of Ang II and cause a decrease in peripheral vascular resistance.¹⁹⁹ Interruption of the binding of Ang II at the tissue level also leads to other effects (beyond vasodilation) that contribute to the antihypertensive effect. Additional mechanisms include the following: (1) augmentation of renal blood flow and reduction of aldosterone release to induce natriuresis and attenuate the compensatory increase in sodium retention that accompanies a fall in BP; (2) direct

depression of tubular sodium reabsorption^{200,201}; (3) improvement of nitric oxide-mediated endothelial function²⁰²; (4) reversal of vascular hypertrophy²⁰²; (5) blunting of SNS activity and presynaptic norepinephrine release; (6) inhibition of postjunctional pressor responses to norepinephrine or Ang II; (7) inhibition of central Ang II-mediated sympathoexcitation and vasopressin release^{203–205}; (8) inhibition of centrally controlled baroreceptor reflexes²⁰⁶; (9) inhibition of central nervous system (CNS) norepinephrine synthesis; (10) inhibition of thirst²⁰⁷; and (11) possible inhibition of RAAS-mediated action on endothelin-1.²⁰⁸ The antihypertensive action of ARBs is dependent on activation of the RAAS and is associated with clinically insignificant increases in circulating levels of Ang II.¹⁹⁹ ARBs also increase bradykinin levels by antagonizing Ang II at its type 1 receptor and diverting Ang II to its counterregulatory type 2 receptor, which potentiates vasodilation.²⁰⁹ ARBs also increase the level of other angiotensin peptides, including angiotensin-(1-7), Ang II, and angiotensin IV, which can act on their respective receptors to modulate vasoconstriction, renal blood flow, and vascular hypertrophy.^{210–215}

CLASS MEMBERS

The ARB class is composed of peptide and nonpeptide analogs that vary in structure, mechanism of receptor inhibition, metabolism, and potency. There are currently eight drugs in clinical use. Many of the newer drugs were developed by modifying losartan, the first biologically active ARB oral agent. These drugs are categorized according to the substitution of carboxylic and other moieties into several groups—the biphenyl tetrazoles (derivatives of losartan), nonbiphenyl tetrazoles, and nonheterocyclic compounds. They are also classified according to their ability to antagonize Ang II.²¹⁶ The competitive (surmountable) antagonists shift the dose-response curve for Ang II-mediated contraction to the right without depressing the maximal response to Ang II. The noncompetitive (insurmountable) antagonists also depress the maximal response to Ang II. The variable effects of ARBs are mediated by differences in the interaction with allosteric binding sites on the receptor, dissociation of the drug-receptor complex, removal of the agonists from tissues, or by the ability to modulate the amount of internalized receptors²¹⁶ (Tables 49.8–49.10).

Table 49.8 Pharmacokinetic Interactions Between Angiotensin II Type 1 (AT₁) Receptor Blockers and Receptor

Agent	AT ₁ Receptor- Receptor Antagonist Dissociation Rate	Affinity (K _d)	Type of AT ₁ Antagonism
Candesartan cilexetil (candesartan)	Slow	280	Noncompetitive
Irbesartan	Slow	5	Noncompetitive
Valsartan	Slow	10	Noncompetitive
Telmisartan	Slow	10	Noncompetitive
Losartan	Fast	50	Competitive
Eprosartan	Fast	100	Competitive

Table 49.9 Pharmacodynamic Properties of Angiotensin Receptor Blockers

Generic Name (Trade Name)	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Eprosartan (Teveten)	200	400–800	800	qd, bid	4	24
Irbesartan (Avapro)	150	150–300	300	qd	4–6, 14	24
Losartan (Cozaar)	50	50–100	100	qd/bid	6	12–24
Valsartan (Diovan)	80	80–160	320	qd	4–6	24
Candesartan (Atacand)	16	8–32	32	qd	6–8	24
Telmisartan (Micardis)	40	40–80	80	qd	3	24
Azilsartan (Edarbi)	40	40–80	80	qd	1.5–3 peak response	≥24
Olmesartan (Benicar)	20	20–40	40	qd	1–2	24

bid, Two times a day; *qd*, once a day.

Table 49.10 Pharmacokinetic Properties of Angiotensin Receptor Blockers

Drug	Absorption (%)	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Eprosartan	>80	13	Yes	4	6	L	F (70%), U (7%)	Inactive
Irbesartan	>80	60–80	No	1.5–2	10–14	L, K	F (65%), U (20%)	Inactive
Losartan	>80	25	No	1	4–9	L, K	F (60%), U (40%)	Active
Valsartan	>80	25	Yes	2–4	6–9	L, K	F (83%), U (13%)	Inactive
Candesartan	—	15	No	2–4	9	I, L, K	F (67%), U (33%)	None
Telmisartan	—	42	Yes	0.5–1	24	L	F	Inactive
Olmesartan	—	26	No	1	13	I	F (50%), U (50%)	Active

F, Feces; I, intestine; K, kidney; L, liver; U, urine.

Biphenyl Tetrazole and Oxadiazole Derivatives

Azilsartan medoxomil is a selective AT₁ receptor blocker that has demonstrated a potent 24-hour sustained antihypertensive effect. At the approved dosage, it reduces systolic BP (SBP) by 12 to 15 mm Hg and diastolic BP (DBP) by 7 to 8 mm Hg.^{217–219} Azilsartan medoxomil is a prodrug that is hydrolyzed to azilsartan in the gastrointestinal tract during absorption. It possesses a unique moiety (5-oxo-1,2,4-oxadiazole) in place of the tetrazole ring that offers a very strong inverse antagonism at the AT₁ receptor.^{220,221} This bond is chemically stronger than that of its predecessors, which may explain its added potency when compared with other members of its class or ACE inhibitors.²²⁰ The estimated oral bioavailability is about 60%; peak plasma concentration is reached after 1.5 to 3 hours following ingestion.²²² Food does not affect the bioavailability of the drug. More than 99% of azilsartan is bound to albumin. The initial starting dose is 20 mg once daily, and it is available in 20-, 40-, and 80-mg tablets. The terminal half-life is 9 hours, and approximately 55% of the parent compound is excreted by the kidney.²²² In a recent study to assess the pharmacokinetics of kidney disease of azilsartan, no dosage adjustment was advised for kidney disease or hemodialysis.²²³ In a report, 17 patients receiving hemodialysis who switched to azilsartan showed a decrease in SBP from 150.9 ± 16.2 to 131.3 ± 21.7 mm Hg ($P < .008$) and a decrease in DBP from 84.1 ± 6.3 to 74.9 ± 8.3 mm Hg.²²⁴

Candesartan cilexetil is an esterified prodrug imidazole that is rapidly and completely converted into the active

7-carboxylic acid candesartan (CV-11974) in the intestinal wall.¹² Candesartan is a selective, nonpeptide, noncompetitive (insurmountable) ARB with the second highest receptor binding affinity and a slow detachment rate from the receptor (Table 49.8). Consequently, the effects are long-lasting and unlikely to be overcome by the upregulation of Ang II that commonly accompanies AT₁ receptor blockade. The initial dose is 16 mg daily, and the usual daily dose is 8 to 32 mg in one or two divided doses. The antihypertensive response occurs initially in 2 to 4 hours, peaks at 6 to 8 hours, and lasts for 24 hours (Tables 49.9 and 49.10).^{216,225} Radioreceptor assays demonstrate the presence of candesartan at the receptor site for longer than predicted periods (from plasma half-life analysis), which correlates with the clinical observation of a sustained effect beyond 24 hours.^{225,226} Maximal response is achieved in 4 weeks. The terminal half-life of candesartan is approximately 9 hours, and it is not affected by kidney failure. No unchanged parent compound is detected in the serum or urine. Candesartan is not dialyzable.

Eprosartan is a nonpeptide selective ARB that was modified to more closely resemble Ang II.²²⁷ It is a noncompetitive antagonist, with a high affinity for the AT₁ receptor (Table 49.8).¹² The initial daily dose is 200 mg (Table 49.9), and the usual daily dose is 200 to 400 mg. Eprosartan is rapidly absorbed, but its absorption is delayed by food (Table 49.10). The initial response occurs in 4 hours and lasts for 24 hours. The elimination half-life is 6 hours. Dosages should be reduced by 50% in patients with kidney failure.

Irbesartan is a nonpeptide-specific imidazolinone derivative of losartan that acts as a noncompetitive AT₁ receptor blocker with a very high receptor binding affinity²²⁸ (Table 49.8).¹² The initial dose is 150 mg daily, and the usual daily dose is 150 to 300 mg. The initial response occurs in 2 hours. The peak response is bimodal; in hypertensive patients, peak responses occur in 4 to 6 hours and 14 hours, corresponding to the peak increases in plasma renin activity and Ang II levels, respectively.²²⁸ With continuous dosing, the maximal effect may not be seen for up to 6 weeks. The duration of action is 24 hours. Irbesartan is not dialyzable.

Losartan potassium is the prototype ARB. The tetrazole moiety on the biphenyl ring accounts for its activity in oral form and its duration of action. It was the first oral active agent, and it is a nonpeptide competitive, selective AT₁ receptor inhibitor, with moderate receptor binding affinity (Table 49.8).^{12,229} The usual starting dosage is 50 mg once daily (Table 49.9). Dosage adjustments should be made at weekly intervals. The antihypertensive efficacy may be improved with divided doses. The usual daily dose is 50 to 100 mg. The potassium content of the 25-, 50-, and 100-mg tablets is 0.054, 0.108, and 0.216 mEq, respectively. The oral bioavailability of losartan is 25%, and it is unaffected by food (Table 49.10). The initial response occurs in 1 hour, and the response peaks at 6 hours and lasts for 24 hours. Only 5% of losartan is recovered unchanged in the urine, which supports extensive metabolism and biliary secretion. Neither the parent drug nor metabolites are removed by dialysis.²²

Olmesartan medoxomil is a nonpeptide selective ARB prodrug that is rapidly and completely bioactivated by hydrolysis to olmesartan during absorption from the gastrointestinal tract.²³⁰ The initial dosage is 20 mg daily, and the usual dosage is 20 to 40 mg daily (Table 49.9). The peak plasma concentration is reached in 1 hour (Table 49.10). The BP-lowering effect lasts for 24 hours and peaks at 2 weeks. Olmesartan is eliminated in a biphasic manner, with a terminal half-life of 13 hours. Dosing and pharmacokinetics have not been studied in dialysis patients.

Nonbiphenyl Tetrazole Derivatives

Telmisartan incorporates a carboxylic acid as the biphenyl acidic group. Telmisartan is a nonpeptide, noncompetitive ARB, with high specificity and receptor affinity.^{12,231} The usual starting dosage is 40 mg daily, and the usual daily dose is 40 to 80 mg. The initial response occurs in 3 hours, and it is dose dependent (Table 49.10). The duration of action is 24 hours but may last up to 7 days after discontinuing the drug. Women typically achieve plasma levels two to three times higher than those of men, but this result is not associated with differences in BP response. Less than 3% of the drug is metabolized in the liver into inactive compounds. The elimination half-life is 24 hours. Telmisartan is not dialyzable, and dosage adjustment is not necessary in patients with kidney disease.

Nonheterocyclic Derivatives

Valsartan is a nonheterocyclic ARB in which the imidazole of losartan is replaced by an acetylated amino acid.¹² Valsartan is a noncompetitive antagonist with high specificity and receptor binding affinity (Table 49.8). The initial starting dosage is 80 mg once daily (Table 49.9), and the usual dosage is 80 to 160 mg daily. The maximal BP response is achieved

after 4 weeks of therapy. The initial response occurs in 2 hours, peaks at 4 to 6 hours, and lasts 24 hours. Valsartan does not undergo significant metabolism. The elimination half-life is 6 to 9 hours, and it is not affected by kidney failure (Table 49.10).

CLASS RENAL EFFECTS

Intrarenal Ang II receptors are widely distributed in the afferent and efferent arterioles, glomerular mesangial cells, inner stripe of the outer medulla, and medullary interstitial cells,²³² as well as on the luminal and basolateral membranes of the proximal and distal tubule cells, collecting ducts, podocytes, and macula densa cells.²³³ Most receptors are of the AT₁ subclass. Circulating and predominantly locally produced Ang II interacts with the receptors; the complex is internalized and Ang II is released into the intracellular compartment, where it exerts its effects. Studies have suggested that most renal interstitial Ang II is formed at sites not readily accessible to ACE inhibition or is formed by non-ACE pathways.

ARBs antagonize the binding of Ang II and cause a number of intrarenal changes. The overall renal hemodynamic responses of AT₁ receptor blockade are variable, depending on the counteracting influences of the decrease in arterial pressure.^{200,234} Decreases in systemic arterial pressure by ARBs may be associated with compensatory activation of the intrarenal SNS, resulting in decreased renal function. This effect is more pronounced in sodium-depleted states because activation of the renin-angiotensin system helps maintain arterial and renal pressure. By contrast, direct intrarenal infusions of ARBs cause an increase in sodium excretion.²³⁵ The enhanced sodium excretion has been shown to be caused by direct inhibition of sodium reabsorption by the proximal tubules, but it may also be caused by hemodynamic changes in medullary blood flow and tubule absorption in distal nephron segments. Because Ang II blockade enhances the ability of the kidneys to excrete sodium, sodium balance can be maintained at lower arterial pressures. Ang II blockade also reduces tubuloglomerular feedback sensitivity by decreasing macula densa transport of sodium chloride to the afferent arteriole.²³⁶ This leads to increased delivery of sodium chloride to the distal segments for excretion, without compensatory changes in GFR.

In addition to the natriuretic and diuretic actions, short-term administration of some ARBs has been observed to induce reversible kaliuresis in salt-depleted normotensive patients in the absence of changes in GFR.²³⁷ However, long-term Ang II receptor blockade does not cause appreciable changes in urinary electrolyte excretion or volume. The kaliuretic effect may be caused by specific intrinsic pharmacologic effects of the losartan molecule.

Another property unique to the losartan molecule is induction of uricosuria.²³⁸ This effect is not observed with ACE inhibitors or other ARBs, including the active metabolite of losartan, and does not appear to be related to inhibition of the RAAS.²³⁸ Losartan has a greater affinity for the urate-anion exchanger than other antagonists, and it inhibits urate reabsorption in the proximal tubule.²³⁹ The uricosuria is associated with a concomitant decrease in serum urate concentrations in normal individuals, hypertensive individuals, and patients with kidney disease including kidney transplant recipients.²³⁹ The effect occurs within 4 hours of drug

administration and is dose dependent. Long-term administration reduces serum urate concentrations by approximately 0.4 mmol/L.²⁴⁰ The clinical implications of this effect are unknown. Concerns that increased uric acid supersaturation might perpetuate renal uric acid deposition have not been borne out clinically, perhaps because losartan simultaneously increases urinary pH, which protects against crystal nucleation.²⁴¹ However, the decrease in the serum urate might be beneficial because it has been suggested that hyperuricemia is a risk factor for kidney disease progression²⁴² and coronary artery disease.²⁴³

Hypertensive patients treated with ARBs—including those with normal and those with impaired kidney function—exhibit renal responses similar to or slightly greater than the responses of patients treated with ACE inhibitors.²⁴⁴ In addition to decreases in SBPs and DBPs, patients demonstrate increases in renal blood flow and decreases in filtration fraction and renal vascular resistance, with no substantial changes in GFR.²⁴⁵ These effects are probably a result of combined decreases in preglomerular and postglomerular resistances. It has been suggested that elevated intrarenal Ang II levels in the presence of AT₁ receptor blockade stimulates AT₂ receptors, which can increase the preglomerular vasodilator actions of bradykinin, cyclic guanosine monophosphate, and nitric oxide.²⁴⁶ ACE inhibitors can potentiate this effect, but the clinical importance of this finding has not been established. Ang II blockade may significantly reduce GFR in underperfused kidneys. Patients with low perfusion pressures, volume depletion, or renovascular disease may experience severe decreases in GFR but less severe decreases than with ACE inhibitors.²⁴⁷ Under conditions of overperfusion, such as in hypertension associated with glomerulosclerosis and nephron loss or diabetes, Ang II blockade is protective. Such patients often have a suboptimal suppression of the RAAS. The lowering of efferent arteriole resistance reduces intraglomerular hydrostatic pressure, which attenuates the progression of kidney injury, and increases renal sodium excretory capacity. In concert with the reduction in systemic arterial pressure, these actions provide more renal protection than other classes of antihypertensive agents, despite equivalent reductions in BP.^{76,248,249}

In healthy and hypertensive patients, ARBs produce dose-dependent increases in circulating Ang II levels and plasma renin activity.²⁵⁰ The increases occur at the peak plasma drug levels and persist for up to 24 hours; they remain elevated with long-term administration. Decreases in plasma concentrations of aldosterone have been reported, but they are variable.²⁵¹ In normal individuals, decreases in aldosterone coincide with the peak interval of ARB activity; in hypertensive patients consuming a fixed-sodium diet, there are no significant changes in plasma aldosterone concentrations relative to baseline.²³⁷ ARBs suppress the Ang II-mediated adrenal cortical release of aldosterone, but these effects appear to be quantitatively less important than the intrarenal suppression of Ang II action. Long-term AT₁ receptor blockade does not appear to induce aldosterone escape.²⁵²

Urinary protein excretion is significantly decreased with administration of ARBs and parallels findings with ACE inhibitor therapy.⁴⁷ Antiproteinuric effects have been described in diabetic and nondiabetic patients, as well as renal transplantation patients.^{47,253} The antiproteinuric effect has a slow onset, and the dose-response curves differ from those of

the antihypertensive effects, in which the maximal effect occurs at 3 to 4 weeks. Whether the antiproteinuric effects are equivalent to or better than those of ACE inhibitors remains to be determined, but it is evident that the suppression of albuminuria is equivalent at all stages of CKD.²⁵⁴ Combining ACE inhibitors and ARBs for renoprotection and proteinuria reduction is no longer recommended owing to the significant development of hyperkalemia, hypotension, and renal hypoperfusion, particularly in patients with underlying CKD.^{97,148,255,256} Like ACE inhibitors, ARBs have nonhemodynamic effects that may contribute to renoprotection, including antiproliferative actions on the vasculature and mesangium, inhibition of transforming growth factor- β (TGF- β),^{30,257} inhibition of atherogenesis²⁵⁸ and vascular deterioration,²⁵⁹ improved superoxide production and nitric oxide bioavailability,²⁶⁰ reduction of collagen formation, reduction of mesangial matrix production, improved vascular wall remodeling, decreased vasoconstrictor effects of endothelin-1, improved endothelial function,²⁵⁹ reduction of oxidative stress and inflammation,²⁶¹ modulation of peroxisome proliferator-activated receptor γ activity, and protection from calcineurin inhibitor injury. ARBs also reduce salt sensitivity by restoring renal nitric oxide synthesis.²⁶² The clinical importance of these effects remains under investigation.

CLASS EFFICACY AND SAFETY

All AT₁ receptor blockers have been demonstrated to lower BP effectively and safely in patients with mild, moderate, and severe hypertension, regardless of age, sex, or ancestry.^{263–265} ARBs are indicated as first-line monotherapy or add-on therapy for hypertension and are comparable in efficacy to other agents.^{266–268} They are safe and effective in patients with CKD (even when used at high dosages), diabetes, heart failure, coronary artery disease, arrhythmias, and left ventricular hypertrophy (LVH)^{269–272} and kidney transplant recipients, and have been shown to protect against hypertensive end-organ damage,^{148,273,274} such as LVH, stroke, ESKD,^{47,275} retinopathy,²⁷⁶ exercise-induced inflammatory and prothrombotic stress, and possibly diabetes and dementia.^{272,277–281} Some but not all studies have shown a weak lowering of the risk for new and recurrent atrial fibrillation because of the beneficial structural and electrical effects on the atria.^{121,122} ARBs have been shown to diminish the rate of persistent atrial fibrillation in patients with preexisting recurrent atrial fibrillation.²⁸² The Prevention Regimen for Effectively Avoiding Second Strokes (PROFESS) study did not show any significant benefit of ARBs for the prevention of recurrent stroke but may have been underpowered to show an effect in its well-treated patient population.²⁸³

Although ARBs may not be the most efficacious agents in terms of BP reduction in African Americans, they are equally or more efficacious in providing target-organ protection and arresting disease progression compared with other antihypertensive agents that do not inhibit the RAAS.²⁸⁴ Moreover, their antihypertensive activity is not attenuated by high-salt diets in African Americans.^{267,284,285} In most patients, ARBs offer BP lowering comparable to that of all other antihypertensive drug classes, with an improved tolerability profile.²⁸⁶

ARBs provide effective control over a 24-hour period and are generally suitable for once-daily dosing.^{287,288} Response rates vary from 40% to 60%. ARBs do not affect the normal circadian BP variation.²⁸⁹ The long onset of action (4–6 weeks)

avoids the first-dose hypotension and rebound hypertension that are commonly observed with other drugs. There is a dose-dependent response with newer agents, but losartan and valsartan have a relatively flat dose-response curve.²⁹⁰ Azilsartan, candesartan, irbesartan, and olmesartan may have the greatest efficacy, with a longer duration of action because of their noncompetitive binding, and telmisartan may have an added advantage by the inhibition of SNS activation through an antioxidant effect in experimental models.^{291,292}

The addition of thiazide or thiazide-type diuretics to ARBs potentiates the therapeutic effect, increases response rates to 70% to 80%, and is more effective than increasing the ARB dosage. ARBs may also abrogate the adverse metabolic effects of thiazides.²⁹³

An enhanced BP-lowering efficacy after 16 weeks of olmesartan-based treatment ($P = .0005$) may be observed in hypertensive patients with CKD-associated sympathetic hyperactivity, as evidenced by a baseline morning home SBP of 165 mm Hg or higher and patients with a morning home pulse rate of 70 beats/min or more in the Home BP Measurement With Olmesartan-Naïve Patients to Establish Standard Target Blood Pressure (HONEST) study.²⁹⁴ Similarly, the addition of mineralocorticoid receptor antagonists (MRAs) to ARBs provides added benefits for BP control and reduction of proteinuria in diabetic patients over the long term but has not been found to be protective from the decline in eGFR.²⁹⁵ As with ACE inhibitor ARB combination therapy, the combination of ARBs with MRAs should be used with caution, with attention to the risks of hyperkalemia and impaired kidney function. Combination therapy with ARBs plus dihydropyridines has additive effects in reducing BP and is well tolerated.²⁹⁶

ARBs may decrease kidney function and elevate serum potassium concentrations; serum chemistries should be checked after initiation or increase in dosage of these drugs. The overall incidence of hyperkalemia from ARBs is 3.3%, similar to that of ACE inhibitors.¹⁴⁸ Participants in the CHARM (Candesartan in Heart Failure-Assessment of Reduction in Mortality and Morbidity; $n = 7599$) program were randomized to standard heart failure therapy plus candesartan or placebo, with recommended monitoring of serum potassium and creatinine levels.²⁹⁷ The authors assessed the incidence and predictors of hyperkalemia over the median 3.2 years of follow-up. Candesartan increased the risk of incident hyperkalemia compared with placebo from 5.2% from 1.8% (difference, 3.4%; $P < .0001$). Risk factors for hyperkalemia in treated patients with symptomatic heart failure include advanced age, male sex, baseline hyperkalemia, impaired kidney function, diabetes, or combined RAAS blockade. As with ACE inhibitors, hyperkalemia associated with ARB use has been treated with patiomer and sodium zirconium cyclosilicate in addition to sodium and calcium polystyrene sulfonate.^{196,197}

ARBs should be stopped immediately at the onset of pregnancy with the first missed menstrual period in the first trimester, as discussed previously in the context of ACE inhibitor use, because ARBs may cause fetal or neonatal death and congenital abnormalities when used during the second and third trimesters of pregnancy.¹³⁸ Breastfeeding is not contraindicated, according to Kidney Disease Improving Global Outcomes (KDIGO) and NICE guidelines; however, low drug levels are detected in breast milk.^{139,140}

Overall, ARBs have neutral metabolic effects and are superior to other hypertensive classes with respect to tolerability. ARBs do not cause hypernatremia or hyponatremia, and hyperkalemia is relatively uncommon. In the ONTARGET study, a serum potassium concentration higher than 5.5 mmol/L was observed in 3.3% of patients who resigned from the study and is comparable to that observed with ACE inhibitor therapy.¹⁴⁸ ARBs have no effect on serum lipids in hypertensive patients but may improve the abnormal lipoprotein profile of patients with proteinuric renal disease and reduce obesity-related morbidity.^{298,299} ARBs have favorable effects on serum glucose concentrations and insulin sensitivity.^{300–302} Clinical trials comparing ARB-based therapy with treatment with other antihypertensive agents in patients with hypertension with and without LVH demonstrated a 25% reduced risk of the development of diabetes in the ARB-treated group.^{303,304} The mechanism for this effect has not been defined.^{277,305} Increased levels of liver transaminases are occasionally reported, but the effects are usually transient, even with continued therapy.¹²

Clinically relevant AEs are not observed more frequently than in placebo-treated patients. Because ARBs do not interfere with kinin metabolism, cough is rare, which is a major clinical advantage.¹⁴⁸ The incidence of cough in patients with a history of ACE inhibitor-induced cough is no greater than in those receiving placebo.³⁰⁶ Similarly, the incidence of angioedema and facial swelling is no greater than that with placebo, but such swelling can occur.³⁰⁷ ARBs are typically associated with a more potent antiinflammatory response than ACE inhibitors.³⁰⁸ The most frequent AEs are headache (14%), dizziness (2.4%), and fatigue (2%), which occur at rates lower than those with placebo.³⁰⁹ ARB therapy not only does not worsen sexual activity, but may also improve it.³¹⁰ Like ACE inhibitors, ARBs may cause minor decreases in the hemoglobin concentration; they may also lower the hemoglobin effectively in posttransplantation erythrocytosis.³¹¹ There have been rare associated cutaneous eruptions.

A spruelike enteropathy associated with olmesartan therapy has been reported.^{312,313} In a recent systematic review of 54 patients,³¹³ the clinical presentation was diarrhea (95%) and weight loss (89%). Less common symptoms were fatigue, nausea, vomiting, and abdominal pain. The patients had been taking olmesartan for 6 months to 7 years. A laboratory examination showed a normochromic, normocytic anemia (45%) and hypoalbuminemia (39%). HLA-DQ2 or HLA-DQ8 was observed in more than 70%. Antibody testing for celiac disease was negative. Duodenal villous atrophy in varying degrees was described in all the reported patients, and they all showed resolution of diarrhea. The US Food and Drug Administration (FDA) issued a drug safety communication on July 13, 2013 (<http://www.fda.gov/Drugs/DrugSafety/ucm359477.htm>).

ARBs have not been shown to increase the risk of cancer. A meta-analysis showed a slightly increased risk of cancer (relative risk [RR], 1.08; 95% confidence interval [CI], 1.01 to 1.15) relative to other antihypertensive agents.³¹⁴ The study's methodology was criticized because some RCTs were not included; had they been, the cancer signal would have disappeared. Subsequent meta-analyses^{315–318} and cohort studies^{319–322} were performed. The conclusion was that there was no evidence that ARBs are associated with cancers of any type in large populations, and one study suggested that

ARBs may actually lower the incidence.³¹⁹ A subsequent systematic review of observational and interventional studies suggested that the use of ACE inhibitors and ARBs may improve cancer outcomes.³²³

Drug interactions with ARBs are uncommon but, as with ACE inhibitors, NSAIDs may blunt the natriuretic effect of ARBs.³²⁴ ARBs have an increased incidence of hypotension and impaired kidney function compared with ACE inhibitors.¹⁴⁸ Acute reversible renal failure has been reported with Ang II receptor blockade therapy in salt-depleted patients.³²⁵ Thus therapy should not be instituted in hypovolemic patients or in the setting of active diuresis. Concomitant use with ACE inhibitors and direct renin inhibitors should be discouraged, especially in patients with a GFR lower than 60 mL/min and is contraindicated in patients with type 2 diabetes mellitus because of an increased risk of hypotension, hyperkalemia, and renal dysfunction.^{97,256,326}

β-ADRENERGIC RECEPTOR ANTAGONISTS

CLASS MECHANISMS OF ACTION

β-Adrenergic receptor antagonists (β-blockers) exert their antihypertensive effects by attenuating sympathetic stimulation through the competitive antagonism of catecholamines at the β-adrenergic receptor.³²⁷ However, the precise mechanism of the antihypertensive effect of β-blockers remains incompletely understood. β₁-Adrenergic receptor blockade has generally been considered responsible for the BP-lowering effect; however, β₂-receptor blockade has an independent antihypertensive effect.³²⁸ Inhibition of β₁-adrenergic receptors in the juxtaglomerular cells in the kidney may inhibit renin release. A direct action on the CNS, with a reduction in CNS sympathetic outflow, may also be involved. Attenuation of cardiac pressor stimuli related to β-blockade may result in baroreceptor resetting. In addition, adrenergic neuron output may be blocked because of the inhibition of β₂-adrenergic receptors at the vascular wall.

β₁-Selective blockers may have a slightly more potent antihypertensive effect than nonselective agents. This effect may be in the range of 2 to 3 mm Hg. It may be that β₂-blockade in some fashion blunts the antihypertensive effects

of β₁-blockade.³²⁹ The β₂ partial agonist activity may mediate peripheral vasodilator effects that could contribute to the antihypertensive action. A β₁-selective antagonist with partial agonist activity at the β₁ receptor may result in less hypotensive effect. The magnitude and clinical significance of these differences are unclear.

In addition to β-adrenergic antagonist properties, certain drugs have antihypertensive effects that are mediated through different mechanisms (Table 49.11), including α₁-adrenergic antagonist activity and effects on nitric oxide-dependent vasodilator action. Partial agonist activity is a property of certain β-blockers that results from a small degree of direct stimulation of the β-adrenergic receptor by the drug, while simultaneously blocking the same receptor to access by stimulating catecholamines.^{329–331} Whether the presence of partial agonist activity is advantageous or disadvantageous remains unclear. Drugs with partial agonist activity slow the resting heart rate less than drugs that lack this pharmacologic effect.³³² The exercise-induced increase in heart rate is similarly blocked by both groups of drugs.³³² However, β-blockers with nonselective partial agonist activity may reduce peripheral vascular resistance and cause less atrioventricular (AV) conduction depression than drugs without partial agonist activity. The specificity of partial agonist activity for β₁ or β₂ receptors may also have a role in the antihypertensive response to a given drug.

β-Blockers may be nonspecific and block β₁- and β₂-adrenergic receptors, or they may be relatively specific for β₁-adrenergic receptors. β₁-adrenergic receptors are found predominantly in heart, adipose, and brain tissue, whereas β₂ receptors predominate in the lung, liver, smooth muscle, and skeletal muscle. Many tissues, however, have both β₁ and β₂ receptors, including the heart, and it is important to realize that the concept of a cardioselective drug is only relative.

β-Blockers differ significantly in gastrointestinal absorption, first-pass hepatic metabolism, protein binding, lipid solubility, penetration into the CNS, and hepatic or renal clearance. β-Blockers that are eliminated primarily by hepatic metabolism have a relatively short plasma half-life; however, the duration of the clinical pharmacologic effect does not correlate well with the plasma half-life in many of these drugs. Water-soluble

Table 49.11 Pharmacologic Properties of β-Adrenergic Antagonists

Generic Name (Trade Name)	β ₁ Selectivity	Partial Agonist Activity	Membrane-Stabilizing Activity	α-Adrenergic Antagonist Activity
Nadolol (Corgard)				
Propranolol (Inderal)			+	
Carteolol (multiple)		+		
Penbutolol (Levitol)		+		
Pindolol (Visken)		+	+	
Labetalol (Trandate)		+		+
Carvedilol (Coreg)			+	+
Atenolol (Tenormin)	+			
Metoprolol (Lopressor)	+			
Betaxolol (Kerlone)	+		+	
Acebutolol (Sectral)	+	+	+	
Celiprolol (none in the United States)	+			+
Bisoprolol (Zebeta)	+			
Nebivolol (Bystolic)	+			

drugs that are eliminated by the kidney may have longer half-lives. Bioavailability varies greatly across the class, as does the degree to which individual agents are dialyzed.³³³

Nonselective β -Adrenergic Antagonists

Nadolol is a nonselective β -blocker without partial agonist activity (Table 49.11). The average adult dosage is 40 to 80 mg given once daily, with a maximum daily dose of 320 mg (Tables 49.12 and 49.13). Nadolol is not appreciably metabolized, and elimination occurs predominantly in the urine and feces. Dosage adjustment is indicated in patients with CKD. Dosage intervals should be increased to 24 to 36 hours, 24 to 48 hours, and 40 to 60 hours in patients with creatinine clearances of 30 to 50, 10 to 30, and less than 10 mL/min/1.73 m², respectively.²² Dosage adjustment is not necessary in patients with hepatic insufficiency. Hemodialysis reduces the serum concentration of nadolol, but specific recommendations for dosage during dialysis are not available.

Propranolol is a noncardioselective β -blocker that has no partial adrenergic activity. The usual daily dosage range is 80 to 320 mg. The drug may be administered in a single daily dose if a long-acting preparation is used. The drug is metabolized by the liver. The major metabolite, 4-hydroxypropranolol, has

β -blocking activity. Renal excretion is less than 1%. Dosage adjustment in patients with CKD is not necessary.²² Patients with liver disease may require variable dosage adjustments and more frequent monitoring.

Timolol is a nonselective β -blocker agent without partial adrenergic activity. The recommended initial dosage of timolol in the management of hypertension is 10 mg twice daily. The maintenance dosage generally ranges from 20 to 40 mg daily. No dosage adjustment is necessary in patients with CKD.²² Because timolol undergoes extensive hepatic metabolism, patients with liver disease may require a dosage adjustment and frequent monitoring. Timolol is not removed by dialysis.

Penbutolol is a nonselective β -blocker³³⁴ with low partial agonist activity. Usually, dosages are 20 to 40 mg given as a single dose or divided twice daily. Hepatic metabolism to inactive metabolites occurs with subsequent renal elimination. The optimal antihypertensive effect is observed at an average of 14 days after initiation of therapy. Dosage adjustments for patients with CKD are not recommended, but adjustment may be required for patients with hepatic insufficiency.²²

Pindolol is a nonselective β -blocker with high partial agonist activity. The usual adult oral dosage is 5 mg twice

Table 49.12 Pharmacokinetic Properties of β -Adrenergic Antagonists

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Nadolol	20–40	No	2–4	20–24	—	U, F	—
Propranolol	16–60	Yes	—	3–4	L	—	—
Timolol	50–90	—	—	2–4	L	U (20%)	—
Penbutolol	100	No	—	17–24	L	U	—
Pindolol	95	No	2	3–11	L	U (40%)	—
Atenolol	40–60	Yes	—	14–16	—	U, F	—
Metoprolol	50	—	1.5–2	3–7	L	U	—
Betaxolol	78–90	No	2–6	12–22	L	U	—
Bisoprolol	90	—	2.3	9.6	L	U	—
Acebutolol	90	No	2–3	3–8	L	U	—

F, Feces; L, liver; U, urine.

Table 49.13 Pharmacodynamic Properties of β -Adrenergic Antagonists

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Nadolol	40	40–80	320	qd	—	—
Propranolol	40	80–320	640	bid	—	—
Timolol	10	20–40	60	bid	—	—
Penbutolol	10	20–40	80	qd, bid	2	20–24
Pindolol	5	10–40	60	qd, bid	—	24
Atenolol	25	50–100	200	qd	3	24
Metoprolol	12.5–50	100–200	400	qd, bid	1	3–6
Betaxolol	10	10–40	40	qd	3	23–25
Bisoprolol	5	2.5–20	20	qd	2–4	24
Acebutolol	400	400–800	1200	qd	3	24

bid, Two times a day; qd, once a day.

daily, with incremental doses of 10 mg every 3 to 4 weeks. The maximum daily recommended dose is 60 mg. Approximately 40% of a dose of pindolol is excreted unchanged in the urine; 60% is metabolized in the liver. The drug half-life increases modestly in patients with impaired kidney function. Dose adjustments do not appear necessary.²² Dose adjustments may be necessary in patients with cirrhosis and advanced CKD.

β_1 -Selective Adrenergic Antagonists

Atenolol is a long-acting β_1 -selective blocker with no partial agonist activity. The usual dosage is 50 to 100 mg once daily. Approximately 50% of the drug is eliminated by the kidneys, and 50% is excreted in the feces. Dosages of more than 100 mg/day are unlikely to produce additional benefits. The time required to achieve the optimal antihypertensive effect is 1 to 2 weeks. In patients with moderate CKD, the dosing interval should be increased to 48 hours, and in patients with advanced CKD, dosing intervals should be increased to 96 hours.²² Atenolol is not significantly metabolized by the liver, and no dosage adjustment is necessary in patients with liver disease. Atenolol is removed by dialysis, and a maintenance dose should be given after a dialysis treatment.

Metoprolol is a β_1 -selective blocker with no partial agonist activity. Extensive hepatic metabolism occurs primarily by the cytochrome P450 (CYP) 2D6 system (CYP2D6), and 3% to 10% of the drug is excreted unchanged in the urine. Metoprolol pharmacokinetics is heavily influenced by the CYP2D6 genotype and metabolizer phenotype, with up to a 15-fold difference in clearance between ultrarapid and poor metabolizers.³³⁵ The initial oral dosage is 12.5 to 50 mg once or twice daily, increasing to 100 to 200 mg twice daily. Sustained-release (SR) preparations may be substituted as a once-daily dose. Metoprolol is extensively removed by hemodialysis.³³³

Betaxolol is a long-acting β_1 -selective blocker³³⁶ with no partial agonist activity. The usual oral dosage for hypertension is 10 to 40 mg once daily. Therapy is typically started at a dosage of 10 mg once daily. Most patients respond to 20 mg once daily. The time to achieve the optimal antihypertensive effect is approximately 1 to 2 weeks. Betaxolol is metabolized predominantly in the liver, with metabolites excreted by the kidney. Approximately 15% of the dose is recovered unchanged in the urine. CKD results in a decrease in betaxolol clearance and titration should begin at 5 mg once daily in these patients.

Bisoprolol is a long-acting β_1 -selective blocker³³⁷ with no partial agonist activity. The usual oral dosage is 2.5 to 20 mg given once daily. Hepatic metabolism occurs with the renal excretion of metabolites; however, 50% of the drug is excreted by the kidney unchanged. In patients with CKD, the initial oral dosage should be 2.5 mg once daily, with careful monitoring of dose titration. The maximum recommended dosage of bisoprolol in patients with CKD is 10 mg/day. Similar dosage reduction is also required for patients with hepatic insufficiency. It is modestly removed during hemodialysis.³³³

Acebutolol is a β_1 -selective blocker with low partial agonist activity. Dosages of 400 to 1200 mg/day are effective in treating hypertension. The drug is metabolized to diacetolol, an active metabolite, with the parent compound being excreted renally and in bile. Diacetolol is excreted mainly by the kidneys. Dosage reduction of 50% to 75% is recommended for patients with advanced CKD.²²

Nonselective β -Adrenergic Antagonists With α -Adrenergic Antagonism or Other Mechanisms of Antihypertensive Action

Labetalol is a nonselective β -blocker³³⁸ with weak partial agonist activity and α_1 -receptor blocking activity (Tables 49.14

Table 49.14 Pharmacokinetic Properties of β -Adrenergic Antagonists With Vasodilatory Properties

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Labetalol	25–40	Yes	1–2	5–8	L	U (50%–60%)	—
Carvedilol	25–35	No	1–1.5	6–8	L	F	—
Celiprolol	30–70	Yes	—	5–6	—	F, U	—
Nebivolol	12–96	No	2.4–3.1	8–27	L	—	—

F, Feces; L, liver; U, urine.

Table 49.15 Pharmacodynamic Properties of β -Adrenergic Antagonists With Vasodilatory Properties

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Labetalol	100	200–800	1200–2400	bid	3	8–12
Carvedilol	6.25	12.5–25	50	bid	4–7	24
Celiprolol	200	200–400	400	qd	—	—
Nebivolol	5	5	40	qd	6	24

bid, Two times a day; qd, once a day.

and 49.15). The drug is approximately equipotent in blocking β_1 - and β_2 -adrenergic receptors. With oral administration, the ratio of α_1 - to β -blocking potency is approximately 1:3. With intravenous administration, the β -blocking potency is more prominent. The usual initial dosages for treatment of hypertension are 100 mg orally twice daily, increasing gradually to a maintenance dosage of 200 to 400 mg twice or thrice daily. The drug is metabolized in the liver, with 50% to 60% of a dose excreted in the urine and the remainder in the bile. Dose adjustment is not required in CKD.²² Chronic liver disease has been demonstrated to decrease the first-pass metabolism of labetalol, and dosage reduction is required in these patients.

Carvedilol is a nonselective β -blocker with peripheral α_1 -blocker activity (Tables 49.14 and 49.15)^{339,340} and no partial agonist activity. The drug is approximately equipotent in blocking β_1 - and β_2 -adrenergic receptors. The ratio of α_1 - to β_1 -blocking activity is estimated to be 1:7.6. There is evidence that the therapeutic actions of carvedilol may depend in part on the endogenous production of nitric oxide, which may improve endothelial dysfunction in hypertensive patients. For the management of hypertension, an initial oral dosage of 6.25 mg twice daily is recommended and may be increased to 12.5 to 25 mg twice daily, if needed. Dose adjustments are not required for patients with CKD. Carvedilol is extensively metabolized in the liver, and dose reductions are suggested for patients with hepatic insufficiency. Carvedilol is not removed by hemodialysis.³³³

Celiprolol is a β -blocker with several unique properties.^{341,342} It is a β_1 -selective blocker with α_2 -receptor blocking activity (Tables 49.14 and 49.15). Celiprolol also causes vasodilation through β_2 -receptor stimulation and possibly nitric oxide, with a subsequent decrease in systemic vascular resistance. In contrast to other β -blockers, celiprolol does not appear to induce bronchospasm or have negative inotropic effects. It does have moderate partial agonist activity. The initial dosage of celiprolol is 200 mg once daily and can be increased to 400 mg once daily. Renal excretion is 35% to 42%. A 50% dosage reduction is suggested in patients with a creatinine clearance of 15 to 40 mL/min/1.73 m². Celiprolol is not recommended for patients with a creatinine clearance less than 15 mL/min/1.73 m². It is not currently available in the United States.

Nebivolol is a long-acting, β_1 -selective blocker (Tables 49.14 and 49.15).^{343–347} The compound is a 1:1 racemic mixture of two enantiomers: D-nebivolol and L-nebivolol. The actions of nebivolol, which are unique and unlike those of other β -blocking agents, are attributable to the individual effects of the isomers. When administered alone, the L-isomer does not produce significant effects on BP, but its presence enhances the antihypertensive effects of the D-isomer. The L-isomer may potentiate the effects of endothelium-derived nitric oxide to induce decreases in BP and peripheral vascular resistance. These effects may improve endothelial dysfunction and potentially influence cardiovascular risks.^{348,349} The L-isomer may also inhibit norepinephrine actions at the presynaptic β receptors. The initial oral dosage is 5 mg once daily. The drug is metabolized in the liver; rapid and slow metabolizers have been identified. The half-life of nebivolol is 8 hours in rapid metabolizers and 27 hours in slow metabolizers. Reduced initial dosages are recommended for patients with CKD.

CLASS RENAL EFFECTS

α - and β -Adrenergic receptors in the kidney mediate vasoconstriction and vasodilation as well as renin secretion. β -Blockers may influence renal blood flow and GFR through their effects on cardiac output and BP in addition to direct effects on intrarenal adrenergic receptors. β -Adrenergic receptors have been localized to the juxtaglomerular apparatus in autoradiographic studies.³⁵⁰ β_2 Receptors predominate in the kidney. The degree of specificity of β -adrenergic blockers for β_1 and β_2 receptors might be expected to influence the effect on kidney function, as might the degree of intrinsic partial agonist activity. In general, the short-term administration of a β -adrenergic blocker usually results in a reduction of GFR and effective renal plasma flow.³⁵¹ This effect is independent of whether the drug has β_1 selectivity or intrinsic partial agonist activity. Nebivolol, carvedilol, and celiprolol, however, have vasodilatory properties and have been shown to increase GFR and renal plasma flow.³⁵² Nebivolol dilates glomerular afferent and efferent arterioles by a nitric oxide-dependent mechanism, in contrast to metoprolol, which had no similar effect.³⁵³ This effect may be mediated by the increased synthesis of vasodilatory nitric oxide. Nadolol, when administered intravenously, has been shown in some studies to increase renal plasma flow and glomerular filtration, whereas oral administration may result in decreased blood flow and GFR. β_1 -Selective drugs, when administered orally, tend to produce smaller reductions in GFR and renal plasma flow. The long-term use of propranolol has been characterized by a 10% to 20% decrease in renal plasma flow and GFR. The degree of reduction in GFR and renal plasma flow is modest and probably not of clinical significance in most cases. Labetalol, with its combined α and β blockade with labetalol, has shown little effect on renal hemodynamics. The fractional excretion of sodium has been observed to decrease by up to 20% to 40% in some studies of the acute renal effects of β blockade.³⁵⁴

CLASS EFFICACY AND SAFETY

β -Blockers provide effective therapy for the management of mild-to-moderate hypertension but their use as first-line therapy was not recommended^{355–357} in the 2017 ACC/AHA guidelines based on results from a contemporaneous systematic review and network meta-analysis. In that review, β -blockers were found to be less effective than CCBs or thiazide diuretics for reducing stroke and cardiovascular risk.³⁵⁸ No significant differences in outcomes were observed in these class-to-class comparisons by age, sex, race, and diabetes mellitus status.

β -Blockers are recommended for patients with specific comorbid conditions.^{327,359–361} For example, the use of β -blockers after an acute MI has been shown to reduce morbidity and mortality,³⁶² regardless of SBP, age, or ejection fraction. Other studies have shown a 20% reduction in total mortality and a 32% to 50% reduction in sudden death with β -blocker therapy in patients who have experienced an MI.³⁶³ Therefore for treatment of hypertension in patients with a history of MI (at least in the last year), β -blockers may be the drugs of choice.^{83,364,365}

Patients with coexisting heart failure and hypertension are another group that benefit from treatment with β -blockers.^{366–368} The Cardiac Insufficiency Bisoprolol Study II demonstrated a 20% reduction in mortality in patients with

moderate heart failure randomly assigned to therapy with a β -blocker.³⁶⁹ Hospitalizations for heart failure and sudden cardiac death were also significantly lower in the β -blocker group. Similar benefits were observed in randomized trial of metoprolol succinate and carvedilol.³⁷⁰ Over the long term, treatment with β -adrenergic blockers improves exercise tolerance, left ventricular geometry, and left ventricular structure and reduces myocardial oxygen demand. The magnitude of heart rate reduction with β -blockers, but not the dose, is significantly associated with the survival benefit in heart failure.³⁷¹ β -Blockers may differ in effects on cardiovascular outcomes. A meta-analysis has suggested that the vasodilatory β -blocker carvedilol has a greater benefit on all-cause mortality in heart failure with reduced ejection fraction (HFrEF) compared with β_1 -selective β -blockers, although this distinction has not been observed in all studies.^{371–373} Genetic polymorphisms affecting the β_1 -adrenergic receptor, the α_2 -adrenergic receptor, and the G protein-coupled receptor kinase have been suggested to modify heart failure risk and the response to β -blocker therapy.³⁷⁴

Agents with β_1 -selectivity or intrinsic sympathetic activity have a therapeutic advantage over nonselective β -adrenergic antagonists in the treatment of patients with bronchospastic airway disease, chronic obstructive pulmonary disease, peripheral vascular disease, and diabetes mellitus.^{330,375,376} Bronchoconstriction is mediated in part by β_2 -adrenergic receptors in the airways. β -Blockade with nonselective agents can lead to increased airway resistance. This increase is less likely to occur with β_1 -selective agents. β_1 -Selectivity is relative, however, and may be less apparent at higher dosages. In general, patients with severe bronchospastic airway disease should not receive β -blockers. In patients with mild-to-moderate disease, β_1 -selective agents may be used cautiously; it has been proposed that they have beneficial effects on airway hyperresponsiveness.³⁷⁷

Symptoms of peripheral artery disease may be exacerbated by β -blocker therapy.³⁷⁸ Cold extremities and absent pulses have been described in patients with severe disease. Raynaud phenomenon has been reported with nonselective β -blockade.³⁷⁹ Blockade of β_2 -receptor-mediated skeletal muscle vasodilation as well as decreased cardiac output may contribute to vascular insufficiency.³⁸⁰ However, a meta-analysis showed that treatment with β -blockers does not worsen intermittent claudication or walking capacity in patients with mild-to-moderate peripheral artery disease,³⁸¹ and current treatment guidelines do not recommend against the use of β -blockers in patients with peripheral artery disease, although the evidence is relatively limited.⁵⁵

The CNS symptoms of sedation, sleep disturbance, depression, and visual hallucinations have been reported with β -blockers. In a review of 15 randomized trials involving more than 35,000 patients, β -blockers were not associated with a significant increase in the risk of reported depressive symptoms (6/1000 patient-years; 95% CI, -7 to 19). β -Blockers were associated with a small but significant annual increase of reported fatigue (18/1000 patient-years; 95% CI, 5–30). These symptoms may be more common with lipid-soluble β -blockers and less common with nebivolol.³⁸²

β -Blockers are associated with weight gain and an increased risk of new-onset diabetes mellitus,^{383,384} perhaps by mediating increases in glycogenolysis and gluconeogenesis from amino acids and glycerol and inhibiting glucose uptake in the

periphery. β -Blockers differ in terms of their effects on glucose metabolism. Nonvasodilating β -blockers such as metoprolol decrease insulin sensitivity and are associated with a worsening of glycemic control, whereas nebivolol does not.^{385,386} In patients with established diabetes mellitus, β -blockers can blunt the effects of epinephrine secretion resulting from hypoglycemia and lead to hypoglycemia unawareness.^{387,388}

Nonselective β -blockers and, to a lesser degree, β_1 -selective agents have been associated with a rise in the serum potassium level.³⁸⁹ Suppression of aldosterone and inhibition of β_2 -linked, sodium-potassium membrane transport in skeletal muscle have been proposed as possible mechanisms.^{390,391} This effect is of limited clinical importance in patients who have normal kidney function and are not taking other medications that might affect serum potassium concentrations.

β -Blockers can affect plasma lipids.³⁹² Long-term use of β -blockers has been associated with an increase in triglyceride levels and a decrease in the level of HDL cholesterol. β -Blockers with increased β_1 selectivity or with partial agonist activity appear to have less effect on the lipid profile. Nonselective β -blockers without partial agonist activity may decrease the HDL cholesterol level by up to 20%; an increase in triglyceride levels of up to 50% has been reported. The effects of β -blockade on lipid metabolism are due primarily to the modulation of lipoprotein lipase activity. Very-low-density lipoprotein (VLDL) cholesterol and triglyceride metabolism is reduced in the setting of unopposed β -adrenergic stimulation of lipoprotein lipase activity. Decreased VLDL metabolism results in decreases in HDL cholesterol levels.

β -Blockers were associated with a small but significant annual increase in reported sexual dysfunction (5/1000 patients; 95% CI, 2–8). None of the AEs differed by lipid solubility.³⁹³

Abrupt withdrawal of β -blockers may be associated with rebound hypertension and worsening angina in patients with coronary artery disease.³⁹⁴ MI has been reported. These withdrawal symptoms may be caused by increased sympathetic activity, which could reflect adrenergic receptor upregulation during long-term sympathetic blockade. Gradual tapering of β -blockers decreases the risk of withdrawal. Withdrawal symptoms have been reported more commonly with abrupt discontinuation of relatively short-acting drugs.³⁹⁵

CALCIUM CHANNEL BLOCKERS

CLASS MECHANISMS OF ACTION

CCBs remain an important therapeutic class of medications for a variety of cardiovascular disorders.^{396–399} Initially introduced in the 1960s as antianginal agents, CCBs are now advocated as first-line therapy for hypertension.^{83,401} The pharmacologic effects of these drugs are related to their ability to attenuate cellular calcium uptake.^{400–404} CCBs do not directly antagonize the effects of calcium; rather, they inhibit the entry of calcium or its mobilization from intracellular stores.

Calcium channels have binding sites for activators and antagonists. The voltage-dependent, L-type calcium channel is a multimeric complex composed of α_1 -, α_2 -, ω -, β -, and γ -subunits.³⁹⁶ These channels have different binding sites for the various CCBs and are regulated by voltage-dependent and receptor-dependent events involving protein phosphorylation and G-protein coupling resulting from, for example,

β -adrenergic stimulation.⁴⁰⁵ Each class of CCB is quantitatively and qualitatively unique; the CCB classes possess different sensitivities and selectivities for binding pharmacologic receptors and the slow calcium channel in various vascular tissues. Even within the dihydropyridine class, there is considerable pharmacologic variability.⁴⁰⁶ This differential selectivity of action has important clinical implications for the use of these drugs and explains why the CCBs vary considerably in their effects on regional circulatory beds, sinus and AV nodal function, and myocardial contractility. The selectivity further explains the diversity of indications for clinical use, ancillary effects, and side effects.⁴⁰⁷

CCBs uniformly lower peripheral vascular resistance in patients, regardless of age, sex, ancestry, salt sensitivity, or comorbid conditions. There are at least three mechanisms through which CCBs lower BP. First, CCBs reduce peripheral vascular resistance by attenuating the calcium-dependent contractions of vascular smooth muscle. Contraction of vascular smooth muscle depends on the total cytosolic calcium concentration, which in turn is regulated by two distinct mechanisms. Depolarization of vascular smooth muscle tissue depends on the inward flux of calcium through voltage-sensitive L-type and T-type calcium channels. Hypertensive patients have an abnormal influx of calcium, which promotes increased peripheral vascular resistance.⁴⁰⁰ Calcium is released from the sarcoplasmic reticulum in response to extracellular calcium influx via a nonvoltage-dependent pathway. Cytosolic calcium binds to calmodulin, initiating a sequence of cellular events that promotes the interaction between actin and myosin and results in smooth muscle contraction. Therefore the importance of the calcium channels lies in their pivotal role in linking cell membrane electrical activity to biologic responses. Calcium influxes through L-type channels from extracellular sources and intracellular sources are both attenuated by CCBs.^{398,401}

Second, CCBs decrease vascular responsiveness to Ang II and the synthesis and secretion of aldosterone.⁴⁰³ CCBs also interfere with α_2 -adrenergic receptor-mediated vasoconstriction and possibly α_1 -adrenergic receptor-mediated vasoconstriction.^{402,408} The maximal vasodilatory response, as measured by forearm blood flow, appears to be inversely related to the patient's plasma renin activity and Ang II concentration. Thus it is possible that there is a greater influence of the calcium influx-dependent vasoconstriction in patients with low-renin hypertension, such as African Americans, which explains the clinical observation that CCBs are often more potent than other agents in these patients.

Finally, CCBs may induce a mild diuresis. It is well-known that the dihydropyridines, in particular, reduce preglomerular resistance and maintain or increase the GFR because of their preferential vasodilatory action on the renal afferent arteriole.⁴⁰⁴ Subsequently, decreased tubular sodium reabsorption and improved renal blood flow and natriuresis are observed. The sodium excretion rate tends to correlate with the reduction in BP.

Antihypertensive activity has not uniformly been demonstrated to be secondary to changes in nitric oxide release. The vasorelaxant properties of nifedipine and verapamil appear to be nitric oxide independent, whereas those of amlodipine are partly nitric oxide dependent.^{409,410} This effect of amlodipine is thought to be mediated by the inhibition of local ACEs and increases in vasodilatory bradykinins.

CLASS MEMBERS

Despite their shared mechanism of action, the CCBs are a very heterogeneous group of compounds. They differ with respect to pharmacologic profile, chemical structure, pharmacokinetic profile, tissue specificity, receptor binding, clinical indications, and side-effect profile (Tables 49.16 and 49.17). Two primary subtypes are distinguished on the basis of their behavior: dihydropyridines and nondihydropyridines. The nondihydropyridines are further divided into two classes – benzothiazepines (diltiazem) and diphenylalkylamines (verapamil). Their distinctly different pharmacologic effects are summarized in Tables 49.16 and 49.17.

Although all CCBs vasodilate coronary and peripheral arteries, the dihydropyridines are the most potent. Because medications in this subclass of CCBs are membrane-active drugs, they exert a greater effect on the peripheral vessels than on myocardial cells, which depend less heavily on the external calcium influx.⁴⁰¹ Their potent vasodilatory action prompts a rapid compensatory increase in sympathetic nervous activity, as mediated by baroreceptor reflexes creating a neutral or positive inotropic stimulus.⁴¹¹ Longer-acting dihydropyridines, however, do not appear to activate the SNS.⁴¹² By contrast, the nondihydropyridines are moderately potent arterial vasodilators but directly decrease AV nodal conduction and have negative inotropic and chronotropic effects, which are not abrogated by the reflex increase in sympathetic tone. Because of their negative inotropic action, their use is contraindicated in patients with HFrEF. As expected, these drugs are more effective at reducing stress-induced cardiovascular responses than dihydropyridines.⁴¹³

A clinically useful classification system for CCBs categorizes them by their duration of action into short-acting and long-acting agents (Tables 49.16 and 49.17). This schema is helpful because the short-acting agents are no longer recommended for the management of hypertension because of their stimulation of the SNS, which may predispose patients to angina, MI, and stroke.⁴¹⁴ The long-acting drugs are commonly divided into three generations. First-generation agents, such as nifedipine, have shorter half-lives and require multiple daily doses. Second-generation agents have been modified into extended-release (ER) formulations, requiring once-daily dosing. The third-generation agents have intrinsically longer plasma or receptor half-lives, possibly related to their greater lipophilicity.⁴¹⁵

Benzothiazepines

Diltiazem hydrochloride is the prototype of the benzothiazepine CCBs. Diltiazem is 98% absorbed from the gastrointestinal tract, but because of extensive first-pass hepatic metabolism, its bioavailability is only 40% compared with intravenous dosing¹² (Tables 49.16 and 49.17). In vivo, the competitively inhibited liver CYP2D6 isoenzyme is the most important metabolic pathway and probably accounts for the substantial proportion of drug interactions that occur with diltiazem.⁴¹⁶ The rates of elimination are lower in older persons and those with chronic liver disease but unchanged in patients with CKD.

Oral forms of diltiazem have been modified to improve delivery and currently include tablets, SR capsules, controlled-diffusion capsules, Geomatrix ER capsules, ER capsules, and buccoadhesive formulations.^{417–419} The usual starting dosage

Table 49.16 Pharmacodynamic Properties of Calcium Channel Blockers

Generic Name (Trade Name)	First Dose (mg)	Usual Daily Dosage (mg)	Maximum Daily Dose (mg)	Peak Response (h)	Duration of Response (h)
Diltiazem (Cardizem)	60	60–120 tid, qid	480	2.5–4	8
Diltiazem SR (Cardizem SR)	180	120–240 bid	480	6	12
Diltiazem CD (Cardizem CD)	180	240–280 qd	480	—	24
Diltiazem XR (Dilacor XR)	180	180–480 qd	480	3–6	24
Diltiazem ER (Tiazac)	180	180–480 qd	480	4–6	24
Amlodipine (Norvasc)	5	5–10 qd	10	30–50	24
Felodipine (Plendil ER)	2.5	2.5 qd	10	2–5	24
Isradipine (DynaCirc)	2.5	2.5–5 bid	20	2–3	12
Isradipine CR (DynaCirc CR)	5	5–20 qd	20	2	7–18
Nicardipine (Cardene)	20	20–40 tid	120	0.5–2	8
Nicardipine SR (Cardene SR)	30	30–60 bid	120	1.4	12
Nifedipine (Procardia, Adalat)	10	10–30 tid, qid	120	0.1	4–6
Nifedipine GITS (Procardia XL)	30	30–90 qd	120	4–6	24
Nifedipine ER (Adalat CC)	30	30–90 qd	120	2–6	24
Nisoldipine (Sular)	20	20–40 qd	60	—	24
Verapamil (Calan, Isoptin)	80	80–120 tid	480	6–8	8
Verapamil SR (Calan SR, Isoptin SR)	120	120–240 bid	480	—	12–24
Verapamil SR Pellet (Verelan)	120	240–480 qd	480	—	24
Verapamil COER-24 (Covera-HS)	180	180–480 qhs	480	>4–5	24
Mibefradil (Posicor)	50	50–100 qd	100	2–4	17–25

bid, Two times a day; *CD*, Controlled-diffusion; *COER*, controlled-onset extended-release; *CR*, controlled release; *ER*, extended release; *GITS*, gastrointestinal therapeutic system; *qd*, once a day; *qhs*, every bedtime; *qid*, four times a day; *SR*, sustained release; *tid*, three times a day; *XR*, extended release.

Table 49.17 Pharmacokinetic Properties of Calcium Channel Blockers

Drug	Oral Absorption (%)	First-Pass Effect	Bioavailability (%)	Peak Blood Level	Elimination Half-Life (h)	Metabolism and Excretion	Protein Binding (%)	Active Metabolites
Diltiazem	98	50%	40	2–3 h	4–6	L, F, U	77–93	Yes
Diltiazem SR	>80	50%	35	6–11 h	5–7	L, F, U	77–93	Yes
Diltiazem CD	95	E	35	12 h	5–8	L, F, U	77–93	Yes
Diltiazem XR	95	E	41	4–6 h	5–10	L, F, U	95	Yes
Diltiazem ER	93	E	40–60	4–6 h	10	L, F, U	95	Yes
Amlodipine	>90	M	88	6–12 h	30–50	L/U	>95	Yes
Felodipine	>90	E	13–18	2.5–5 h	11–16	L/U	>95	No
Isradipine	>90	E	15–25	2–3 h	8	L, F, U	>95	No
Isradipine CR	>90	E	15–25	7–18 h	—	L, F, U	>95	No
Nicardipine	>90	E	35	0.5–2 h	8.6	L, F, U	>95	No
Nicardipine SR	>90	E	35	1–4 h	—	L, F, U	>95	No
Nifedipine	>90	20%–30%	60	<30 min	2	L, U	98	Yes
Nifedipine GITS	>90	25%–35%	86	6 h	—	L, U	98	Yes
Nifedipine ER	>90	25%–35%	86	2.5–5 h	7	L, U	98	Yes
Nisoldipine	>85	E	4–8	6–12 h	10–22	L, F, U	99	No
Verapamil	>90	70%–80%	20–35	1–2 h	2.8–7.4	L, F, U	85–95	Yes
Verapamil SR	>90	70%–80%	20–35	5–6 h	4–12	L, F, U	85–95	Yes
Verapamil SR pellet	>90	70%–80%	20–35	7–9 h	12	L, F, U	85–95	Yes
CODAS Verapamil	>90	70%–80%	20–35	11 h	—	L, F, U	85–95	Yes

CD, Controlled-diffusion; *CODAS*, chronotherapeutic oral drug absorption system; *CR*, controlled release; *E*, extensive; *ER*, extensive release; *F*, feces; *GITS*, gastrointestinal therapeutic system; *L*, liver; *M*, minimal; *SR*, sustained release; *U*, urine; *XR*, extended release.

for the drug in tablet form is 180 mg/day in three divided doses, and the drug may be titrated to a total dosage of 480 mg/day (Table 49.16).

Diphenylalkylamine

Verapamil hydrochloride, the oldest CCB, is the prototype diphenylalkylamine derivative. Verapamil inhibits membrane transport of calcium in myocardial cells, particularly the AV node, and smooth muscle cells, which renders it antiarrhythmic, antihypertensive, and a negative inotrope. The drug is available for oral administration as film-coated tablets containing 40, 80, or 120 mg of racemic verapamil hydrochloride.¹² The usual daily dose is 80 to 120 mg three times (Table 49.16). The elimination half-life increases with long-term administration and in older patients with CKD (Table 49.17).

The SR caplets are available in scored 120-, 180-, and 240-mg forms. The usual antihypertensive dose is equivalent to the total daily dose of immediate-release tablets and can be given as 240 to 480 mg/day. An adequate antihypertensive response may be improved by divided twice-daily dosing.

The SR pellet-filled verapamil capsules are gel-coated capsules with an onset of action of 7 to 9 hours that is not affected by food. The peak concentrations are approximately 65% of those of immediate-release tablets, but the trough concentrations are 30% higher. The usual daily dose is 240 to 480 mg.

The controlled-onset, ER and chronotherapeutic oral drug absorption system (CODAS) tablets have unique pharmacologic properties and deliver verapamil 4 to 5 hours after ingestion. A delay coating is inserted between the outer semipermeable membrane and active inner drug core. As the delay coating expands in the gastrointestinal tract, the pressure causes drug from the inner core to be released through laser-drilled holes in the outer membrane, making this formulation ideal for nighttime dosing by providing maximal plasma levels in the early morning hours, from 6 AM to noon, and minimizing nighttime diurnal BP variations.⁴²⁰ A buccal gel formulation of verapamil that provides SR of the drug up to 6 hours has been reported.⁴²¹

Of the 13 known metabolites of verapamil, norverapamil is the only one with cardiovascular activity; it has 20% of the potency of the parent compound. Renal excretion accounts for 70% of clearance and occurs within 5 days. The remainder is excreted in the feces. Clearance decreases with increasing age and decreasing weight.⁴²² With long-term administration, there is a significant increase in bioavailability, possibly as a result of saturation of hepatic enzymes. Dose adjustment is necessary in patients with liver disease but not CKD. However, verapamil should be used with caution in patients who ingest large amounts of grapefruit juice or patients taking concurrent AV nodal blocking agents.⁴²³

Dihydropyridines

Nifedipine is a dihydropyridine CCB that causes decreased peripheral resistance, with no clinically significant depression of myocardial function. Because of the reflex sympathetic stimulation triggered by vasodilation, nifedipine has no tendency to prolong AV conduction or sinus node recovery or slow the sinus rate. Clinically, there is usually a small increase in heart rate and cardiac index. The labeling for immediate-release nifedipine capsules has been revised to

recommend against using this dosage form for the management of hypertension.^{12,424} In older persons, use of the immediate-release form has been associated with a more than threefold increase in mortality compared with the use of other antihypertensive agents, including other CCBs.⁴²⁵ In most patients, immediate-release nifedipine causes a modest hypotensive effect that is well tolerated. However, in occasional patients, the hypotensive effect is profound and has resulted in MI, stroke, and death.⁴²⁴ This effect appears to be more pronounced in patients also taking β -blockers.⁴²⁶ Consequently, its use should be reserved for short periods, but not in the setting of acute syndromes. The usual adult dosage is 10 to 30 mg three times daily, and the dose can be titrated weekly (Table 49.16). Nifedipine is rapidly and fully absorbed, and drug levels are detectable within 10 minutes of ingestion. Peak levels are achieved within 30 minutes, and the half-life is 2 hours. There is no clinical advantage to ingestion using the technique of bite and swallow or bite and hold sublingually.

Nifedipine is extensively metabolized in the liver and then excreted in the urine. Most of the population is reported to metabolize the drug rapidly. Because nifedipine is 98% protein bound, the dosage should be adjusted in patients with hepatic insufficiency or severe malnutrition.

The ER tablets of nifedipine are available in 30-, 60-, and 90-mg doses. These tablets consist of an outer semipermeable membrane surrounding an active drug core.⁴²⁷ The core is composed of an inner active drug layer surrounded by an osmotically active, inert layer that forces the dissolution of the drug core as it swells from gastrointestinal juice absorption. The drug is then slowly and steadily released over 16 to 18 hours. This method of delivery is termed the “gastrointestinal therapeutic system” formulation. The ER form should not be bitten or divided. The time to peak concentration is 6 hours, and plasma levels remain steady for 24 hours. The bioavailability of the ER tablet is 86% compared with that of immediate-release forms, and tolerance does not develop.⁴²⁷ Of the metabolites, 80% are excreted in the urine. The remainder is excreted in the feces, along with the outer semipermeable membrane shell. The usual adult maintenance dosage is 30 to 90 mg/day. Conversion from the immediate-release form to ER tablets can be done on an equal-milligram basis.

A similar ER formulation is composed of a coat and core.⁴²⁸ The outer layer contains a slow-release form of nifedipine; the inner core is a fast-release preparation. Peak concentrations are reached within 2.5 to 5 hours, and there is a second peak after 6 to 12 hours as the inner core is released. When the drug is administered in this way, the half-life is extended from 2 to 7 hours. The usual daily dose is 30 to 90 mg, and the dose should be titrated by 30-mg increments in 7 to 14 days for maximal effect. Because of the unique delivery system, which provides a rapid-release core, peak plasma concentrations are not always reliable. Ingestion of three 30-mg tablets simultaneously, but not two, results in a 29% higher peak plasma concentration than the ingestion of a single 90-mg tablet. Consequently, two tablets may be substituted for 60 mg, but the substitution of three 30-mg tablets to make 90 mg is not recommended.¹²

Amlodipine besylate is unique among the dihydropyridine CCBs. It appears to bind to dihydropyridine and nondihydropyridine sites to produce peripheral arterial vasodilation without significant activation of the SNS.²² The parent

compound has substantially slower but more complete absorption than others in the class (Table 49.17). After ingestion, amlodipine is almost completely absorbed, peak plasma concentrations are achieved in 6 to 12 hours, and the clinical response can be detected at 24 hours. The mean peak serum levels are linear, age independent, and achieved after 7 to 8 days of continuous dosing.⁴²⁹ The elimination half-life is long, ranging from 30 to 50 hours, and is prolonged in older adults. The long half-life permits once-daily dosing; the hypotensive response may last up to 5 days,⁴³⁰ 90% of amlodipine is metabolized in the liver, and 10% is excreted unchanged. The metabolites are excreted primarily in the urine, but no dosage adjustment is necessary with renal impairment. The minimum effective dose is 2.5 mg, particularly in older patients. Most patients require a dosage of 5 to 10 mg/day.

Benidipine is a long-acting dihydropyridine CCB that is available currently in Asia for the management of mild-to-moderate hypertension. It has several unique mechanisms of action⁴³¹: It has a high vascular selectivity and inhibits L-, N-, and T-type calcium channels. The usual dosage is 2 to 4 mg once daily, but the dosage can be increased to 4 mg twice daily for those with angina pectoris.

Felodipine is a dihydropyridine CCB that is administered in ER tablets of 2.5, 5, and 10 mg (Table 49.16).¹² Felodipine is almost completely absorbed from the gastrointestinal tract, with a time to peak concentration of 2 to 5 hours (Table 49.17). There is extensive first-pass hepatic metabolism. Bioavailability is influenced by food. Large meals and the flavonoids in grapefruit juice increase the bioavailability by approximately 50%.⁴³³ The overall half-life is 11 to 16 hours. Felodipine is metabolized in the liver to inactive metabolites, most of which are excreted in the urine. The usual daily dose is 2.5 to 10 mg, and titration can be instituted at 2-week intervals. The dosage should be adjusted for liver disease but not for CKD.

Isradipine is a dihydropyridine CCB that is effective alone or in combination with other antihypertensive agents for the management of mild-to-moderate hypertension¹² (Tables 49.16 and 49.17). Isradipine is rapidly and almost completely absorbed after oral administration. Extensive first-pass hepatic metabolism reduces bioavailability to less than 25%. The hypotensive effect peaks at 2 to 3 hours for the regular release form. The drug is active for 12 hours; however, the full antihypertensive response does not occur until 14 days. The usual dosage is 2.5 to 5 mg two to three times daily. The onset of action of the SR formulation is achieved in 2 hours and lasts for 7 to 18 hours. The usual daily dose of the controlled-release tablet is 5 to 20 mg. Isradipine is extensively protein bound. The elimination half-life is biphasic, with a terminal half-life of 8 hours. Dosage adjustment is unnecessary in liver disease or CKD.

Manidipine is a third-generation dihydropyridine CCB that is structurally related to nifedipine.^{434,435} The usual adult dosage is 10 to 20 mg once daily. Dosage should be adjusted at 2-week intervals. Manidipine is highly protein bound and extensively metabolized in the liver. Metabolism is impaired by grapefruit juice⁴³⁶; 63% of the drug is excreted in the feces. The peak plasma concentration occurs after 2 to 3.5 hours, with an elimination half-life of 5 to 8 hours. Dose adjustment is not necessary in CKD. Manidipine may be less likely to cause significant ankle edema than amlodipine.⁴³⁷

Nicardipine hydrochloride is a dihydropyridine CCB available as 20- and 40-mg immediate-release gelatin capsules or 30-, 45-, and 60-mg SR capsules.¹² The usual dosage is 20 to 40 mg three times daily for the immediate-release form and 30 to 60 mg twice daily for the SR preparation. When conversion is made to the SR form, the previous daily total of immediate-release drug should be administered on a twice-daily regimen. Titration should be instituted at least 3 days after administration. Nicardipine is well absorbed orally but has only 35% systemic bioavailability because of its extensive first-pass hepatic metabolism. The time to peak concentration is 30 minutes to 2 hours for immediate-release capsules and 1 to 4 hours for SR forms. The elimination half-life is 8.6 hours. Nicardipine is 100% oxidized in the liver to inactive pyridine metabolites. There is no evidence of microsomal enzyme induction. Metabolites are excreted primarily in the urine and feces. The parent compound is not dialyzable. Dosage adjustments are necessary with liver disease but not CKD.

Nisoldipine is a dihydropyridine CCB that is formulated as ER tablets of 10, 20, 30, and 40 mg (Tables 49.16 and 49.17).¹² The initial starting dose is 20 mg, and the usual maintenance dose is 20 to 40 mg given once daily, which can be titrated at weekly intervals. The bioavailability of nisoldipine is low and variable (4%–8%). The coat core design affords a full 24-hour effect after oral administration. The drug reaches therapeutic concentrations in 6 to 12 hours, and absorption is slowed by high-fat meals. The elimination half-life ranges from 10 to 22 hours. Nisoldipine is metabolized in the liver and intestine. Variable hepatic blood flow induced by the drug probably contributes to its pharmacokinetic variability. Most of the metabolites are excreted in the urine and the remainder in the feces. Dose adjustments are necessary with liver disease but not CKD.

Lacidipine is a second-generation dihydropyridine CCB that is available in tablet form. It is reported to be unusually potent and long acting, possibly because it diffuses deeper into lipid bilayer membranes. A unique attribute of this drug is its apparently greater vascular selectivity, but the clinical relevance of this remains unclear.⁴³⁸ The usual dosage is 4 to 6 mg once daily, and the dose should be titrated at 2- to 4-week intervals.⁴³⁹ The duration of action is 12 to 24 hours. The elimination half-life is 12 to 19 hours. The parent compound is converted 100% by the liver into inactive fragments that are excreted primarily in the feces (70%) and kidney. Dosage adjustment is necessary in older persons and in patients with liver disease but not CKD.

Lercanidipine is a dihydropyridine CCB whose molecular design imparts greater solubility within the arterial cellular membrane bilayer, conferring a tenfold higher vascular selectivity than that of amlodipine.⁴³⁹ In contrast to amlodipine, lercanidipine has a relatively short half-life but a long-lasting effect at the receptor and membrane levels and is associated with significantly less peripheral edema.⁴³⁹ The drug is administered at a starting dose of 10 mg and increased to 20 mg daily as needed. It has a gradual onset of action, and its effects last for 24 hours.⁴⁴⁰ Lercanidipine also appears to dilate the efferent renal arteriole.⁴⁴¹

CLASS RENAL EFFECTS

All CCBs exert natriuretic and diuretic effects.^{442,443} Experimental studies and studies in humans with hypertension have

indicated that the increase in sodium excretion is, in part, independent of vasodilatory action or changes in GFR, renal blood flow, or filtration fraction. This effect is probably the result of changes in renal sodium handling that can potentiate the antihypertensive vascular effect. In normal persons, CCBs acutely increase sodium excretion, frequently in the absence of changes in BP. In hypertensive persons, the short-term administration of CCBs uniformly increases sodium excretion 1.1- to 3.4-fold; the magnitude of the increase is not related to the decrease in BP.⁴⁴²

The natriuretic effect appears to persist in the long term. Long-term administration of CCBs to hypertensive patients results in a cumulative sodium deficit that is abruptly reversed with the discontinuation of the drug. Natriuresis frequently occurs 3 to 6 hours after the morning dose.⁴⁴³ The net negative sodium balance levels off after the first 2 to 3 days of administration but persists for the duration of therapy.⁴⁴⁴ There are no significant changes in long-term body weight, serum concentrations of potassium, urea nitrogen, catecholamines, or GFR. Moreover, stimulation of renin release and aldosterone does not occur to an appreciable degree. It has been postulated that the natriuresis induced by CCBs increases distal sodium delivery to the macula densa, suppressing renin release. Because Ang II mediates aldosterone synthesis by way of cytosolic calcium messengers, CCBs blunt this response as well.⁴⁴⁵

The mechanism whereby CCBs induce natriuresis appears to be direct inhibition of renal tubular sodium and water absorption. Dihydropyridines increase urinary flow rate and sodium excretion without changing the filtered water and sodium load. Studies have suggested that CCBs may diminish sodium uptake at the amiloride-sensitive sodium channels.⁴⁴⁶ Inhibition of water reabsorption occurs distally to the late distal tubule. Proximal tubular sodium reabsorption may be inhibited by higher dosages. One possible mediator of this effect is atrial natriuretic peptide. In human studies, CCBs augment atrial natriuretic peptide release and potentiate its action at the level of the kidney. Other potential mediators are under investigation. How much the natriuretic effects contribute to the antihypertensive response is unknown, but unlike effects of other vasodilators, the changes attenuate the expected adaptive changes in sodium handling.

The renal hemodynamic effects of CCBs are variable and depend primarily on which vasoconstrictors modulate the renal vascular tone.⁴⁴⁷ Experimentally, CCBs improve GFR in the presence of the vasoconstrictors norepinephrine and Ang II, as well as others, by preferentially attenuating afferent arteriolar resistance.⁴⁴⁸ The efferent arteriole appears to be refractory to these vasodilatory effects. Patients with primary hypertension appear to be more sensitive to the renal hemodynamic effects of CCBs than normotensive patients, and this effect is more pronounced with more advanced CKD.⁴⁴⁹ Short-term administration of CCBs results in little change in, or augmentation of, GFR and renal plasma flow, no change in the filtration fraction, and reduction of renal vascular resistance. Long-term administration is not associated with significant changes in renal hemodynamics. The response is maximal in the presence of Ang II, which selectively causes postglomerular vasoconstriction. Clinically significant changes are counteracted by the reduction in renal perfusion pressure coincident with a reduction of BP.

The long-term effects of CCBs on kidney function are variable.^{450,451} In hypertensive patients, the effects on renal hemodynamics vary. Some patients exhibit no change in GFR, whereas others have an exaggerated increase in GFR and renal plasma flow.⁴⁴⁷ Even normotensive patients with a family history of hypertension have an exaggerated hemodynamic response.⁴⁵²

The effects of CCBs on proteinuria also vary with respect to the specific drug and the degree of BP reduction achieved.⁴⁵³ Some dihydropyridines increase protein excretion by up to 40%. It is not clear whether this increase is a result of hemodynamic vasodilation at the afferent arteriole, resulting in increased glomerular capillary pressure (because CCBs directly impair renal autoregulation), changes in glomerular basement membrane permeability, or increased intrarenal Ang II. By contrast, felodipine, diltiazem, verapamil, and others do not appear to have this effect and may lower protein excretion, possibly by also decreasing the efferent arteriolar tone and glomerular pressure.⁴⁵⁴ Clinical implications of these differential effects remain to be determined.

Large clinical trials underscore this controversy. In African Americans with hypertension and mild-to-moderate CKD, treatment with an ACE inhibitor demonstrated superior renoprotective effects compared with amlodipine.⁴⁵⁵ This effect was independent of BP reduction and was more evident in proteinuric patients; it was also suggestive in patients with less than 300 mg of protein/day at baseline. Hypertensive patients with diabetic nephropathy also fared worse with amlodipine than with ARB therapy.²⁴⁸ Patients experienced higher rates of progression of CKD and all-cause mortality in the amlodipine- and placebo-treated groups. This effect was also independent of the achieved BP. However, it should be emphasized that coadministration of a dihydropyridine and an ARB does not abrogate the ARB's protective effect on kidney function.⁷⁶ It has been postulated that the selective dilation of the afferent arteriole favors an increase in glomerular capillary pressure that perpetuates progression.

CLASS EFFICACY AND SAFETY

All CCBs are considered among first-line antihypertensive agents and appear to be equally efficacious and safe.^{397,456} In contrast to other vasodilators, CCBs attenuate the reflex increase of neurohormonal activity that accompanies a reduction in BP and in the long term, they inhibit or do not change the sympathetic activity.^{457,458} The longer-acting agents produce sustained BP reductions of 16 to 28 mm Hg systolic and 14 to 17 mm Hg diastolic, with no appreciable development of tolerance. The CCBs are effective in young, middle-aged, and older patients with white coat hypertension and mild, moderate, or severe hypertension.^{459–462} Their efficacy may be determined by genetic polymorphisms.⁴⁶³ CCBs are equally efficacious in men and women, in patients with a high or low plasma renin activity regardless of dietary salt intake, and in African American, white, and Hispanic patients.⁴⁶⁴ Effects of CCBs are diminished in smokers.⁴⁶⁵ CCBs are effective and safe in patients with hypertension and coronary artery disease,⁴⁶⁶ as well as in patients with ESKD.⁴⁶⁷ CCBs also reduce adverse cardiovascular events and slow the progression of atherosclerosis in normotensive patients with coronary artery disease.⁴⁶⁸

The use of CCBs is contraindicated in patients with HFrEF (except, perhaps, amlodipine or felodipine). These drugs

should not be used as first-line antihypertensive agents in patients with heart failure, a history of MI, or unstable angina.⁴⁵⁵

Among the different categories, dihydropyridines appear to be the most powerful for reducing BP but may also be associated with more pronounced activation of baroreceptor reflexes.⁴⁶⁹ Dihydropyridines induce a more prominent shift in the sympathovagal balance that favors sympathetic predominance compared with nondihydropyridines.⁴¹³ In general, however, compared with other vasodilators, CCBs attenuate the reflex increase in sympathetic activity – increased heart rate, cardiac index, and plasma norepinephrine levels and renin activity.

Verapamil and, to a lesser extent, diltiazem exert greater effects on the heart and have less vasoactivity. These drugs typically reduce the heart rate, slow AV conduction, and depress contractility (Table 49.18). Generally, they should not be used together with β -blockers because of increased risk for heart block. The second- and third-generation CCBs consist of pharmacologically manipulated formulations, whose half-lives are progressively longer.⁴¹⁵

CCBs are not associated with significant impairments in glycemic control or sexual dysfunction.³⁹⁷ The rapid antihypertensive action of CCBs may encourage patient adherence. Orthostatic changes do not occur because venoconstriction remains intact. AEs are usually transient and are the direct result of vasodilation. Hypotension is most common with intravenous administration. The most common AE of the dihydropyridines is peripheral edema⁴⁷⁰; it is dose related and thought to be the result of uncompensated precapillary vasodilation, which causes increased intracapillary hydrostatic pressure. The edema is not responsive to diuretics but improves or resolves with the addition of an ACE inhibitor or ARB, which preferentially vasodilates postcapillary beds and reduces intracapillary hydrostatic pressure.⁴⁷¹ Other AEs related to vasodilation include headache, nausea, dizziness, and flushing, and occur more commonly in women. The nondihydropyridines verapamil and isradipine more commonly cause constipation and nausea. The gastrointestinal effects are directly related to the inhibition of calcium-dependent smooth muscle contraction–reduced peristalsis and relaxation of the lower esophageal sphincter. Another common AE of the dihydropyridines is gingival hyperplasia, which is exacerbated in patients who are also taking cyclosporine. Dihydropyridines lead to the accumulation of gingival inflammatory B-cell infiltrates as stimulated by bacterial plaque, immunoglobulins, and folic acid, which causes the growth of the gingiva.⁴⁷² This growth can be controlled

with regular periodontal treatment and reversed with discontinuation of the drug.⁴⁷³

CCBs are notable among antihypertensive agents because of their metabolic neutrality. Because the calcium influx across β -cell membranes helps regulate insulin release,⁴⁷⁴ CCBs might predispose to low insulin levels. At typical therapeutic levels, CCBs have no effect on serum glucose concentrations, insulin secretion, or insulin sensitivity in persons with and without diabetes mellitus. The use of CCBs was not significantly associated with incident diabetes compared with other antihypertensive agents in a meta-analysis of RCTs.⁴⁷⁵ The association with diabetes was lowest for ACE inhibitors and ARBs, followed by CCBs, β -blockers, and diuretics.⁴⁷⁵ Furthermore, it was recently demonstrated that CCBs can even prevent diabetes and increase β -cell survival in vitro. The mechanism behind the β -cell mass destruction is the human islet cell protein TXNIP (thioredoxin-interacting protein), which is upregulated by hyperglycemia. Orally administered verapamil resulted in a reduction of TXNIP expression and β -cell apoptosis, enhanced endogenous insulin levels, and rescued mice from streptozotocin-induced diabetes. Verapamil also promoted β -cell survival and improved glucose homeostasis and insulin sensitivity in ob/ob mice.⁴⁷⁶ CCBs do not increase triglyceride or LDL cholesterol and do not reduce HDL cholesterol. CCBs do not precipitate hyponatremia, hyperkalemia, hypokalemia, or hyperuricemia. Therefore they are suitable agents for patients with dysmetabolic syndromes or diabetes.

Properties beyond their antihypertensive actions make the CCBs particularly useful in certain clinical situations. CCBs not only lower arterial pressure but also have variable effects on cardiac function. All CCBs are vasodilators and increase coronary blood flow. With the exception of the short-acting dihydropyridines, most CCBs reduce heart rate, improve myocardial oxygen demand, improve ventricular filling, diminish ventricular arrhythmias, reduce myocardial ischemia, and conserve contractility,^{477,478} making them suitable for patients with angina or diastolic dysfunction.⁴⁶⁶ In short-term use, CCBs improve diastolic relaxation; when administered over a long term, they reduce left ventricular wall thickness,⁴⁷⁹ may prevent the development of hypertrophy, and may improve arterial compliance.^{480–482} These effects may be crucial in hypertensive patients because LVH is one of the strongest risk predictors for cardiovascular morbidity and mortality.⁴⁸³ Verapamil may also be used for secondary cardioprotection to reduce reinfarction rates in patients who are intolerant of β -blockers (unless they have concomitant heart failure)⁴⁸⁴ and in patients with chronic headaches.⁴⁸⁵ The BP-independent

Table 49.18 Hemodynamic Effects of Calcium Channel Blockers

Class	Arterio- lar Dilation	Coro- nary Dilation	Cardiac After- load	Cardiac Con- tract- ility	Myocar- dial O ₂ Demand	Cardiac Output	AV Con- duction	SA Auto- maturity	Heart Rate: Short Term/ Long Term	Activation of Baroreceptor Reflexes
Dihydropyridines	↑↑↑	↑↑↑	↓↓	↔	↓	↓ or ↔	↔	↔	↑/↑	↑ or ↔
Diltiazem	↑↑	↑↑↑	↓	↓	↓	↔	↓	↓↓	↓/↓ or ↔	↔
Verapamil	↑↑	↑↑	↓	↓↓	↓	↔	↓↓	↓	↓/↓ or ↔	↔

AV, Atrioventricular; SA, sinoatrial.

inhibition of atherogenesis by CCBs may be another indication to use a CCB, particularly in high-risk patients, such as those with diabetes and ESKD.^{486,487}

The use of CCBs may prevent or slow the decline of dementia. In the Systolic Hypertension in Europe (SYST-EUR) trial,⁴⁸⁸ nitrendipine use was associated with a lower risk of incident dementia. The potential benefit of CCBs on dementia has also been suggested in observational studies.^{489,490}

In general, the antihypertensive effects of CCBs are enhanced more in combination with β -blockers or ACE inhibitors than in combination with diuretics.^{491–494} Perhaps this reflects the fact that CCBs themselves have intrinsic diuretic activity. It has been theorized that this particular combination (a dihydropyridine and ACE inhibitor) maximizes precapillary and postcapillary vasodilation to lower peripheral vascular resistance.

The combination of dihydropyridine CCBs with β -blockers is efficacious and even desirable in selected patients. CCBs have the potential to blunt the AEs associated with β -blockade, such as vasoconstriction, and β -blockers have the ability to attenuate the increased sympathetic stimulation induced by dihydropyridine CCBs. By contrast, concomitant therapy with β -blockers and nondihydropyridine CCBs is ill advised because they may have additive effects in suppressing heart rate, AV node conduction, and cardiac contractility (Table 49.19). This combination may be particularly dangerous in patients with ESKD because of the effects of hyperkalemia on cardiac conduction.

Drug interactions are not uncommon (Table 49.19). Concurrent use of a CCB and amiodarone exacerbates sick sinus syndrome and AV block. Diltiazem, verapamil, and nifedipine have been shown to increase the levels of cyclosporine (including the microemulsion formulation), tacrolimus, and sirolimus by 25% to 100%.⁴⁹⁵ This interaction may be clinically useful for reducing the dosage and cost associated with immunosuppressive therapy. Frequent monitoring of serum concentrations of calcineurin inhibitor is recommended. By contrast, nifedipine and isradipine exert no

effect on serum concentrations of calcineurin inhibitors and can be used safely. Diltiazem is a potent inhibitor of CYP3A4, which is responsible for the metabolism of methylprednisolone. Coadministration of diltiazem and methylprednisolone resulted in a more than a 2.5-fold increase in the steroid blood level and enhanced adrenal suppressive responses.⁴⁹⁶ Coadministration of diltiazem also increased nifedipine levels by 100% to 200%.⁴⁹⁷ This combination has additive antihypertensive efficacy and appears to be safe.⁴⁹⁸ Concomitant administration of CCBs with the digitalis glycosides resulted in up to a 50% increase in serum digoxin concentrations because of reduced renal clearance of digoxin, an effect that appears to be dose dependent.⁴⁹⁹ Insofar as dihydropyridine CCBs partially suppress aldosterone synthesis, they provide an attractive alternative for patients who cannot tolerate blockade of the RAAS.⁵⁰⁰

Several issues regarding the inherent safety of CCBs have come under scrutiny. CCBs may be associated with an increased risk of gastrointestinal hemorrhage, particularly in older persons.⁵⁰¹ Diltiazem inhibits platelet aggregation *in vitro*,⁵⁰² but the clinical relevance of this finding has not been substantiated. Nonetheless, it is prudent to use caution when coadministering CCBs with NSAIDs because NSAIDs may exacerbate the risk of bleeding and may antagonize the antihypertensive effects of CCBs.^{503,504}

There have been concerns regarding a possible relation between the long-term use of CCBs and breast cancer.^{132,505} A population case-control study of breast cancer showed that the use of CCBs for more than 10 years was associated with ductal breast cancer (odds ratio [OR], 2.4; 95% CI, 1.2 to 4.9; $P = .04$) and lobular breast cancer (OR, 2.6; 95% CI, 1.3 to 5.3; $P = .01$), respectively. However, two recent large population-based observational studies demonstrated no association between CCB use and breast cancer.^{506,507}

Short-acting CCBs have been associated with a small increased risk of MI in meta-analyses^{424,508} when compared with other agents. It has been speculated that the disadvantageous activation of the RAAS and SNS induced by the

Table 49.19 Drug-Drug Interactions With Calcium Channel Blockers

Calcium Channel Blocker	Interacting Drug	Result
Verapamil	Digoxin	Digoxin level $\uparrow$ by 50%–90%
Diltiazem	Digoxin	Digoxin level $\uparrow$ by 40%
Verapamil	β -Blockers	AV nodal blockade, hypotension, bradycardia, asystole
Verapamil, diltiazem	Cyclosporine-tacrolimus and sirolimus	Cyclosporine level $\uparrow$ by 25%–100%
Verapamil, diltiazem	Cimetidine	Verapamil and diltiazem levels $\uparrow$ by decreased metabolism
Verapamil	Rifampin-phenytoin	Verapamil level $\downarrow$ by enzyme induction
Dihydropyridines	Amiodarone	Exacerbation of sick sinus syndrome and AV nodal blockade
Dihydropyridines	α -Blockers	Excessive hypotension
Dihydropyridines	Propranolol	Increases propranolol level
Dihydropyridines	Cimetidine	Increased area under the curve and plasma level of calcium channel blocker
Nicardipine	Cyclosporine	Cyclosporine level $\uparrow$ by 40%–50%
Amlodipine	Cyclosporine	Cyclosporine level $\uparrow$ by 10%
Felodipine	Flavonoids	Bioavailability $\uparrow$ by 50%
Diltiazem	Methylprednisolone	Methylprednisolone $\uparrow$ 2.5-fold
Nifedipine	Diltiazem	Nifedipine level $\uparrow$ 100%–200%

AV, Atrioventricular.

short-acting agents may predispose to myocardial ischemia. Currently, there is no evidence to prove the existence of additional beneficial or detrimental effects of CCBs on coronary disease events, including fatal or nonfatal MIs and other deaths from coronary heart disease. Because of a potential risk, however, as well as for simplicity and improved patient adherence, longer-acting agents should be considered over short-acting CCBs for the management of hypertension.

CENTRAL ADRENERGIC AGONISTS

CLASS MECHANISMS OF ACTION

Central adrenergic agonists act by crossing the blood–brain barrier and have a direct agonist effect on α_2 -adrenergic receptors located in the midbrain and brain stem.^{494,509,510} Binding to the I_1 imidazoline receptors in the brain may also play a role in the inhibition of central sympathetic output.^{511–516} Drugs in this class bind to the α -adrenergic or I_1 imidazoline receptors with some degree of specificity (Table 49.20). Moxonidine and rilmenidine have a 30-fold greater specificity for the I_1 imidazoline receptor than the α_2 receptor. Clonidine, by contrast, exhibits a fourfold greater specificity for the I_1 imidazoline receptor than for the α_2 receptor. The central AEs are thought to be largely related to α_2 -receptor binding. Moxonidine and rilmenidine have reduced central side effects because of the lower activity at the α_2 receptor relative to other agents.^{516,517} In addition to decreasing the total sympathetic outflow, binding to these receptors results in increased vagal activity. A reduction in catecholamine release and turnover, as evidenced by decreased biochemical markers

of noradrenergic activity, such as plasma concentrations of norepinephrine, correlated with the magnitude of BP lowering.

Stimulation of both receptor types is probably mediated through the same neuronal pathways.⁵¹⁷ The classic α_2 -receptor agonists, such as clonidine and α -methyldopa (acting through its active metabolite, α -methylnorepinephrine), result in vasodilation in the resistance vessels and thus a reduction in peripheral vascular resistance. Despite vasodilator action, reflex tachycardia generally does not occur, probably as a result of peripheral sympathetic inhibition.

The selective I_1 receptor agonists moxonidine and rilmenidine are predominantly arterial vasodilators that lead to a reduction in peripheral vascular resistance.⁵¹² Moxonidine is associated with a reduction in plasma renin activity. The central α_2 -adrenergic agonists may also stimulate peripheral α_2 -adrenergic receptors. This effect predominates at high drug concentrations. These receptors mediate vasoconstriction, which may result in a paradoxical increase in BP.⁵⁰⁹ Overall, these drugs generally result in a decrease in peripheral vascular resistance, slowing of the heart rate, and either no change or a mild decrease in cardiac output.^{517,518} Orthostatic hypotension is generally not a feature of these drugs. The pharmacokinetic and pharmacodynamic properties of these drugs are shown in Tables 49.21 and 49.22.

CLASS MEMBERS

Methyldopa is a methyl-substituted amino acid that is active after conversion to an active metabolite. This active metabolite, α -methylnorepinephrine, accumulates in the CNS and is selective for α_2 -adrenergic receptors. The initial dosage of methyldopa in hypertension is 250 mg two to three times daily. This dose may be increased at intervals of not less than 2 days until a therapeutic response is achieved. The usual maintenance dosage is 500 mg to 2 g daily in two to four doses. The maximum recommended daily dose is 3 g. An initial response occurs within 3 to 6 hours after dosing. The peak response occurs at 6 to 9 hours. The drug is approximately 50% metabolized by the liver. The drug half-life is increased in patients with CKD. Excretion in the urine is largely in the form of an inactive metabolite. The dosing interval should be increased to every 12 to 24 hours in patients with advanced CKD. Approximately 60% of methyldopa is removed with hemodialysis. A supplemental dose is recommended after dialysis treatment.

Table 49.20 Receptor Binding of Centrally Acting Antihypertensives

Drug	Receptor
Clonidine	α_2 , I_1
α -Methyldopa	α_2
Guanabenz	α_2
Guanfacine	α_2
Rilmenidine	$I_1 > \alpha_2$
Moxonidine	$I_1 > \alpha_2$

I_1 , Imidazole receptor; α_2 , α_2 -adrenergic receptor.

Table 49.21 Pharmacokinetic Properties of Central Adrenergic Agonists

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Clonidine (Catapres)	50	—	—	6–23	L	F (30%–50%) U (24%)	Methyldopa-o-sulfite
α -Methyldopa (ALDOMET)	65–96	—	1.5–5	6–23	L	F (22%) U (65%)	—
Guanabenz (Wytensin)	75	—	2–5	7–10	L	F (16%)	—
Guanfacine (Tonex)	80	—	1–4	17	L	U (40%–75%)	—
Rilmenidine (Hyperium)	80–90	No	2	2–3	L	U (90%)	—
Moxonidine (Physiotens)	100	No	0.5–3	2	L	U (90%)	—

F, Feces; L, liver; U, urine.

Table 49.22 Pharmacodynamic Properties of Central Adrenergic Agonists

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Clonidine	0.1	0.3–0.9	2.4	bid, tid	2–4	6–10
α -Methyldopa	250	250–500	3000	bid, tid, qid	6–9	24–48
Guanabenz	4	16–32	64	bid	2–4	10–12
Guanfacine	1	1–3	3	qd	6	24
Moxonidine	0.1–0.2	0.2–0.3	0.6	bid	1.5–4	48–72
Rilmenidine	1	1–2	2	qd, bid	1–2	10–12

bid, Two times a day; *qd*, once a day; *qid*, four times a day.

Clonidine is a central-acting α -adrenergic agonist.^{151,513,519} The usual oral dosage is 0.1 mg two to three times daily, adjusted as necessary in 0.1- to 0.2-mg increments. The usual maintenance dosage is 0.3 to 0.9 mg daily in two to three divided doses. Total doses more than 1.2 mg daily are usually not associated with a greater effect. The onset of activity is 30 to 60 minutes after an oral dose. The peak antihypertensive activity occurs within 2 to 4 hours. The duration of the antihypertensive effect is 6 to 10 hours. The half-life of the absorbed drug is 6 to 23 hours. Hepatic metabolism to inactive metabolites is followed by renal excretion. Transdermal patches are available and may be applied on a once-weekly basis. The drug half-life with the transdermal patch is approximately 20 hours after removal of the patch. With a transdermal patch, steady-state drug levels are reached within approximately 3 days. Dose adjustment is not needed for patients with any degree of CKD including ESKD. Approximately 5% of clonidine body stores are removed after a 5-hour hemodialysis session.

Guanabenz is an orally active, central α_2 -adrenergic agonist.⁵⁰⁹ The usual starting dosage for the management of hypertension is 4 mg twice daily. Doses may be increased by 4 to 8 mg/day at 1- to 2-week intervals. The onset of antihypertensive activity usually occurs within 60 minutes, and the activity lasts approximately 10 to 12 hours. The drug is highly protein bound and extensively metabolized. Less than 1% of the unchanged drug is excreted in the urine. The half-life of the drug is 7 to 10 hours. Dose adjustment in patients with CKD is not necessary but may be necessary in patients with severe hepatic insufficiency. Because of extensive protein binding, drug removal by dialysis is minimal.

Guanfacine is a centrally acting antihypertensive drug with actions similar to those of clonidine.⁵²⁰ Effective dosages are 1 to 3 mg daily. Peak levels are noted between 1 and 4 hours. The drug half-life is approximately 17 hours. The drug is 70% protein bound. It is metabolized in the liver, with renal excretion of 40% to 75% as an unchanged drug. Limited data are available on dosing in CKD, but dosage adjustments do not appear warranted.

Moxonidine is a central I_1 imidazole and α_2 -receptor agonist.^{151,521,522} Serum concentration peaks are reached within 30 to 180 minutes; 90% of the dose is excreted through the urine within 24 hours, and 50% of this is as unchanged drug. The average half-life is 2 hours. For the management of hypertension, the starting dosage is 0.2 to 0.4 mg/day. The dose may be increased after several weeks to 0.2 to 0.3 mg twice daily. The maximum daily dose is 0.6 mg. Selectivity

for the I_1 imidazole receptor results in fewer central AEs, such as dry mouth and sedation, compared with those of clonidine. Drug clearance is delayed in patients with impaired kidney function. Single doses of 0.2 mg and a maximum daily dosage of 0.4 mg should not be exceeded in patients with CKD.

Rilmenidine is a centrally acting imidazole receptor and α_2 -adrenergic receptor agonist.^{151,515,523–527} Rilmenidine binds preferentially to central I_1 imidazole receptors in the brain stem. At higher doses, rilmenidine can bind and activate central α_2 -adrenergic receptors. Antihypertensive effects occur within 1 hour after a single 1-mg dose. The duration of action is 10 to 12 hours. The concentration after oral dosing peaks at approximately 2 hours. Steady-state plasma levels are reached by day 3. Rilmenidine is eliminated primarily unchanged in the urine. The usual oral dosage is 1 mg once or twice daily. Dose reductions are required for patients with impaired kidney function. In patients with advanced CKD, the dose should be decreased to 1 mg every other day.

CLASS RENAL EFFECTS

Central α_2 - and I_1 imidazole receptor agonists have little if any clinically important effect on renal plasma flow, GFR, or the RAAS. The fractional excretion of sodium is unchanged. Body fluid composition and weight are not altered. A water diuresis may be associated with the use of guanabenz through inhibition of the central release of vasopressin or altered renal responsiveness to vasopressin. These agents may result in decreased renal vascular resistance, as mediated by a decrease in preglomerular capillary resistance related to decreased levels of circulating catecholamines.

CLASS EFFICACY AND SAFETY

The antihypertensive efficacy of this class of drugs has been confirmed in large numbers of patients. These agents provide effective monotherapy for hypertension.⁵⁰⁹ Combination with a diuretic is associated with additive effects. Drugs in this class are effective for young and old patients, and the effects do not differ in different racial or ethnic groups. Moxonidine and rilmenidine have been associated with decreased plasma glucose concentrations and may improve insulin sensitivity. These drugs may also decrease total cholesterol, LDL, and triglyceride levels^{511,528,529} and may play a role in the management of metabolic syndrome. They may also be of benefit in patients with congestive heart failure. Treatment with rilmenidine and moxonidine reverses LVH and improves

arterial compliance. This effect was associated with a reduction in plasma levels of atrial natriuretic peptide.

Stimulation of α_2 -adrenergic receptors in the CNS induces several AEs of these drugs, including sedation and drowsiness. The most common AE related to α_2 -adrenergic activation is dry mouth caused by a decrease in salivary flow. This decrease is attributed to centrally mediated inhibition of cholinergic transmission. Clonidine in high doses may precipitate a paradoxical hypertensive response related to the stimulation of postsynaptic vascular α_2 -adrenergic receptors.⁵⁰⁹ Methyldopa use has been associated with a positive result on the direct Coombs test in patients with and without hemolytic anemia.⁵⁰⁹ Because of a long history of safe use during pregnancy, methyldopa remains a common therapeutic agent for hypertensive disorders of pregnancy and for essential hypertension during pregnancy.^{494,510} The α_2 -adrenergic agonists are associated with sexual dysfunction and may produce gynecomastia in men and galactorrhea in men and women.

Abrupt cessation of α_2 -adrenergic blockers may result in rebound hypertension, which occurs 18 to 36 hours after the cessation of short-acting agents.⁵³⁰ Patients may experience tachycardia, tremor, anxiety, headache, nausea, and vomiting. This syndrome may be related to downregulation of the α_2 -adrenergic receptors in the CNS in the setting of long-term use of these agents. These agents have a higher specificity for the I_1 receptor and appear to produce significantly fewer CNS effects, such as dry mouth and drowsiness. Rebound hypertension secondary to abrupt withdrawal has not been associated with moxonidine or rilmenidine.

CENTRAL AND PERIPHERAL ADRENERGIC NEURONAL BLOCKING AGENT

MECHANISMS OF ACTION AND CLASS MEMBER

Reserpine, a *Rauwolfia* alkaloid, reduces BP by decreasing the activity of central and peripheral noradrenergic neurons. Reserpine blocks norepinephrine and dopamine uptake into the storage granules of noradrenergic neurons. The result is norepinephrine depletion. A similar effect is seen in central dopaminergic and serotonergic neurons. At the dosages currently used to treat hypertension, the major effect of the use of reserpine is in the CNS. Reserpine results in a rapid reduction in cardiac output, heart rate, and peripheral vascular resistance. Enhanced vagal activity may also be involved. Tolerance to the antihypertensive effects of reserpine does not occur.

Reserpine is used at initial dosages of 0.1 to 0.25 mg daily.⁵³¹ Approximately 40% of an oral dose is absorbed. The half-life is 50 to 100 hours. Extensive hepatic metabolism occurs; 1% is recovered as unchanged compound in the urine. The maximal clinical effect is observed 2 to 3 weeks after initiation of therapy. No dosage adjustment is necessary for patients with renal insufficiency. Dosage supplementation is not required after hemodialysis.

RENAL EFFECTS

The GFR and renal plasma flow are not affected by reserpine therapy. Renal vascular resistance may be reduced, perhaps mediated by decreased sympathetic stimulation of vascular α -adrenergic receptors. Significant effects on the RAAS have not been observed. Renal handling of sodium and potassium is unchanged.

EFFICACY AND SAFETY

Reserpine provides effective therapy as a single agent or in combination with HCTZ or HCTZ and hydralazine.^{531,532} This has been observed in numerous large and small trials, including the Veterans Administration Cooperative Study on Antihypertensive Agents, Hypertension Detection and Follow-up Program, and the Multiple Risk Factor Intervention Trial. Reserpine used in combination with a diuretic has shown comparable efficacy to combinations of β -blockers and diuretics. In these studies, the dose of reserpine was between 0.1 and 0.3 mg daily, which is many times lower than the doses used in the 1960s that led to reserpine's reputation as having a poor side-effect profile. The most common AE of reserpine is nasal congestion, which is reported in 6% to 20% of patients. Unlike other AEs, nasal congestion does not appear to decrease at lower drug dosages and is thought to be related to the cholinergic effects of the drug. Increased gastric motility and gastric acid secretion can occur; however, the incidence of dyspepsia or peptic ulcer disease with reserpine therapy is not greater than that with other antihypertensive drug treatments. Inability to concentrate, sedation, sleep disturbance, and depression have been reported. Other AEs include weight gain, increased appetite, and sexual dysfunction.

DIRECT-ACTING VASODILATORS

CLASS MECHANISMS OF ACTION

The direct-acting vasodilators reduce SBPs and DBPs by decreasing peripheral vascular resistance. These drugs act directly on vascular smooth muscle with the selective vasodilation of the arteriolar resistance vessels and have little or no effect on the venous capacitance vessels.⁵³³ There is no effect on the functioning of carotid or aortic baroreceptors. The vasodilating effects are thought to involve inhibition of calcium uptake into the cells. Decreases in arterial pressure are associated with a decrease in peripheral resistance and a reflex increase in cardiac output. Sodium and water retention are promoted secondary to the stimulation of renin release and possibly by direct effects on renal tubules. The arteriolar dilation produced by these drugs causes a decrease in cardiac afterload,⁵³³ and the absence of venodilation leads to an increase in venous return to the heart, which produces an elevated preload. These combined effects result in increased cardiac output.⁵³³ The pharmacokinetic and pharmacodynamic properties of these drugs are shown in [Tables 49.23](#) and [49.24](#).

CLASS MEMBERS

The initial oral dose of hydralazine for hypertension should be 10 mg four times daily, increasing to 50 mg four times daily over several weeks. Patients may require doses of up to 300 mg/day. Dosing can be changed to twice daily for maintenance. The drug may also be used as an intravenous bolus injection or as a continuous infusion. The elimination half-life is 1.5 to 8 hours and varies with the acetylation rate in the liver. Slow and fast acetylators have been described. The onset of action is approximately 1 hour. In patients with mild-to-moderate CKD, the dosing interval should be increased to every 8 hours. In patients with advanced CKD, the dosing interval should be increased to every 8 to 24 hours. No dose

Table 49.23 Pharmacokinetic Properties of Direct-Acting Vasodilators

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Hydralazine (Apresoline)	20–50	No	1–2	1.5–8	L	U (3%–14%) F (3%–12%)	—
Minoxidil (Loniten)	90–100	—	1	4.2	L	U (90%) F (3%)	Glucuronide

F, Feces; L, liver; U, urine.

Table 49.24 Pharmacodynamic Properties of Direct-Acting Vasodilators

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Hydralazine	10	200–400	400	bid, qid	1	3–8
Minoxidil	2.5	10–20	40	qd, qid	4–8	10–12

bid, Two times a day; qd, once a day; qid, four times a day.

supplement is required after hemodialysis or peritoneal dialysis (Table 49.7).

Minoxidil is more potent than hydralazine. For severe hypertension, the initial recommended starting dose is 2.5 mg as a single daily dose, increasing to 10 to 20 or 40 mg in single or divided doses. Minoxidil is usually used in conjunction with salt restriction and diuretics to prevent fluid retention. Concomitant therapy with a β -adrenergic blocking agent is often required to control tachycardia related to minoxidil use. The onset of the antihypertensive effect is within 30 to 60 minutes. The peak response occurs at 4 to 8 hours. The drug is 90% metabolized by the liver. The glucuronide metabolite has reduced pharmacologic effects but accumulates in patients with ESKD. Renal excretion is 90%. Dose adjustments may be required for patients with advanced CKD, although the mean daily doses required to control BP are similar in patients with normal and impaired kidney function (Table 49.7).

The effectiveness of minoxidil in patients with resistant hypertension has been well established.^{534,535} Minoxidil is frequently a therapy of last resort in patients with CKD unresponsive to other therapies. It must nearly always be used in combination with a β -blocker and a loop diuretic to prevent tachycardia and fluid retention.

CLASS RENAL EFFECTS

Hydralazine and minoxidil both increase the juxtaglomerular cell secretion of renin, which is associated with increased Ang II and aldosterone levels. Long-term use is associated with the return of plasma aldosterone levels to baseline. Retention of salt and water may be attributed to direct drug effects on the proximal convoluted tubule. Renal vascular resistance is decreased in association with a relaxation of resistance vessels.⁵³³ GFR and renal plasma flow are preserved.

Hydralazine may reduce BP compared with placebo in patients with essential hypertension; however, these results are based on before and after studies, not on RCTs. It is

more commonly used in practice to reduce afterload in heart failure among patients with reduced left ventricular function.

CLASS EFFICACY AND SAFETY

Minoxidil is commonly reserved for severe or intractable hypertension. Because of its propensity to cause salt and water retention, administration with a loop diuretic is usually required. Hypertrichosis is a common AE. Pericarditis and pericardial effusions have been described.⁵³⁶ An increase in left ventricular mass has been reported, which may be related to adrenergic hyperactivity.

In addition to adrenergic activation and fluid retention, long-term treatment with hydralazine has been associated with the development of systemic lupus erythematosus. Generally, this syndrome occurs early in therapy, but can develop after many years of treatment. It has been estimated that between 6% and 10% of patients receiving high doses of hydralazine for longer than 6 months develop hydralazine-induced lupus.⁵³⁷ It is seen most frequently in women and rarely in African Americans. This syndrome occurs primarily in slow acetylators and is reversible when hydralazine is discontinued, but months may be required for complete clearing of symptoms. Hydralazine has been frequently used to treat pregnancy-associated hypertension in view of its relatively low teratogenicity.

ENDOTHELIN RECEPTOR ANTAGONISTS

Endothelin is among the most potent endogenous vasoconstrictors known.⁵³⁸ It also enhances mitogenesis and induces extracellular matrix formation. Endothelin is also thought to be involved in vascular remodeling and end-organ damage under several different cardiovascular conditions.⁵³⁹ As a result, endothelin receptor antagonists have been studied as targets for treatment of hypertension. However, concerns about their safety and tolerability have limited their use largely to treatment of pulmonary hypertension.

CLASS MECHANISM OF ACTION AND CLASS MEMBERS

The two primary receptor sites, endothelin type A (ET-A) and endothelin type B (ET-B), can be selectively blocked with different chemicals, or both sites can be blocked simultaneously. Bosentan, a mixed ET-A/ET-B receptor antagonist, was studied in a large placebo-controlled trial lasting 4 weeks and demonstrated a dose-dependent reduction in BP compared with placebo when bosentan was dosed from 100 mg once daily up to 1000 mg twice daily.⁵⁴⁰ The mean reduction in DBP of 5.8 mm Hg across all dosages at or above 5 mg/day was almost identical to that seen with 20 mg of enalapril. The most common AEs were peripheral edema, flushing, headache, and some alterations in liver enzyme levels.

CLASS RENAL EFFECTS

Endothelin receptor antagonists have been used to study the role of endothelin in the development of acute CKD in different experimental models.⁵⁴¹ Clinical studies using endothelin receptor blockers have shown improvement in proteinuria. This effect is independent of the antiproteinuric effects of inhibition of the RAAS and may help relieve renal injury.⁵⁴² What is unknown is whether the selective blockage of the ET-A receptor over the ET-B receptor will not only control BP but also mitigate renal ischemia and reduce proteinuria.

MODERATELY SELECTIVE PERIPHERAL α_1 -ADRENERGIC ANTAGONISTS

CLASS MECHANISMS OF ACTION

The nonselective agents phentolamine and phenoxybenzamine have an occasional role in hypertension management. Phentolamine is administered parenterally, and the longer-acting agent phenoxybenzamine has been used orally for the management of hypertension associated with pheochromocytoma.⁵⁴³ Phenoxybenzamine is a moderately selective, peripheral α_1 -adrenergic antagonist. Its specificity for the α_1 -adrenergic receptor is 100 times greater than that for the α_2 -adrenergic receptor.

CLASS MEMBERS

Phenoxybenzamine is a long-acting α -adrenergic blocking agent. This agent irreversibly and covalently binds to α receptors only. β Receptors and the parasympathetic system are not affected by phenoxybenzamine. The total peripheral resistance is decreased, and cardiac output increases with phenoxybenzamine. Phenoxybenzamine is also believed to inhibit the uptake of catecholamines into adrenergic nerve terminals and extraneural tissues. The usual oral dose of phenoxybenzamine for the treatment of pheochromocytoma is initially 10 mg twice daily, with the dose gradually increased every other day to dosages ranging between 20 and 40 mg two or three times daily. The final dosage should be determined by the BP response. Phenoxybenzamine may be administered with a β -blocking agent if tachycardia becomes excessive during therapy. The pressor effects of a pheochromocytoma must be controlled by α -blockade before β -blockers are initiated. With oral use, the pheochromocytoma symptoms decrease after several days. The oral bioavailability is 20% to 30%. The drug is extensively metabolized by the liver.

Phenoxybenzamine should be administered cautiously to patients with renal impairment. Specific dosage recommendations are not available.

Phentolamine is an α -adrenergic blocking agent that produces peripheral vasodilation and cardiac stimulation, with a resulting decrease in BP in most patients. The drug is used parenterally. The usual dose is 5 mg, repeated as needed. The onset of activity with intravenous dosing is immediate. The drug is not absorbed well orally; its half-life is 19 minutes. Phentolamine is metabolized by the liver, with 10% excreted in the urine as unchanged drug.

CLASS RENAL EFFECTS

Phenoxybenzamine has no clear effect on the RAAS. Blood volume and body weight are not altered. Salt and water retention do not occur. GFR and effective renal plasma flow would be expected to increase. Renal vascular resistance probably decreases in proportion to the degree of blockade of α -adrenergic receptors.

CLASS EFFICACY AND SAFETY

Phenoxybenzamine is used primarily as an agent to counteract the excessive α -adrenergic tone associated with pheochromocytoma. Tachycardia may result from an α -adrenergic blockade, which unmasks β -adrenergic effects with epinephrine-secreting tumors. This may be controlled with concurrent use of a β -adrenergic antagonist. α -Adrenergic blockade must be initiated before β -adrenergic blockade to avoid paradoxical hypertension. AEs of phenoxybenzamine are sedation, weakness, nasal congestion, hypertension, and tachycardia.

PERIPHERAL α_1 -ADRENERGIC ANTAGONISTS

CLASS MECHANISMS OF ACTION

Drugs of the peripheral α_1 -adrenergic antagonist class, including doxazosin, prazosin, and terazosin, are selective antagonists of the postsynaptic α_1 -adrenergic receptor. These drugs, which blunt the increases in arteriolar and venous tone mediated by norepinephrine released from sympathetic nerve terminals, act at the α_1 -adrenergic receptor located postjunctionally in the blood vessel wall. The affinity of these drugs for the α_2 receptor is very low. Because of the selective α_1 action, there is no interference with the negative feedback control mechanisms that are mediated by the prejunctional α_2 receptors. As a result, the reflex tachycardia associated with the blockade of the presynaptic α_2 receptor decreases substantially. The pharmacokinetic and pharmacodynamic properties of these drugs are shown in [Tables 49.25](#) and [49.26](#).

CLASS MEMBERS

Doxazosin is a selective long-acting α_1 -adrenergic antagonist. The initial antihypertensive dosage is 1 mg daily. This dose can be titrated up to a maximum of 16 mg daily. The maximal antihypertensive effect is seen 4 to 8 hours after a single dose. The drug is highly plasma protein bound and extensively metabolized. Most of the administered dose is excreted in the feces. The estimated half-life ranges from 9 to 22 hours. Doxazosin pharmacokinetics are not altered in patients with impaired kidney function. The drug should be used with caution in patients with advanced liver dysfunction.

Prazosin is a selective α_1 -adrenergic antagonist that is structurally related to doxazosin. Oral dosing is 3 to 20 mg/

Table 49.25 Pharmacokinetic Properties of Peripheral α_1 -Adrenergic Antagonists

Drug	Bioavailability (%)	Affected by Food	Peak Blood Level (h)	Elimination Half-Life (h)	Metabolism	Excretion	Active Metabolites
Doxazosin (Cardura)	62–69	No	2–5	9–22	L	F (63%–65%) U (1%–9%)	—
Prazosin (Minipress)	—	No	1–3	2–4	L	F	—
Terazosin (Hytrin)	90	Yes	1	12	L	F (45%–60%) U (10%)	—

F, Feces; *L*, liver; *U*, urine.

Table 49.26 Pharmacodynamic Properties of Peripheral α_1 -Adrenergic Antagonists

Drug	Initial Dose (mg)	Usual Dose (mg)	Maximum Dose (mg)	Interval	Peak Response (h)	Duration of Response (h)
Doxazosin	1	8	16	qd, qid	4–8	24
Prazosin	1	3–20	20	bid, qid	0.5–1.5	10
Terazosin	1	5	20	qd, bid	3	24

bid, Two times a day; *qd*, once a day; *qid*, four times a day.

day. Full therapeutic effects are seen within 4 to 8 weeks after initiation of therapy. Peak serum levels are reached 1 to 3 hours after an oral dose. The drug is highly protein bound with an elimination half-life of 2 to 4 hours. There is extensive hepatic metabolism followed by renal excretion of a very small amount of unchanged drug. Dose adjustment is not required for patients with CKD. Patients with significant liver disease may require dose adjustment and more frequent monitoring.

Terazosin is a selective, long-acting α_1 -adrenergic antagonist that has structural similarities to prazosin and doxazosin. The initial dosage is 1 mg orally at bedtime, with titration to 5 mg daily. Doses of 10 to 20 mg orally have been given. Peak serum levels after oral administration occur within 1 hour. The half-life is approximately 12 hours. Terazosin is extensively metabolized in the liver and eliminated primarily through the biliary tract. Pharmacokinetics are not affected by CKD, and dose adjustment is not required. Patients with severe hepatic insufficiency may require dose adjustments.

CLASS RENAL EFFECTS

GFR and renal blood flow are maintained during long-term treatment. In some studies, there was a slight increase in renal blood flow. Renal vascular resistance may be reduced, perhaps mediated by a reduction in preglomerular capillary resistance related to inhibition of α_1 -mediated vasoconstriction. Urinary protein excretion has been reported to be reduced. The RAAS is not significantly affected by specific α_1 -adrenergic antagonists. Extracellular fluid volume has been reported to be increased, and fractional excretion may be decreased.

CLASS EFFICACY AND SAFETY

Comparative clinical studies of the efficacy of α_1 -adrenergic blockers have shown that the antihypertensive responses are

similar to those elicited by other antihypertensive drugs.⁸⁸ The availability of ER formulations of these drugs has improved tolerability. These drugs have been shown to increase insulin sensitivity⁵⁴⁴ to exert potentially beneficial effects on lipid metabolism.^{92,545–548} They can cause a modest reduction in total and LDL cholesterol and a small increase in HDL cholesterol levels. This metabolic benefit may be linked to the beneficial effect on insulin responsiveness, leading to increased peripheral glucose uptake.

There are also potential AEs of α_1 -adrenergic receptor blockers. In the Antihypertensive and Lipid-Lowering Treatment To Prevent Heart Attack Trial (ALLHAT), patients randomized to receive doxazosin as their initial antihypertensive drug were found to have poorer BP control than those receiving a chlorthalidone-based treatment.⁵⁴⁹ No difference was seen in the primary outcomes of fatal coronary heart disease or nonfatal MI in patients, but patients randomized to doxazosin had higher rates of stroke and congestive heart failure.⁵⁵⁰ Data from ALLHAT have largely relegated doxazosin and other peripheral α_1 -adrenergic antagonists to second-line agents in the management of hypertension.

α_1 -Adrenergic receptor blockers can also cause a first-dose orthostatic hypotension effect, resulting in lightheadedness, palpitations, and occasionally syncope. This is related to the drug effect on the venous capacitance vessels, which results in venous dilation and inadequate venous return. It may occur when peak drug levels are reached 30 to 90 minutes after the first dose; it can be minimized by initiating therapy with a small dose taken at bedtime.⁵⁵¹ This effect can be exacerbated in patients with underlying autonomic insufficiency.

α_1 -Adrenergic antagonists are also used for the symptomatic management of prostatic hypertrophy. Prostatic smooth muscle has significant α_1 -adrenal receptor expression. Blockade of these receptors results in smooth muscle

relaxation within the prostate.^{552,553} Terazosin has been marketed in combination with amlodipine for male patients with hypertension and lower urinary tract symptoms.⁵⁵⁴

Renin Inhibitors

The earliest renin-inhibiting compounds in development included aliskiren, zankiren, and remikiren. However, because of problems with oral bioavailability, only aliskiren has been approved for the treatment of hypertension.

CLASS MECHANISM OF ACTION AND CLASS MEMBER

Aliskiren is an oral nonpeptide, low-molecular-weight renin inhibitor used for the management of hypertension (Fig. 49.2).^{555,556} Renin inhibition interferes with the first and rate-limiting step in the renin enzyme cascade, the interaction of renin with its substrate angiotensinogen. The renin blockade step is an attractive target for hypertension therapeutics, in large part because of the remarkable specificity of renin for its substrate.⁵⁵⁷ This specificity reduces the likelihood of unwanted interactions and possible AEs. In addition, unlike ACE inhibitors or ARBs, which lead to a reactive increase in renin and associated angiotensin peptides, direct renin inhibition renders the RAAS quiescent. Although aliskiren has low inherent bioavailability (2.6%), it is potent in reducing BP and has an effective half-life of 40 hours.⁵⁵⁸ The drug is not actively metabolized by the liver and is primarily excreted in the urine, with most of it as unchanged drug. CYP enzymes are not involved in the metabolism of aliskiren. Thus clinically significant CYP inhibition by aliskiren is unlikely.⁵⁵⁹

CLASS RENAL EFFECTS

Experimental animal studies suggested that aliskiren may provide renal protection by reducing BP and urinary albumin excretion.^{560–563} Clinical trial data in humans indicated that aliskiren reduces proteinuria in conjunction with decreasing BP.⁵⁵⁷ In a study involving 600 patients with hypertension, diabetes, and nephropathy, the administration of aliskiren, 300 mg/day, facilitated a 20% incremental reduction in proteinuria compared with placebo in patients being treated concurrently with losartan, 100 mg/day. The baseline BP was 135 mm Hg with losartan treatment before random assignment to aliskiren therapy and did not change with the addition of aliskiren.³²⁶ Renal vascular response curves for ACE inhibition or renin inhibition at the top of the dose-response curve indicated greater increase in renal blood flow with the renin inhibitor, despite similar changes in BP.⁵⁵⁷ However, despite these encouraging clinical observations, a subsequent clinical trial, ALTITUDE, did not demonstrate an incremental benefit of aliskiren when used with an ACE inhibitor or ARB in patients with diabetic kidney disease when examining cardiovascular and renal endpoints.²⁵⁶ From a safety standpoint, there were more adverse events in the patients receiving both classes of RAAS-blocking drugs.

CLASS EFFICACY AND SAFETY

Clinical trials of aliskiren have demonstrated a dose-dependent efficacy in reducing SBP and DBP. In an 8-week, double-blind, placebo-controlled trial, aliskiren was studied in dosages from 150 to 600 mg/day.⁵⁶³ The placebo-corrected reduction in sitting SBP was approximately 10 to 11 mm Hg for both the

300- and 600-mg doses. Also evaluated in the same study was irbesartan, an ARB, at a dosage of 150 mg/day, which effectuated BP reduction comparable to that of aliskiren at a dosage of 150 mg/day. Other clinical studies have confirmed the antihypertensive efficacy of aliskiren compared with placebo or with other active therapies, such as the ARB losartan. As expected, BP reduction was comparable with that produced by the active agents. The only observed difference was suppression of plasma renin activity and Ang I and Ang II levels with aliskiren but increased levels of Ang I and Ang II with ARB therapy.

The tolerability of aliskiren is comparable to that of placebo, with a relatively low incidence of adverse events at all dosages tested in the 150- to 600-mg range.⁵⁶³ Aliskiren treatment was comparable in tolerability to treatment with an ARB or placebo, with a low discontinuation rate and no statistical difference in the incidence of different types of adverse events, except for some diarrhea at the 600-mg dose.

Aliskiren has also been studied in combination with HCTZ, CCBs, ARBs, and ACE inhibitors in the management of hypertension.^{564–567} In one small clinical trial, all patients received aliskiren 150 mg once daily for 3 weeks. Patients who had a daytime BP above 130/80 mm Hg, as measured by ambulatory BP monitoring, were given HCTZ, 25 mg daily, for an additional 3 weeks. Adding 25 mg of HCTZ resulted in an additional 10 mm Hg of SBP reduction. Not surprisingly, there was no difference in plasma renin activity between the patients receiving aliskiren with HCTZ and those receiving aliskiren alone. Adding aliskiren to amlodipine, valsartan, or ramipril provides incremental and statistically significant improvements in BP reduction. However, the ALTITUDE study did raise safety concerns about using aliskiren with other RAAS blocking drugs due specifically to declining kidney function and hyperkalemia.²⁵⁵ The additional BP reduction may or may not be related to the neutralization of plasma renin activity that occurs with the addition of the renin inhibitor.

SELECTIVE ALDOSTERONE RECEPTOR ANTAGONISTS

CLASS MECHANISM AND CLASS MEMBERS

Spironolactone and eplerenone are members of the class of MRAs. The mineralocorticoid receptor forms part of the steroid–thyroid–retinoid–orphan receptor family of nuclear transactivating factors.⁵⁶⁸ When unbound, these receptors are in an inactive multiprotein complex of chaperones. On binding of aldosterone, the chaperones are released, and the receptor hormone complex is translocated into the nucleus, where it binds to hormone response elements on DNA and interacts with transcription initiation complexes, which ultimately modulate gene expression.⁵⁶⁹ In the kidney, mineralocorticoid receptors are located primarily in the epithelial cells of the distal nephron. These receptors bind physiologic glucocorticoids and mineralocorticoids with a similar affinity. Activation of mineralocorticoid receptors by aldosterone results in the activation of epithelial sodium channels, which leads to a rapid increase in sodium and water reabsorption and promotes the tubular secretion of potassium.^{570,571} A persistent increase in sodium balance does not occur, even with continued stimulation of mineralocorticoid receptors by aldosterone. The mechanism of the aldosterone escape phenomenon has not been fully elucidated.

There is evidence indicating the presence of biologic activity of mineralocorticoid receptors in nonepithelial tissues.⁵⁷² These receptors have been identified in blood vessels of the heart and brain and may be involved in vascular injury and repair responses.^{520,573} Aldosterone mediates fibrosis and collagen formation through the upregulation of Ang II receptor responsiveness.^{519,572} Aldosterone increases sodium influx in vascular smooth muscle and inhibits norepinephrine uptake in vascular smooth muscle and myocardial cells.⁵⁷⁴ Aldosterone also directly participates in vascular smooth muscle cell hypertrophy.

Spironolactone and eplerenone are both inhibitors of the mineralocorticoid receptor. Spironolactone is a prodrug that is metabolized in the liver and excreted in the urine. The half-life of the prodrug is 1.3 to 2 hours whereas the half-life of the active metabolites ranges from 14 to 17 hours.

Eplerenone is approximately 24 times less potent than spironolactone but is more specific for blocking mineralocorticoid receptors with little agonist activity at estrogen and progesterone receptors.⁵⁷⁵ Therefore eplerenone is associated with a lower incidence of gynecomastia, breast pain, and impotence in men and diminished libido and menstrual irregularities in women. The time to peak concentration is 1 to 2 hours. No significant accumulation occurs with multiple-dose administration, but up to 4 weeks may be required for the full antihypertensive effects to be evident. It appears to be well absorbed, but absolute (oral vs. intravenous) data are unavailable; specific data on protein binding and metabolism are also unavailable. The elimination half-life of eplerenone is 4 to 6 hours. Bioavailability is 69%, with approximately 50% protein binding. It is metabolized primarily by the hepatic CYP3A4 system to inactive metabolites excreted two-thirds in the urine and one-third in the feces.

CLASS RENAL EFFECTS

Selective aldosterone receptor antagonism may have benefits for the kidney, independently of its effects on BP. Experimental and clinical studies have demonstrated that Ang II may be the primary mediator of the RAAS associated with the progression of renal disease.^{576,577} The relative importance of aldosterone in this cascade has been the subject of experimental and clinical studies. Hyperaldosteronism and adrenal hypertrophy are common observations in remnant kidney models and correlate with progressive loss of renal function.⁵⁷⁶ Hypertension, proteinuria, and structural injury are less prevalent in subtotaly nephrectomized rats that have undergone adrenalectomy, even when administered large doses of replacement glucocorticoids.⁵⁷⁸ Other investigators have demonstrated that aldosterone infusion can reverse the renal protective effects of an ACE inhibitor in stroke-prone, spontaneously hypertensive rats.⁵⁷⁹ Interestingly, in this model, the kidney injury induced by aldosterone was independent of increases in BP, suggesting a toxic tissue effect of aldosterone. Other experimental studies have indicated that aldosterone receptor antagonism can prevent the development of proteinuria.⁵⁷⁹

Even though selective aldosterone receptor antagonists have no observable effects on glomerular hemodynamics, therapy with these drugs may provide an incremental effect in protecting the kidney when added to ACE inhibitors or ARBs by inhibiting the effects of aldosterone that persist, despite treatment with the latter drugs. Studies in patients with diabetic nephropathy or other glomerular diseases have

demonstrated that the addition of spironolactone to ACE inhibition markedly reduces albuminuria.^{580,581}

CLASS EFFICACY AND SAFETY

These medications can be very effective for BP control, particularly in patients with resistant hypertension. In the PATHWAY-2 trial, a double-blind, placebo-controlled, crossover trial, 285 patients with resistant hypertension received spironolactone, doxazosin, bisoprolol or placebo as the fourth agent.^{581a} During spironolactone therapy versus placebo, the participants' average home systolic BP decreased by 8.7 mm Hg (95% CI, 7.7 to 9.7 $P < .001$). Spironolactone was also superior to doxazosin and bisoprolol. The drug was generally well tolerated; the rates of discontinuations due to renal impairment, hyperkalemia, and gynecomastia were not increased with spironolactone relative to other treatments and placebo. Collectively, these data suggest that spironolactone at 25 to 50 mg/day is an effective treatment option for individuals with resistant hypertension (uncontrolled on three or more medicines including a diuretic).

Eplerenone is less potent than spironolactone and has a shorter half-life (3–4 hours), leading to reduced antihypertensive efficacy and a requirement for twice-daily dosing.⁵⁸² A study of 52 patients has demonstrated the efficacy of the selective aldosterone blocker eplerenone at a dose of 50 to 100 mg daily in patients with resistant hypertension. After eplerenone treatment, the clinical BP was reduced by 18/8 mm Hg compared with baseline ($P < .0001$ for SBPs and DBPs).⁵⁸³ Individuals treated with aldosterone antagonists require careful monitoring for hyperkalemia.

A meta-analysis of RCTs has shown a positive effect of these agents on changes in the cardiac structure and left ventricular function in patients with heart failure.⁵⁸⁴ The role of aldosterone antagonists in treating hypertensive obese patients with sleep apnea has been reported, representing approximately 20% of all individuals with sleep apnea. The aldosterone production relates directly to the adipocyte.^{585–588} In the PATHWAY-2 trial, patients with resistant hypertension (defined as uncontrolled BP despite three antihypertensive agents) were randomized to receive add-on treatment with spironolactone, doxazosin, bisoprolol, or placebo. Spironolactone was the most effective, reducing SBP by approximately 4 mm compared with the other active antihypertensive agents.

When using MRAs, patients must be monitored for hyperkalemia, particularly among those with CKD, and those without CKD using ACE inhibitor or ARBs. MRAs are not recommended for use among patients with eGFR lower than 30 mL/min/1.73 m².

TYROSINE HYDROXYLASE INHIBITOR

MECHANISMS OF ACTION

Metyrosine, the only drug in the tyrosine hydroxylase inhibitor class, blocks the rate-limiting step in the biosynthetic pathway of catecholamines. Metyrosine inhibits tyrosine hydroxylase, the enzyme responsible for the conversion of tyrosine to dihydroxyphenylalanine. This inhibition results in decreased levels of endogenous catecholamines. In patients with pheochromocytomas, metyrosine reduces catecholamine biosynthesis by up to 80%, resulting in a decrease in total peripheral vascular resistance. Heart rate and cardiac output

increase because of the vasodilation. It is used primarily for the management of pheochromocytoma.

CLASS MEMBER

The recommended initial dosage of metyrosine is 250 mg four times a day orally. The dose may be increased by 250 to 500 mg every day until a maximum of 4 g/day is given. Following oral absorption, metyrosine is eliminated primarily unchanged in the urine. The half-life is 7.2 hours. Dose reduction is appropriate in patients with CKD.

RENAL EFFECTS

Little information is available on the renal effects of metyrosine. On the basis of its mechanism of action, which would counteract the renal effects of excessive circulating catecholamines, renal plasma flow and glomerular filtration would be expected to increase and renal vascular resistance would be expected to decrease.

EFFICACY AND SAFETY

Metyrosine is used in the preoperative or intraoperative management of pheochromocytoma. Hypertension and reflex tachycardia may result from vasodilation. These effects can be minimized by volume expansion. AEs include sedation, changes in sleep patterns, and extrapyramidal signs. Metyrosine crystals have been noted in the urine in patients receiving high dosages. Patients taking metyrosine should maintain a generous fluid intake; some patients have occasionally experienced diarrhea.

SELECTION OF ANTIHYPERTENSIVE DRUG THERAPY

DETERMINATION OF BLOOD PRESSURE GOAL

Numerous factors confound the management of hypertension, which is often a lifelong, progressive, asymptomatic disease process.

The purpose of treating BP elevation is to primarily decrease the risk of cardiovascular events and improve survival and quality of life.

Clinicians must ask themselves three major questions:

- How low should the BP go?
- What drugs should be used?
- What are the best strategies for facilitating the attainment of the target BP?

The recent ACC/AHA guidelines defined normal BP as less than 120/80 mm Hg, stage 1 hypertension as SBP of 130 to 139 mm Hg or DBP of 80 to 89 whereas stage 2 hypertension was defined by SBP of 140 mm Hg (or higher) or DBP of 90 mm Hg (or higher) (Table 49.27).⁵⁵ These BP thresholds significantly revised the previous BP cut-offs recommended by the Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood pressure (JNC7).⁸³ Apart from metaanalyses of observational studies that suggested an excess risk of stroke, MI, heart failure, and cardiovascular death with SBP greater than 120 mm Hg, the landmark SPRINT trial demonstrated that compared with the standard SBP goal of less

Table 49.27 Categories of Blood Pressure in Adults

BP Category	SBP		DBP
Normal	<120 mm Hg	And	<80 mm Hg
Elevated	120–129 mm Hg	And	<80 mm Hg
Hypertension			
Stage 1	130–139 mm Hg	Or	80–89 mm Hg
Stage 2	>140 mm Hg	Or	>90 mm Hg

Individuals with SBP and DBP in 2 categories should be designated to the higher BP category.

BP, Blood pressure (based on an average of ≥ 2 readings obtained on ≥ 2 occasions); DBP, diastolic blood pressure; SBP, systolic blood pressure.

From Whelton PK, Carey RM, Aronow WS, et al. ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on clinical practice guidelines. *Circulation*. 2018;138:e426–e483.

than 140 mm Hg, the intensive SBP goal of less than 120 mm Hg resulted in 25% lower risk of a cardiovascular composite (nonfatal MI, acute coronary syndrome, nonfatal stroke, or cardiovascular death or hospitalized heart failure) and a 27% lower risk of all-cause mortality.^{18,20,589}

However, the American College of Physicians/American Academy of Family Physicians (ACP/AAFP) guidelines still recommended an SBP goal of less than 150 mm Hg in persons aged 60 years or older and less than 140 mm Hg in persons with cardiovascular risk.⁵⁹⁰ The concerns for the widespread adoption into routine clinical practice of SPRINT findings include questions on generalizability to the elderly, in persons with diabetes or advanced CKD, potential AEs including on the kidneys, and the effects of lowering SBP on DBP. These issues are discussed sequentially later.

The SPRINT trial included a subgroup of 2636 adults aged 75 years or older (mean baseline age 79.9 years). In a prespecified analysis of this subgroup, treatment to the intensive SBP goal was associated with a 34% (95% CI, 15%–49%) lower risk of the primary composite outcome relative to a standard SBP target.⁵⁹¹ By contrast, the incidence of serious adverse events (hazard ratio [HR], 0.99; 95% CI, 0.89–1.11) and injurious falls (HR, 0.91; 95% CI, 0.65–1.29) in the intensive and standard SBP groups were similar in the older SPRINT subgroup. Furthermore, in the older patient subgroup, the benefits of intensive SBP lowering did not differ substantially by baseline frailty status, and indeed, participants who were the frailest at baseline had the most pronounced reduction in risk of the primary composite cardiovascular outcome with the intensive SBP target. Similarly, in the Hypertension in the Very Elderly Trial (HYVET), frailty also did not modify the benefits of BP lowering.⁵⁹²

The current ACC/AHA guideline recommends a BP target less than 130/80 mm Hg for persons with diabetes. The issue of BP targets in adults with diabetes has been controversial because the primary results of the ACCORD BP trial did not show evidence of beneficial effects of intensive SBP lowering

(SBP goal <120 mm Hg) compared with standard SBP goal (<140 mm Hg) in 4733 adults with type 2 diabetes mellitus. Differences in BP measurement techniques, differences in the achieved SBP separation,^{593,594} and differences in selection criteria have been proposed as potential explanations for the discrepant findings between SPRINT and ACCORD BP. Perhaps, the most widely cited reason is a lack of statistical power in ACCORD BP.^{594,595} However, this reason is unlikely because the number of events and event rates were actually higher in the ACCORD BP compared with SPRINT.

Of note, in addition to randomization to the intensive or standard SBP goals, ACCORD BP participants were randomized in a 2 × 2 factorial design to either standard glucose lowering (target hemoglobin A1c 7.0%–7.9%) or intensive glucose lowering (glycosylated hemoglobin A1c <6.0%).⁵⁹⁶ The intensive glucose lowering arm was discontinued after 3.5 years because of an increased risk of all-cause death (HR, 1.35; 95% CI, 1.04–1.76)⁵⁹⁶ but the two BP arms were completed over 5 years and were not terminated prematurely. An alternative explanation for the discrepant findings of ACCORD BP and SPRINT is that an interaction between the intensive glycemia intervention and the intensive SBP-lowering intervention in ACCORD BP may have masked the potential beneficial effects of the SBP intervention. This hypothesis is supported by a recent reanalysis of ACCORD BP and SPRINT data.⁵⁹⁷ Intensive SBP lowering decreased the hazard of the composite CVD endpoint similarly in SPRINT (HR, 0.75; 95% CI, 0.64–0.89) and the ACCORD BP standard glycemia arm (HR, 0.77; 95% CI, 0.63–0.95, interaction $P = .87$). However, the effect of intensive SBP lowering on the composite CVD endpoint in the ACCORD BP intensive glycemia arm (HR, 1.04; 95% CI, 0.83–1.29) was significantly different from SPRINT (interaction $P = .02$).

Intensive SBP lowering in SPRINT resulted in a 3.5-fold higher hazard of incident CKD among participants without CKD at baseline⁵⁹⁷ and a 65% higher risk of AKI.^{597,598} Although the cardiovascular and survival benefits of intensive SBP lowering appear to outweigh the risk of these kidney events during the trial, the long-term effects on kidney outcomes and kidney function remain to be determined.

Despite the success of SPRINT, lowering of DBP as a consequence of the SPRINT intensive therapy intervention has been cited as a cause for concern in adopting SPRINT findings in routine clinical practice.^{599,600} There are a number of reports of J-shaped curves indicating associations of both low and high DBPs with worse CVD outcomes. The concept was first reported by Stewart in 1979⁶⁰¹ and subsequently championed by Cruickshank in the 1980s.⁶⁰² An appraisal of studies conducted in the 1980s concluded that low on-treatment DBP levels (i.e., <85 mm Hg) were associated with increased risk of cardiac events.⁶⁰³ These findings were supported by subsequent cohort (observational) studies.^{604,605} Secondary analyses of achieved BPs in RCTs of BP-lowering treatments^{606–608} or non-BP-lowering interventions⁶⁰⁹ also suggested that a low achieved DBP was associated with worse CVD outcomes.

Biological plausibility for the J-curve phenomenon has been proposed.^{602,603} As most of the ventricular myocardial perfusion occurs during diastole, particularly in persons with LVH (increased oxygen demand) and coronary artery disease (decreased oxygen supply), lower DBP could potentially lead

to myocardial hypoperfusion and associated damage. Indeed, in the Atherosclerosis Risk In Communities (ARIC) cohort, lower DBP was associated with increased serum concentrations of cardiac troponin T, a marker of myocardial injury.⁶⁰⁰

Based on the aforementioned observational studies of cohort or postrandomization achieved BP data sets, strong causal inferences on lowering DBP have been drawn.^{600,603–607,609} For instance, depending upon the observed threshold in the given study, various lower bounds of DBP have been proposed below which DBP lowering is considered deleterious.

Although the existence of the J-curve phenomenon has been disputed by some,^{610–612} the central question is whether the J-curve phenomenon observed in some of the aforementioned studies reflects a causal effect of lower DBP on CVD outcomes. It is conceivable that underlying processes (such as increased arterial stiffness) that lead to a decline in DBP rather than the level of DBP per se might be the reason for the observed associations of worse outcomes with lower DBP. Statistical modeling and multivariable regression are typically employed in observational data analyses, but residual confounding could still explain observed results. The most direct and valid method of testing the J-curve hypothesis is to actively intervene to lower DBP, particularly in those with DBP lower than 70 mm Hg; if lowering DBP is deleterious below a certain DBP level, one would expect that the effects of lowering SBP on CVD outcomes and death would be modified by baseline level of DBP.

In SPRINT, irrespective of the randomized treatment, baseline DBP had a U-shaped association with the hazard of the primary CVD outcome.⁶¹³ However, in randomized comparisons between the intensive and standard treatment groups, the effect of the intensive SBP intervention on the primary outcome was not modified by baseline DBP. Results were similar for all-cause death and kidney events; in other words, the beneficial effect of intensive SBP lowering on survival and the deleterious effect on incident CKD were present and similar in magnitude irrespective of the baseline DBP. Therefore there was no evidence that lowering SBP in those with low DBP worsened CVD outcomes. A recent causal mediation analysis of achieved low DBP in SPRINT also supports this observation.⁶¹⁴

Thus the ACC/AHA guidelines of SBP goal of less than 130 mm Hg is supported by evidence.

The promptness of achieving BP control has long been thought to be an important strategy for the clinical care for patients with hypertension. Clinical trials and observational studies have indicated that initiating treatment with a two-drug combination of medications results in a more rapid achievement of target BP goals compared with monotherapy. A retrospective analysis of the Valsartan Antihypertensive Long-term Use Evaluation (VALUE) trial has indicated that earlier BP control results in a significant reduction in the 5-year risk of cardiovascular events, regardless of the type of medication used.⁶¹⁵ A matched cohort study has demonstrated that initial combination therapy reduces the risk of cardiovascular events in hypertensive patients.³⁶⁵ This reduction was thought to be related to an earlier achievement of target BP as the main contributor of risk reduction.

Decisions about which drug(s) should be used for a given patient require careful consideration and individualization. As addressed later, the appropriate pharmacotherapy may depend on age, sex, race, obesity, and associated cardiovascular

or renal disease. Clinical trials in patients with vascular disease, heart disease, or kidney disease have demonstrated the important therapeutic advantage of drugs that block the RAAS—ACE inhibitors or ARBs^{76,248,249,616–621} to prevent progression of cardiovascular or kidney disease as part of a multidrug regimen to lower BP. In the aggregate, RAAS inhibitors provide an approximately 20% RR reduction benefit compared with other therapies. These drugs should be part of every antihypertensive regimen in patients with heart disease or kidney disease unless there are specific contraindications.⁶¹⁸ RAAS-blocking drugs are also some of the best-tolerated therapies available for treating hypertension. Although these drugs provide important risk reduction opportunities, they are not a substitute for achieving BP control.

SINGLE-PILL COMBINATION THERAPY

Most available antihypertensive drugs, when appropriately dosed, reduce SBP by approximately 8 to 10 mm Hg (Fig. 49.3). For many patients, three or four drugs may be required. Ideally, medications that are long-acting, can be taken once daily, are well tolerated, and preferably work well with other medications to facilitate BP control should be used. In addition, there has been a marked increase in the number of single-pill, fixed-dose combination antihypertensive drugs that are available in the marketplace, developed in large

part to facilitate adherence by reducing the complexity of the antihypertensive regimen (Table 49.28).⁶²²

The recommended first-line agents include thiazide diuretics, CCBs, and ACE inhibitors or ARBs. Chlorthalidone is a more potent antihypertensive than HCTZ. The advantage of a low-dose thiazide or thiazide-like drug is that it nearly doubles the antihypertensive effects of the parent drug without adding toxicity to the regimen (Fig. 49.4).⁶²³ Drugs that block the RAAS system, such as ACE inhibitors or ARBs in combination with chlorthalidone or HCTZ, are a very effective combination to control BP. Single-pill combinations of an ACE inhibitor or ARB with CCB are also available. Clinical studies have demonstrated that these drugs are also additive in their ability to lower DBPs and SBPs.^{624–628} Moreover, use of an ACE inhibitor or ARB in combination with a CCB abrogates the development of pedal edema, a common AE of CCBs.⁶²⁹ Two triple single-pill combinations (ARB, CCB, and HCTZ) have been approved for the treatment of hypertension.⁶³⁰ An ARB has also been formulated with the thiazide-like diuretic chlorthalidone. Some single-pill combinations are now approved by the U.S. FDA for initial treatment in patients with moderate-to-severe hypertension.

Considerations for physicians about how to consolidate and simplify pharmacotherapy to control BP are of great interest, given the complexity of the current multidrug regimens that many patients require. Administering four drugs

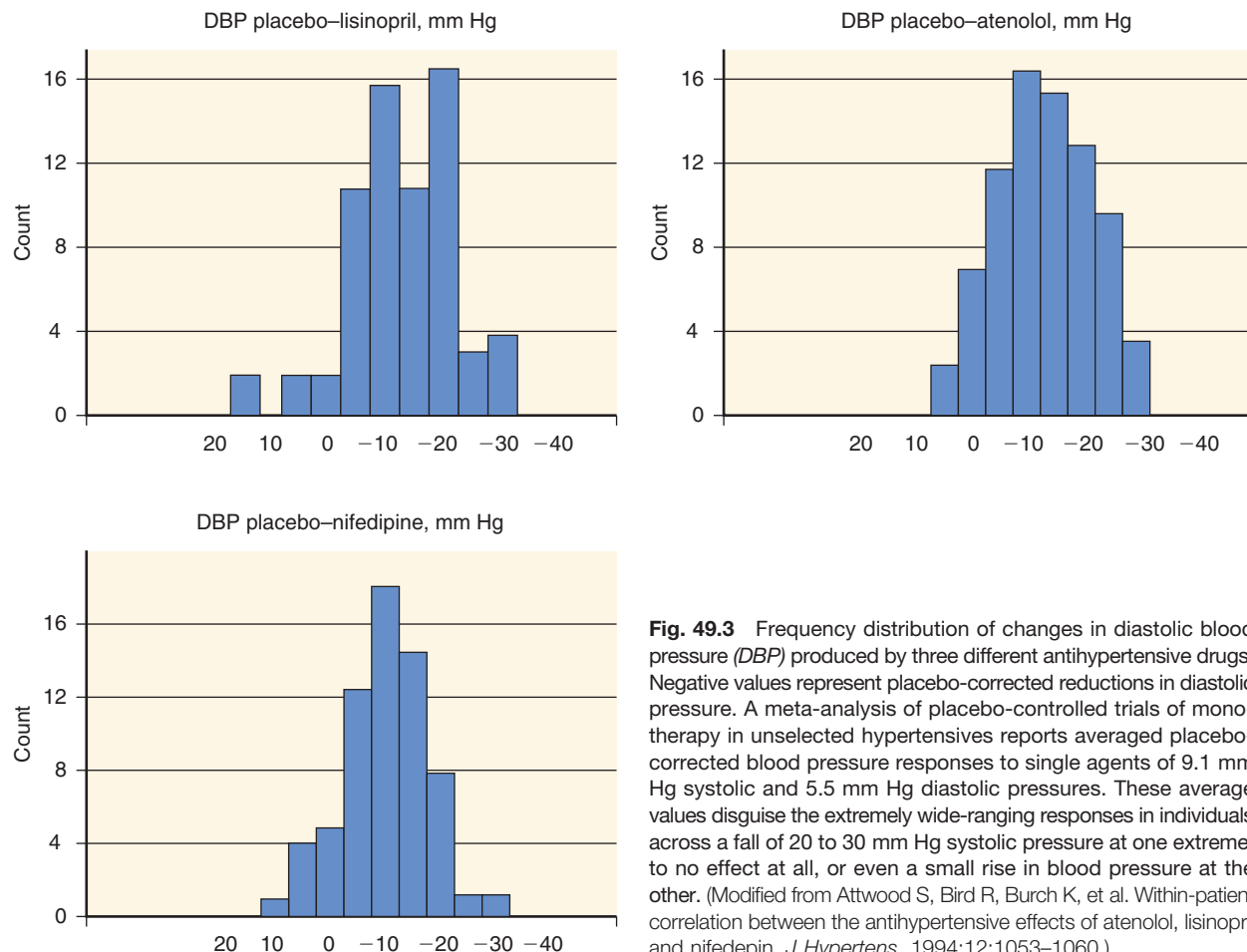


Fig. 49.3 Frequency distribution of changes in diastolic blood pressure (DBP) produced by three different antihypertensive drugs. Negative values represent placebo-corrected reductions in diastolic pressure. A meta-analysis of placebo-controlled trials of monotherapy in unselected hypertensives reports averaged placebo-corrected blood pressure responses to single agents of 9.1 mm Hg systolic and 5.5 mm Hg diastolic pressures. These average values disguise the extremely wide-ranging responses in individuals across a fall of 20 to 30 mm Hg systolic pressure at one extreme, to no effect at all, or even a small rise in blood pressure at the other. (Modified from Attwood S, Bird R, Burch K, et al. Within-patient correlation between the antihypertensive effects of atenolol, lisinopril and nifedipine. *J Hypertens*. 1994;12:1053–1060.)

Table 49.28 Single-Pill Combinations Available for the Treatment of Hypertension⁶²²

Type of Combination	First Drug (Doses)	Second Drug (Doses)	Third Drug (Doses)
Dual Combinations			
Thiazide/K sparing diuretic	HCTZ (25/50)	Triamterene (37.5/75)	
	HCTZ (25/50)	Spironolactone (25/50)	
	HCTZ (25/50)	Amiloride (2.5/5)	
	Furosemide (20)	Spironolactone (50)	
ACE inhibitor/diuretic	Captopril (25/50)	HCTZ (15/25)	
	Enalapril (10/20)	HCTZ (12.5/25)	
	Lisinopril (10/20)	HCTZ (12.5/25)	
	Cilazapril (5)	HCTZ (12.5)	
	Fosinopril (10/20)	HCTZ (12.5)	
	Quinapril (10/20)	HCTZ (12.5)	
	Benazepril (5/10/20)	HCTZ (6.25/12.5/25)	
	Moexipril (7.5/15)	HCTZ (12.5/25)	
	Ramipril (2/5)	HCTZ (12.5/25)	
	Ramipril (5)	Piretanide (6)	
	Zofenopril (30)	HCTZ (12.5)	
ACE inhibitor/CCB	Perindopril (2.5/5/10)	Indapamide (0.625/1.25, 2.5)	
	Perindopril (5/10)	Amlodipine (5/10)	
	Benazepril (10/20/40)	Amlodipine (2.5/5/10)	
	Enalapril (5)	Diltiazem (180)	
	Enalapril (10)	Nitrendipine (20)	
	Enalapril (10/20)	Lercanidipine (10)	
	Ramipril (2.5/5)	Felodipine ER (2.5/5)	
	Trandolapril (1/2/4)	Verapamil (180, 240)	
ARB/diuretic	Delapril (10)	Manipipine (30)	
	Losartan (50/100)	HCTZ (12.5/25)	
	Valsartan (80/160)	HCTZ (12.5/25)	
	Irbesartan (150/300)	HCTZ (12.5/25)	
	Candesartan (8/16/32)	HCTZ (12.5/25)	
	Telmisartan (40/80)	HCTZ (12.5)	
	Eprosartan (600)	HCTZ (12.5/25)	
	Olmesartan (20/40)	HCTZ (12.5/25)	
ARB/CCB	Azilsartan (40)	Chlorthalidone (12.5/25)	
	Valsartan (80/160)	Amlodipine (5/10)	
	Telmisartan (40/80)	Amlodipine (5/10)	
	Olmesartan (20/40)	Amlodipine (5/10)	
	Candesartan (8)	Amlodipine (5) ^a	
	Irbesartan (100/150)	Amlodipine (5/10) ^a	
Renin inhibitor/diuretic	Aliskiren (150/300)	HCTZ (12.5/25)	
Renin inhibitor/CCB	Aliskiren (150/300)	Amlodipine (5/10)	
β-Blocker/diuretic	Atenolol (50/100)	Chlorthalidone (25)	
	Atenolol (50)	HCTZ (25)	
	Metoprolol (50/100)	HCTZ (25/50)	
	Bisoprolol (2.5/5/10)	HCTZ (6.25)	
	Bisoprolol (1)	Chlorthalidone (25)	
	Nadolol (40/80)	Bendroflumethiazide (5)	
	Oxprenolol (120)	Chlorthalidone (20)	
	Pindolol (10)	Clopamide (5)	
	Propranolol (40/80)	HCTZ (25)	
	Propranolol LA (80/160)	HCTZ (50)	
	Timolol (10)	HCTZ (25)	
β-Blocker/CCB	Atenolol (25/50)	Nifedipine (10/20)	
	Metoprolol (50/100)	Felodipine (5/10)	
α-Blocker/diuretic	Methyldopa (250)	HCTZ (15)	
	Clonidine (0.1/0.2/0.3)	Chlorthalidone (15)	
	Reserpine (0.1)	HCTZ (10)	
Triple Combinations			
ARB/CCB/diuretic	Valsartan (160/320)	Amlodipine (5/10)	HCTZ (12.5/25)
	Olmesartan (20/40)	Amlodipine (5/10)	HCTZ (12.5/25)
	Telmisartan (40/80)	Amlodipine (5/10)	HCTZ (12.5/25)
ACE inhibitor/CCB/diuretic	Perindopril (5/10)	Amlodipine (5/10)	Indapamide (1.25/2.5)
Renin inhibitor/CCB/diuretic	Aliskiren (150/300)	Amlodipine (5/10)	HCTZ (12.5/25)

^aSome dosages may be available only in certain countries. The list is not exhaustive.

ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CCB, calcium channel blocker; ER, extended release; HCTZ, hydrochlorothiazide; K, potassium; LA, long acting.

From Burnier M. Antihypertensive combination treatment: state of the art. *Curr Hyperten Rep*. 2015;17:51.

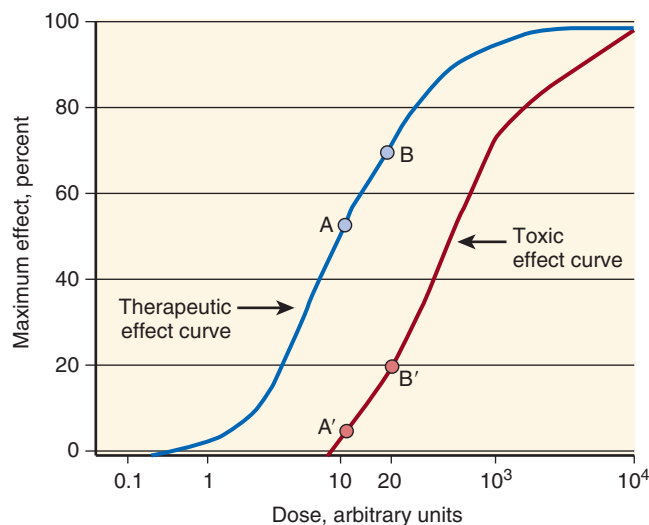


Fig. 49.4 Dose relationship between therapeutic effect and toxicity with antihypertensive drugs. For the initial therapy of hypertension, a strategy to reduce side effects and enhance the tolerability of multiple antihypertensive drugs in combination has been identified using low doses of more than one antihypertensive agent with different combined modes of action to minimize side effects. With a low dose of drug A, a partial therapeutic effect is obtained, and adverse effects (A') are minimal. If the dose is raised to B, a greater therapeutic effect will be accompanied by more adverse effects (B'). If a low dose of a second drug is added with its own minimal side effects, an extra effect will be obtained without more adverse effects, which will remain at A'. (Modified from Epstein M, Bakris G. Newer approaches to antihypertensive therapy. Use of fixed-dose combination therapy. *Arch Intern Med.* 1996;156:1969–1978.)

in two pills or three drugs in a single pill is possible with available fixed-dose combinations. This goal is important because many patients require eight to ten or more medications to control their various medical problems, including diabetes, dyslipidemia, and angina. Hypertension is a disease that is largely asymptomatic and, consequently, to optimize adherence, the therapeutic approach should be simple, effective, and well tolerated.

CHOICE OF APPROPRIATE AGENTS

This section considers the initial therapy for various types of patients depending on factors such as age, sex, race/ethnicity, obesity, and coexistent cardiovascular or renal disease. These suggestions are primarily generalizations based on clinical experience and should not be viewed as rigorous guidelines. Because each patient is different, variation in the approach is frequently necessary (Fig. 49.5).

TREATMENT OF OLDER PATIENTS

Older patients frequently have vascular stiffening, leading to an increase in SBP, decrease in diastolic pressure, and wider pulse pressure.⁶³¹ Management of isolated systolic hypertension in older patients frequently requires multiple drugs. A careful assessment of the metabolic and excretory routes of the drugs, as well as possible drug–drug interactions, is recommended because older patients frequently have impaired metabolic function. As shown by the SPRINT

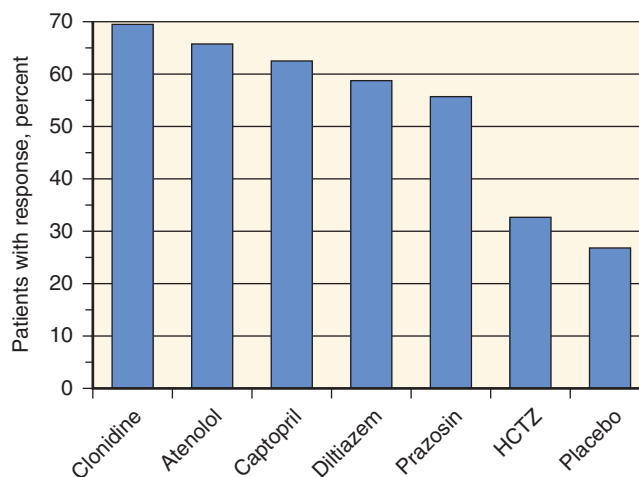


Fig. 49.5 Antihypertensive response to different drugs for white individuals younger than 60 years of age. Hydrochlorothiazide (HCTZ) appears to be the least effective as a single agent. (From Materson BJ, Reda DJ, Cushman WC, et al. Single-drug therapy for hypertension in men. A comparison of six antihypertensive agents with placebo. The Department of Veterans Affairs Cooperative Study Group on Antihypertensive Agents. *N Engl J Med.* 1993;328:914–921.)

experience, BP control can be safely achieved in elderly individuals, even in those who are frail.⁵⁹¹

Older patients have a higher likelihood of orthostasis than younger patients, due in part to a reduction in cardiovascular baroreceptor reflex function. As many as 18% of untreated older patients with hypertension have a decrease in SBP of more than 20 mm Hg after standing for 1 to 3 minutes.⁶³² Older patients may also have pseudohypertension, which may interfere with a true determination of BP.⁶³³ Consequently, three-position BP measurements should always be used during the initiation and titration of medications. Older patients are also more likely to have hypertrophic cardiomyopathy with impaired diastolic function than are younger patients, which may impair cardiac output.⁶³⁴

Older patients may benefit from the use of a vasodilators such as CCBs, α -blockers, or ACE inhibitors and ARBs (Table 49.29). They are much better tolerated in the lower half of their dosing range and are quite effective even in the presence of a high-salt diet, perhaps owing to their natriuretic effects⁶³⁵ or intrinsic vasodilatory effects.^{636,637} α -Blockers may be useful in older men with benign prostatic hyperplasia because they facilitate prostatic urethral relaxation and improve urinary stream. ACE inhibitors and ARBs are well tolerated, and their efficacy is enhanced when they are combined with thiazide and thiazide-type diuretics.

Thiazide and thiazide-type diuretics may also have a vasodilating effect.⁵⁵ The ACC/AHA guidelines recommend chlorthalidone as the preferred thiazide as it is more potent than HCTZ. An ongoing pragmatic point-of-care clinical trial will randomize veterans aged 65 years and older to receive chlorthalidone or HCTZ to assess whether there is a difference in cardiovascular events.⁶³⁸ The initial dose of chlorthalidone is 12.5 mg/day and if needed, can be increased to 25 mg/day. At doses more than 25 mg/day, the risk of side effects is higher with a lower incremental BP benefit. Hence, instead of increasing chlorthalidone to more than 25 mg another

Table 49.29 Considerations for Initial Therapy in Older Patients

Clinical Observation	Pharmacologic Considerations
Decreased vascular compliance and peripheral vascular resistance	Use vasodilator (e.g., HCTZ, ACE inhibitor, ARB, CCB, α -blocker).
Isolated systolic hypertension and wide pulse pressure	Use vasodilator (e.g., HCTZ, CCB).
Reduction of cardiovascular baroreflex function with blood pressure lability	Avoid sympatholytics and volume depletion. Use β -blockers cautiously.
Orthostatic hypertension during recumbency	Consider using short-acting medications (<8-h duration) at bedtime.
Reduced metabolic capability	Adjust all medications for renal and hepatic function; start at half-dose.
Prostatic hypertrophy	Use α -blocker.
>20 mm Hg from systolic goal	Use fixed-dose combination therapy (ACE inhibitor/CCB, ARB/CCB, ACE inhibitor/HCTZ, ARB/HCTZ, β -blocker/HCTZ).

ACE, Angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; CCB, calcium channel blocker; HCTZ, thiazide diuretic.

agent should be added if BP control is not achieved with 25 mg/day of chlorthalidone.

TREATMENT BASED ON SEX

Sex differences may help to guide the selection of antihypertensive therapy (Table 49.30).⁶³⁹ Men and women benefit equally from a more intensive control of BP, which results in a reduction in the risk of cardiovascular events.⁶⁴⁰ In general, men have a lower resting heart rate, longer left ventricular ejection fraction time, and higher pulse pressure when stressed than women.⁶³⁹ Women tend to have lower peripheral vascular resistance and a relatively higher blood volume for overall size than men⁶³⁹; they also have a lower likelihood of coronary disease before menopause. However, when menopause occurs, or in the presence of diabetes, women have the same risk of coronary disease as men.⁶³⁹ Vasodilation is always a good choice for treatment because elevated peripheral vascular resistance is almost always involved in BP elevation, regardless of sex. Thiazide diuretics, ACE inhibitors, ARBs, and CCBs are all effective treatments. Many patients require two or more of these drugs, and fixed-dose combinations can be used.

Women should avoid the use of ACE inhibitors and ARBs in pregnancy because of their possible teratogenic effects. CCBs may delay labor. Optimal therapy in a pregnant woman remains α -methyldopa, hydralazine, or β -blockers because they have a safety record with a minimal risk of teratogenic effects on the fetus (Table 49.6).

Women experience higher rates of osteoporosis and osteopenia than do men. In women (and men) with

Table 49.30 Considerations for Initial Therapy Based on Gender

Clinical Observation	Pharmacologic Considerations
Men have lower resting heart rate, longer left ventricular ejection time, and higher stressed pulse pressure compared with women.	Use vasodilator (e.g., HCTZ, ACE inhibitor, ARB, CCB).
Women have lower total peripheral resistance and greater blood volume compared with men.	Use drugs that provide vasodilation, heart rate reduction; may need diuresis (e.g., HCTZ, ACE inhibitor, ARB, β -blocker, CCB).
Postmenopausal women more frequently have coronary artery disease with atypical chest pain.	Use drugs that counter angina and reduce heart rate (β -blocker, CCB).
Osteoporosis	Antagonize calciuria (HCTZ).
Pregnancy	Avoid teratogenic drugs (ACE inhibitor, ARB). Avoid drugs that may cause ureteroplacental insufficiency (loop diuretics). Optimal choices are α -methyldopa, hydralazine, and β -blockers.
Women report more pedal edema with CCBs and cough with ACE inhibitors than men.	Adjust dosage or discontinue drug.
More than 20 mm Hg from systolic goal	Use fixed-dose combination therapy (ACE inhibitor/CCB, ARB/CCB, ACE inhibitor/HCTZ, β -blocker/HCTZ, ARB/HCTZ).

ACE, Angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; CCB, calcium channel blocker; HCTZ, thiazide diuretic.

osteoporosis, thiazide and thiazide-like diuretics may antagonize calciuria and facilitate bone mineralization.⁶⁴¹

Women experience more cough with ACE inhibitors and more pedal edema with CCBs than men.⁶⁴² These differences in AEs may require adjustment in the dose or switching of medications. Interestingly, despite sex differences in the underlying pathophysiologic mechanisms of high BP, there does not appear to be a substantial difference in the response rate to similar doses of commonly used antihypertensive drugs.

RACE AND ETHNICITY

African Americans frequently present with hypertension at an earlier age, have more substantial elevations in BP, and earlier development of target-organ damage, than matched persons of white race.^{643–645} Racial differences in the response to antihypertensive medications have been demonstrated in numerous clinical trials.^{636,643} The mechanisms for these differences are not yet elucidated but appear to be independent of dietary salt or potassium intake. Some investigators have suggested possible genetic differences in renal sodium handling, but

clinical trials have not been conclusive. Irrespective of the mechanism, African Americans frequently display BP salt sensitivity.^{646,647} A careful assessment of the dose-response for different medication classes, with adjustment for differences in dietary sodium consumption and body mass index among races and ethnic groups, has not been performed.

In general, thiazide diuretics and CCBs have more robust antihypertensive effects at lower dosages in African Americans than other commonly used therapeutic classes.^{636,637,648} Drugs that block the RAAS are effective in African Americans, but higher dosages are frequently required to achieve the same degree of BP reduction as observed in non-African Americans.²⁸⁴ As in most population groups, elevated peripheral vascular resistance contributes to BP elevation. Some investigators have suggested that African Americans have a modest volume component contributing to BP elevation that may also contribute to antihypertensive drug resistance.⁶⁴⁶ It is not uncommon for multiple drugs to be required to reach the target BP. Consequently, fixed-dose combinations may be most useful in this population group as part of a strategy to simplify the approach.

Hispanics and Asians do not appear to have different hypertensive responses to commonly used drugs compared with whites.⁶⁴³ However, Caribbean Hispanics who have a higher proportion of African ancestry may respond to commonly prescribed antihypertensive agents in a way that is characteristic of non-Hispanic blacks.⁶⁴⁷

OBESSE PATIENTS

Obese patients with hypertension frequently have other medical problems that complicate the management of hypertension (Table 49.31).⁸⁶ These patients tend to have a hyperdynamic circulation, increased peripheral vascular resistance, expanded plasma volume, and a greater sensitivity to the influence of dietary salt in increasing BP.

Table 49.31 Considerations for Initial Therapy in Obese Patients With Hypertension

Clinical Observation	Pharmacologic Considerations
Hyperdynamic circulation	Reduce heart rate and sympathoadrenal outflow (β -blocker).
Increased peripheral vascular resistance	Use vasodilator (e.g., HCTZ, ACE inhibitor, ARB, CCB).
Salt sensitivity	Use natriuretic (HCTZ, ACE inhibitor, ARB, CCB). Have patient reduce salt intake.
Expanded plasma volume	Use diuretic (HCTZ). Have patient reduce salt intake.
Hypoventilation	Order sleep study to evaluate need for positive pressure ventilation at night.
>20 mm Hg from systolic goal	Use fixed-dose combination therapy (ACE inhibitor/CCB, ARB/CCB, ACE inhibitor/HCTZ, β -blocker/HCTZ, ARB/HCTZ).

ACE, Angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; CCB, calcium channel blocker; HCTZ, thiazide diuretic.

β -Blockers may be helpful in diminishing sympathoadrenal drive. Vasodilators such as ACE inhibitors, ARBs, and CCBs are useful for reducing peripheral vascular resistance. Combinations of these drugs may also be helpful. Because of the tendency toward expanded plasma volume, thiazide or thiazide-like diuretics can be helpful because they provide an opportunity to produce vasodilation and mild volume reduction. Frequently, obese patients require multiple drugs to achieve BP goals. β -Blockers may increase the likelihood of weight gain and may compromise glucose tolerance.^{636,649} Aldosterone receptor blockers may be preferred in obese, hypertensive individuals requiring treatment with multiple medications because of adipocyte production of aldosterone.^{585–588}

TREATMENT OF PATIENTS WITH CARDIOVASCULAR OR KIDNEY DISEASE

Cardiovascular Disease

Patients with hypertension and cardiac diseases such as stable ischemic heart disease, HFrEF, and heart failure with preserved ejection fraction (HFpEF) frequently require multiple agents for the joint control of hypertension and heart disease. The choice of specific antihypertensive medications will generally be guided by first-line therapies. For example, for patients with a history of stable ischemic heart disease, the first choice of agents should include a β -blocker and an ACE inhibitor or ARB, given evidence from randomized trials indicating a benefit in terms of cardiovascular morbidity and mortality. Other agents such as CCBs and diuretics can be added as needed to achieve the BP goal. Patients with HFrEF should also be treated with β -blockers, ACE inhibitors or ARBs, and MRA, whereas nondihydropyridine calcium channels should generally be avoided, based on data from randomized clinical trials. Less is known about optimal treatment choices in HFpEF, but the current guidelines⁵⁵ recommend management of volume overload with diuretics, followed by the addition of ACE inhibitors or ARBs and β -blockers as needed to achieve a target SBP of less than 130 mm Hg.

Kidney Disease

In patients with kidney disease (Table 49.32), BP control is more difficult to achieve because these patients not only have increased vascular resistance but also frequently have elevated blood volume.⁶⁵⁰ The understanding of renal autoregulation provides some insight into the appropriate levels of BP control and the relative importance of different types of antihypertensive drugs in preserving kidney function.

Glomerular circulation operates optimally at one-half to two-thirds of the systemic BP.⁶⁵¹ Preglomerular vasoconstriction is necessary to decrease the systemic pressure to glomerular capillary pressure levels that are optimal for filtration yet low enough to avoid mechanical injury to the filtering apparatus.^{651,652} The efferent glomerular arteriole also serves an important purpose—it vasoconstricts during situations of diminished effective arterial blood volume to maintain adequate pressure for glomerular filtration. With the development of vascular disease, the afferent glomerular arteriole does not vasoconstrict properly, which permits the transmission of systemic BP into the glomerulus. A clinical clue that could indicate the failure of autoregulation is the presence of microalbuminuria or protein in the urine. Under these

Table 49.32 Considerations for Initial Therapy in Patients With Renal Disease

Clinical Observation	Pharmacologic Considerations
Increased blood volume (common in glomerular diseases)	Reduce blood volume
Decreased blood volume (common in tubular diseases)	Salt supplementation may be needed.
Increased peripheral vascular resistance	Use vasodilator (ACE inhibitor, CCB, ARB).
Proteinuria	Reduce proteinuria (ACE inhibitor, ARB, NDCCB; recommended target systolic blood pressure ≤ 130 mm Hg).
Diabetes with proteinuria	Control blood pressure and glycemia (ACE inhibitor if type 1 diabetes, ARB if type 2; recommended target systolic blood pressure <130 mm Hg).
>20 mm Hg from systolic goal	Use fixed-dose combination therapy (ACE inhibitor/CCB, ARB/CCB, ACE inhibitor/HCTZ, β -blocker/HCTZ, ACE inhibitor/CCB). Use of HCTZ depends on renal function.

All medications adjusted according to renal function.
ACE, Angiotensin-converting enzyme; *ARB*, angiotensin II receptor blocker; *CCB*, calcium channel blocker; *HCTZ*, thiazide diuretic; *NDCCB*, nondihydropyridine calcium channel blocker.

circumstances, systemic BP should be reduced more substantially to minimize the risk of mechanical injury to the glomerulus.

Drugs that block the RAAS, such as ACE inhibitors^{616,617} and ARBs,^{620,624} provide a more consistent opportunity to reduce the progression of kidney disease as part of an intensive BP-lowering strategy than other commonly used antihypertensive drugs. The benefit of these drugs resides, in part, on their ability to facilitate efferent glomerular arteriolar dilation by antagonizing the effects of Ang II as they lower BP.⁶⁵³ Renin inhibitors have similar clinical effects on reducing both BP and proteinuria to those of ACE inhibitors and ARBs, but have not been studied alone in renal protection trials.⁶⁵⁴ Overall, there is a more consistent reduction in systemic and glomerular capillary pressures with RAAS-blocking drugs. Additional medications can be added to these drugs to achieve better BP control and help reduce glomerular capillary pressure and proteinuria. Sufficient doses of diuretics to control the blood volume should also be used. When kidney function declines to a range of eGFR of 30 mL/min/1.73 m² or below, volume reduction is more amenable to the provision of loop diuretics as opposed to thiazide or thiazide-type diuretics, although chlorthalidone and metolazone retain antihypertensive efficacy.

Some investigators have questioned the safety of using CCBs in patients with kidney disease, given their preferential effects on dilating the afferent glomerular arteriole.⁵² Some studies have demonstrated that CCBs can increase proteinuria,

despite lowering BP.⁶⁵⁵ However, there is no clinical evidence that these drugs are detrimental and worsen the progression of kidney disease if they are given with ACE inhibitors or ARBs, which dilate the efferent glomerular arteriole. If anything, the lower BP achieved using these drugs in combination may provide better protection against the loss of kidney function, as was observed in the ACE inhibitors and CCBs for hypertension (ACCOMPLISH) study.⁶⁵⁶

Antiproteinuric strategies should be considered in patients with kidney disease because a reduction in proteinuria by specific antihypertensive drugs, such as ACE inhibitors or ARBs, correlates with slowing the progression of kidney disease.^{657,658} However, dual RAAS blockade, despite reducing proteinuria, has not provided incremental benefit on kidney disease progression and is associated with a higher risk for declining kidney function and hyperkalemia.^{95,255} For further discussion of drugs for the treatment of hypertension associated with kidney disease, see Chapter 59.

STRATEGIES FOR SELECTING THE OPTIMAL COMBINATION ANTIHYPERTENSIVE THERAPY

A meta-analysis involving 40,000 treated patients evaluated the usefulness of low-dose combination treatments separately and in combination.⁶⁵⁹ The average BP reduction achieved with low-dose therapy was only 20% in 354 randomized, double-blind, placebo-controlled trials of thiazides, β -blockers, ACE inhibitors, ARBs, and CCBs. Thiazides and CCBs cause side effects infrequently at a half-standard dose (2% and 1.6%, respectively), whereas side effects occur commonly at standard doses (9.9% and 8.3%, respectively). β -Blockers, however, cause an equivalently high number of side effects at both dosage levels (5.5% vs. 7.5%; $P = .04$). The only side effect observed with ACE inhibitors was cough (3.9%), which did not vary with dosage. ARBs were not associated with an excess of side effects at a standard dose. The reduction in BP was additive; however, the prevalence of AEs was not additive. Therefore combinations of two or three drugs at low doses may be preferable to one or two drugs at standard doses. In the absence of cough, ACE inhibitors and ARBs can be used at higher doses because of the lack of AEs associated with these drugs. In a subsequent meta-analysis, the authors quantified the effect of a combination of drugs from the four drug classes on BP reduction. The extra BP reduction achieved by combining two drugs from different classes is approximately five times greater than that achieved when the dosage of one drug is doubled.⁶⁶⁰ This confirms what might be expected based on knowledge of the dose–effect relationship between the therapeutic effects of antihypertensive drugs and their toxicity (Fig. 49.4).

The rationale for fixed-dose combination therapy is as follows⁶⁶¹:

- To combine drugs acting on different systems, thus initiating a pharmacologic attack on two systems with greatly enhanced impact compared with monotherapy.
- To combat counterregulatory responses that interfere with BP reduction.

For example, CCBs activate the SNS and RAAS, which can attenuate BP-lowering effects.^{662,663} This action can be buffered by RAAS blockers. The antihypertensive effect of an RAAS

blocker can be enhanced by the negative sodium balance created by the diuretic and natriuretic properties of diuretics or CCBs.⁶⁶⁴

- To decrease the pill burden and enhance adherence to prescribed therapy.

Many individuals with BPs that are 20 mm Hg (systolic) and 10 mm Hg (diastolic) above the target pressure will require more than one drug. Clinical trials have documented that achieving BP targets necessitates the use of multiple drugs. In the ALLHAT study, for example, only one-third of patients achieved their target BP with monotherapy after 5 years.⁶⁶⁵

- To decrease BP variability.

BP has been shown to be less variable with combination therapy versus monotherapy.⁶⁶⁶ Variability in visit-to-visit SBPs was shown to be a strong predictor of stroke and MI.⁶⁶⁷ CCBs and diuretics were found to be the most useful agents for reducing this variability, whereas β -blockers increased the variability.⁶⁶⁷

- To decrease AEs.

Combining two antihypertensive drugs at lower doses may actually reduce AEs compared with higher doses of both drugs. Studies have shown that the combination of an ACE inhibitor and ARB with a CCB reduces the incidence of CCB-related peripheral edema by as much as 38%.⁶⁶⁸ Combining an RAAS blocker with a thiazide reduces the occurrence of diuretic-induced hypokalemia.

In the TRIple Pill vs. Usual care Management for Patients with mild-to-moderate Hypertension (TRIUMPH) trial,⁶⁶⁹ 700 patients with hypertension from 11 urban hospitals in Sri Lanka were randomly assigned to receive a once-daily fixed dose triple combination pill containing 20 mg of telmisartan, 2.5 mg of amlodipine, and 12.5 mg of chlorthalidone versus usual care. A higher proportion of patients in the triple combination pill arm achieved target BP at 6 months (70% vs. 55%; $P < .001$). There were no significant differences in the proportion of patient withdrawal from BP-lowering therapy because of adverse events (6.6% vs. 6.8%). Limitations have been identified with the single-pill combination approach, however, and are listed in Table 49.33.

PREFERRED COMBINATIONS

Renin-Angiotensin-Aldosterone System Inhibitors and Calcium Channel Blockers

The ACCOMPLISH trial demonstrated the superiority of an ACE inhibitor and CCB in single-pill form versus an ACE inhibitor plus HCTZ. The ACCOMPLISH trial compared BP reduction and cardiovascular outcomes achieved with two combination therapies, benazepril-amlodipine and benazepril-HCTZ in a single tablet. The trial included more than 11,000 participants, and the mean BP was nearly equivalent between the benazepril-amlodipine and benazepril-HCTZ groups (131.6 and 73.3 vs. 132.5 and 74.3, respectively). The outcome in this study was time to reach a primary endpoint—a composite of cardiovascular events or death. The trial was terminated early (after a mean of 36 months) because the benazepril-amlodipine group had

Table 49.33 Advantages and Limits of the Use of Single-Pill Combinations in Hypertension

Advantages

- Reduction of the pill burden
- Simplification of the treatment schedule
- Increased adherence to therapy (better long-term persistence)
- Improved efficacy with reduced incidence of side effects
- Better prevention of cardiovascular events (to be demonstrated prospectively)

Limits

- Reduction of the prescription flexibility
- Difficulty to identify the precise cause of an unexpected side effect
- Difficulty to memorize the exact content of the single-pill combinations
- Risk of a more pronounced rebound hypertension in case of repeated omissions
- Risk of acute hypotension when restarting a triple combination after interruptions
- Increased cost versus free combinations of generics

From Burnier M. Antihypertensive combination treatment: state of the art. *Curr Hyperten Rep.* 2015;17:51.

reached a lower event rate (9.6%) than the benazepril-HCTZ group (11.8%; RR, 19.6%; HR, 0.80; $P < .001$).

Renin-Angiotensin-Aldosterone System Inhibitors and Thiazide-Type Diuretics

By depleting intravascular volume, diuretics activate the RAAS system, which causes vasoconstriction as well as salt and water retention. The RAAS blockade attenuates the counterregulatory response and causes additional BP reduction. This combination has several advantages—the RAAS mitigates diuretic-induced hypokalemia and glucose intolerance. In the HYVET trial, indapamide, a thiazide-type diuretic, combined with the ACE inhibitor perindopril reduced the incidence of stroke (by 30%) and heart failure (by 64%).

ACCEPTABLE COMBINATIONS

β -Blockers and Thiazide-Like Diuretics

There are multiple commercially available β -blocker-diuretic formulations (e.g., bisoprolol, metoprolol, propranolol, timolol with HCTZ, and atenolol with chlorthalidone). β -Blockers and diuretics have both been shown to reduce cardiovascular endpoints in RCTs; however, meta-analyses have suggested that they are less effective than other agents.^{355,670} The addition of diuretics increases the effectiveness of β -blockers in individuals of African ancestry and in other persons with low-renin hypertension. These drug classes have several overlapping AEs, including the development of glucose intolerance, new-onset diabetes, fatigue, and sexual dysfunction. In 2007, the European Society of Hypertension warned against the use of this combination for patients with metabolic syndrome or for those at high risk of diabetes.⁶⁷¹

A meta-analysis has demonstrated a 35% risk of new-onset diabetes with diuretics compared with non- β -blocker antihypertensives and a 31% risk of new-onset diabetes with diuretics compared with nondiuretic antihypertensives.⁶⁷² Given the possibility of adverse metabolic consequences, it seems reasonable to avoid the use of this combination for initial therapy without other compelling indications.⁶⁷³

Calcium Channel Blockers and Thiazide-Like Diuretics

The combination of a CCB and diuretic provides a partially additive antihypertensive effect.⁶⁷⁴ CCBs increase kidney sodium excretion, albeit not to the same extent as diuretics. This combination performed well in the VALUE trial, in which HCTZ was administered to participants randomized to amlodipine as a second step compared with the valsartan arm.⁶⁷⁵ AEs included peripheral edema and hypokalemia as well as a slight increase in the incidence of new-onset diabetes.

Calcium Channel Blockers and β -Blockers

Additive BP reduction can be achieved with the combination of a β -blocker and a dihydropyridine CCB. In one study, the antihypertensive effects of combination ER metoprolol succinate and ER felodipine produced dose-related BP-lowering effects over a wide dose range; moreover, the incidence of edema and AEs was low. Nondihydropyridine CCBs such as verapamil and diltiazem should not be given together with β -blockers because of dual negative chronotropic effects, which may result in heart block or severe bradycardia.

Dual Calcium Channel Blockade

In a meta-analysis of six studies consisting of 153 patients, dual CCB therapy (dihydropyridine CCB with verapamil or diltiazem) was shown to have additive antihypertensive effects without an increase in adverse events.⁶⁷⁶ Long-term prospective clinical trials are required to assess the long-term safety of this combination and to obtain cardiovascular outcome data. Dual CCB therapy should not be used as an alternative treatment for resistant hypertension until outcome data are available.

UNACCEPTABLE OR INEFFECTIVE COMBINATIONS

Dual Renin-Angiotensin Blockade

In the ALTITUDE trial, aliskiren or placebo was added to ACE inhibitors or ARBs, which inhibit RAAS, in persons with CKD and type 2 diabetes.⁹⁶ The trial was halted early because of an increased incidence of hypotension, AKI, hyperkalemia, and stroke. In the ONTARGET trial, there were more adverse events, including declining kidney function, with a combination of telmisartan and ramipril than with the individual drugs, and the combination did not improve cardiovascular outcomes.¹⁴⁸ The 2013 Veterans Affairs Nephropathy in Diabetes (VA NEPHRON-D)⁹⁵ study enrolled 1448 patients with type 2 diabetes and diabetic nephropathy to receive losartan 100 mg daily plus either lisinopril or placebo. That trial was terminated early because of safety concerns, as patients in the dual ACE inhibitor/ARB arm had higher rates of hyperkalemia and AKI, and no benefits were observed in the primary endpoint of reduction in eGFR, ESKD, or death.

β -Blockers and Central Adrenergic Agonists

β -Blockers and central adrenergic agents (e.g., clonidine, α -methyldopa) both interfere with the SNS. No studies have explored the potential additive antihypertensive effects of these agents. However, combination therapy can result in bradycardia and heart block, particularly in elderly patients. Moreover, rebound hypertension with abrupt discontinuation would be another concern.

BEDTIME ANTIHYPERTENSIVE DOSING VERSUS MORNING DOSING

Shifting one nondiuretic antihypertensive medicine from morning to evening can reduce the total 24-hour mean BP and restore the normal nocturnal BP decrease. This failure of the BP to decrease at night, termed “nondipping,” is a cardiovascular disease risk factor.⁶⁷⁷ After a median follow-up of 5.4 years, patients who took at least one BP-lowering medication at bedtime had an adjusted risk for total cardiovascular events (a composite of death, MI, angina pectoris, revascularization, heart failure, arterial occlusion of lower extremities, occlusion of the retinal artery, and stroke) that was approximately one-third that of patients who took all medications on awakening (adjusted HR, 0.31; 95% CI, 0.21 to 0.46; $P \leq .001$).⁶⁷⁸ Similar results were demonstrated in patients with CKD.⁶⁷⁹

RESISTANT HYPERTENSION

“Resistant hypertension” is a term used to characterize high BP that fails to respond to what the clinician thinks is an adequate antihypertensive regimen (Table 49.34).^{680,681} Resistant hypertension has also been defined as the inability to reach SBP goal of less than 130 mm Hg, despite the use of three optimally dosed drugs, one of which is a diuretic, or the need for four or more medications to reach the desired SBP goal of less than 130 mm Hg.⁶⁸² Before a diagnosis of resistant hypertension, it is important to exclude “pseudo-resistance” by verifying adherence to the medication regimen, ensuring that BPs are measured accurately, and excluding white coat hypertension.

A variety of factors can interfere with the ability of what is deemed to be appropriate antihypertensive therapy to normalize BP. Perhaps most important is nonadherence to the therapy regimen. Nonadherence is common and is one of the most serious problems that interfere with attaining the BP goal. This problem has many sources, including inadequate education, poor clinician–patient relationship, lack of understanding of side effects, and complexity of multidrug regimens. The health care provider should make every effort to determine whether adherence to therapy is part of the problem before pursuing other potential explanations for resistant hypertension. If nonadherence is eliminated, a methodologic approach can be used to help diagnose the cause of resistant hypertension and then correct it.

A checklist for accurate measurement of office BP has been proposed by the ACC/AHA guidelines on hypertension (Table 49.35).⁵⁵ Common reasons for falsely elevated office BP readings is not following the 5-minute rest, improper positioning of the patient, and improper positioning or size of BP cuff. If office BP is elevated despite proper technique, white coat hypertension where the out-of-office BP readings are acceptable but in-office BP readings are high needs to

Table 49.34 Causes of Resistant Hypertension

Pseudoresistance
White coat hypertension or office elevations
Pseudohypertension in older patients
Use of small cuff on very obese arm
Nonadherence to therapeutic regimen
Volume overload
Drug-related causes
Antihypertensive drug dosage too low
Wrong type of diuretic
Inappropriate combinations of antihypertensive drugs
Drug actions and interactions
Sympathomimetics
Nasal decongestants
Appetite suppressants
Cocaine
Caffeine
Oral contraceptives
Adrenal steroids
Licorice (may be found in chewing tobacco)
Cyclosporine, tacrolimus
Erythropoiesis-stimulating agents and erythropoietin
Antidepressants
Nonsteroidal antiinflammatory drugs
Concomitant conditions
Obesity
Sleep apnea
Ethanol intake >1 oz (30 mL)/day
Anxiety, hyperventilation
Secondary causes of hypertension
Renovascular hypertension
Primary aldosteronism
Pheochromocytoma
Hypothyroidism
Hyperthyroidism
Hyperparathyroidism
Aortic coarctation
Renal disease

be excluded with home BP monitoring or 24-hour ambulatory BP measurements. Again, proper technique of home BP measurements is critical as proposed by the ACC/AHA guidelines. Although 24-hour ambulatory measurements are very useful to exclude white coat hypertension and to get accurate measurements outside the office, the need to wear the device for 24 hours, the need to bring back the device to the clinic, costs, and insurance coverage are issues that might limit the practical use of these devices in all patients.

Corresponding values of SBP/DBP for clinic, home BP monitoring, daytime, nighttime, and 24-hour ambulatory blood pressure monitoring (ABPM) measurements are presented in Table 49.36.⁵⁵ Underlying lifestyle factors that contribute to resistant hypertension need to be addressed. These include obesity, physical inactivity, excessive alcohol consumption, smoking, and high salt intake.

Volume overload is an important and common cause of resistant hypertension. It may be related to excessive salt intake or to the inability of the kidney to excrete an appropriate salt and water load because of endocrine abnormalities or intrinsic kidney disease.

High dietary salt intake offsets the antihypertensive activities of all antihypertensive medications.⁶³⁶ Some patients are more

salt sensitive than others. Salt sensitivity is common in patients of African ancestry.⁶⁴⁷ This sensitivity is also a more common problem in patients with CKD and heart failure. A careful clinical examination coupled with the judicious use of a thiazide-type or loop diuretic (depending on the level of kidney function) is critical in achieving an ideal blood volume to restore the antihypertensive efficacy of most classes of drugs. It is also appropriate to consider educating the patient about avoiding foods that are high in salt content, such as processed foods.

Drug-related causes of resistant hypertension are common and need to be carefully assessed in each patient. Oral contraceptives, immunosuppressants (e.g., cyclosporine, tacrolimus), erythropoiesis-stimulating agents, and some antidepressants can increase BP. Perhaps the most common drugs that cause resistant hypertension are over-the-counter (OTC) preparations of sympathomimetics, such as nasal decongestants, appetite suppressants, and NSAIDs.⁶⁸³ Caffeine and licorice may also increase BP. Unfortunately, patients may not always recognize OTC preparations as medications. Therefore careful questioning specifically focusing on these types of medications should be routine during the evaluation for refractory hypertension. In addition, alcohol use, smoking, and cocaine use can be complicating factors that interfere with the ability of medications to lower BP.

Some medications may interfere with the antihypertensive activity of other drugs. For example, NSAIDs interfere with the antihypertensive activity of diuretics and ACE inhibitors.⁶⁸⁰ Interestingly, only the antihypertensive activity of CCBs appears to be unaffected by the effects of NSAIDs.⁶⁸⁴ Drug-drug interactions that can interfere with drug absorption, drug metabolism, or the pharmacodynamics of the concomitantly administered drugs can also interfere with antihypertensive activity.

Secondary causes of hypertension should also be considered in the evaluation of resistant hypertension. These causes can be divided into renal, endocrine, and other (principally sleep apnea). Kidney disease is the most common cause of secondary hypertension. A chemistry profile and urinalysis facilitate the diagnosis. Renal artery stenosis is often considered a common cause of secondary hypertension. Although renal artery disease caused by fibromuscular dysplasia in young women often responds to interventional procedures, revascularization of atheromatous renovascular diseases often seen in elderly persons with widespread atherosclerosis has not been shown to improve BP control, progression of CKD (because of ischemic nephropathy),⁶⁸⁵ or cardiovascular outcomes compared with conservative treatments in three large RCTs.⁵⁵ According to the ACC/AHA guidelines, in the setting of significant atherosclerotic renal artery stenosis (>60%) with inability to control BP with medical management, stenting of the atherosclerotic renal artery could be considered but evidence is weak, as shown by a class IIb recommendation (weak) and C level evidence (observational studies). Therefore whether renal vascular assessment with Doppler ultrasonography or a direct imaging technique should be part of the workup for resistant hypertension is debatable.

Associated endocrine abnormalities include primary hyperaldosteronism, Cushing disease, pheochromocytoma, hypothyroidism or hyperthyroidism, and hyperparathyroidism. Rarely, aortic coarctation can be a cause of resistant hypertension. With the realization that hyperaldosteronism explains

Table 49.35 Checklist for Accurate Measurement of Blood Pressure

Key Steps for Proper BP Measurements	Specific Instructions
Step 1: Properly prepare the patient	1. Have the patient relax, sitting in a chair (feet on floor, back supported) for >5 min. 2. The patient should avoid caffeine, exercise, and smoking for at least 30 minutes before measurement. 3. Ensure patient has emptied his/her bladder. 4. Neither the patient nor the observer should talk during the rest period or during the measurement. 5. Remove all clothing covering the location of cuff placement. 6. Measurement made while the patient is sitting or lying on an examining table do not fulfill these criteria.
Step 2: Use proper technique for BP Measurements	1. Use BP measurement device that has been validated, and ensure that the device is calibrated periodically. 2. Support the patient's arm (e.g., resting on a desk). 3. Position the middle of the cuff on the patient's upper arm at the level of the right atrium (the midpoint of the sternum). 4. Use the correct cuff size, such that the bladder encircles 80% of the arm, and note if a larger-or-smaller-than-normal cuff size is used. 5. Either the stethoscope diaphragm or bell may be used for auscultatory readings.
Step 3: Take the proper measurements needed for diagnosis and treatment of elevated BP/hypertension	1. At the first visit, record BP in both arms. Use the arm that gives the higher reading for subsequent readings. 2. Separate repeated measurements by 1–2 min. 3. For auscultatory determination, use a palpated estimate of radial pulse obliteration pressure to estimate SBP. Inflate the cuff 20–30 mm Hg above this level for an auscultatory determination of the BP level. 4. For auscultatory readings, deflate the cuff pressure 2 mm Hg per second, and listen for Korotkoff sounds.
Step 4: Properly document accurate BP readings	1. Record SBP and DBP. If using the auscultatory technique, record SBP and DBP as onset of the first Korotkoff sound and disappearance of all Korotkoff sounds, respectively, using the nearest even number. 2. Note the time of most recent BP medication taken before measurements.
Step 5: Average the readings	Use an average of ≥ 2 readings obtained on ≥ 2 occasions to estimate the individual's level of BP.
Step 6: Provide BP readings to patient	Provide patients with the SBP/DBP readings both verbally and in writing.

BP, Blood pressure; DBP, diastolic blood pressure; SBP, systolic blood pressure.

From Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on clinical practice guidelines. *Circulation*. 2018;138:e426–e483. With permission.

at least 15% to 20% of resistant hypertension, all patients with resistant hypertension should be screened for occult hyperaldosteronism by measuring the ratio of plasma aldosterone level to plasma renin activity. In a retrospective study, more widespread use of the plasma aldosterone concentration-to-plasma renin activity ratio in hypertensive patients resulted in a 1.3- to 6.3-fold increase in the annual detection rate for primary aldosteronism (1%–2% before screening and 5%–10% after screening). Although these results are complicated by selection bias, they suggest that screening for primary hyperaldosteronism should be considered for patients with resistant hypertension, despite the fact that it is unusual to find an adenoma.⁶⁸⁶ Often, patients with subtle hyperaldosteronism respond to the addition of a selective aldosterone receptor blocker to their antihypertensive regimen.

Observational data suggest an association of sleep apnea with hypertension. However, the effects of continuous positive airway pressure (CPAP) on BP is only modest; for instance, in a meta-analysis of 16 RCTs,⁶⁸⁷ compared with controls, CPAP resulted in a mean net change in SBP of –2.46 mm

Hg (95% CI, –4.31 to –0.62) and a mean net change in DBP of –1.83 mm Hg (95% CI, –3.05 to –0.61). In a more recent RCT including 2717 patients with obstructive sleep apnea, CPAP treatment did not improve cardiovascular outcomes. However, other studies demonstrated that CPAP treatment improves quality of life and daytime somnolence.⁶⁸⁸ Therefore CPAP treatment in sleep apnea might be indicated for quality of life rather than improving BP or cardiovascular outcomes.

Strategies to control BP in patients with refractory hypertension should first address issues related to adherence, simplification of the medical regimen, and determination of whether side effects play a role. Subsequently, one can evaluate the medications and try to choose those that work well with one another to facilitate an almost additive antihypertensive response. As previously noted, most drugs reduce SBP by approximately 8 to 10 mm Hg. Consequently, it is not unusual for patients who are 40 or 50 mm Hg from their target SBP to require multiple medications. Lifestyle issues should also be emphasized. Weight loss, dietary salt restriction, and mindfulness-based stress reduction and other relaxation

Table 49.36 Corresponding Values of Systolic Blood Pressure/Diastolic Blood Pressure for Clinic, Home Blood Pressure Monitoring, Daytime, Nighttime, and 24-Hour Ambulatory Blood Pressure Monitoring Measurements

Clinic	HBPM	Daytime ABPM	Nighttime ABPM	24-Hour ABPM
120/80	120/80	120/80	100/65	115/75
130/80	130/80	130/80	110/65	125/75
140/90	135/85	135/85	120/70	130/80
160/100	145/90	145/90	140/85	145/90

ABPM, Ambulatory blood pressure monitoring; HBPM, home blood pressure monitoring.

From Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on clinical practice guidelines. *Circulation*. 2018;138:e426–e483. With permission.

techniques should be considered, as for patients with milder forms of hypertension.

One should also be careful to ensure that volume excess is controlled and that there are no drug–drug interactions or clinical situations that would promote diuretic resistance, such as excessive salt intake, impaired drug bioavailability, impaired diuretic secretion by the proximal tubule, increased protein binding in the tubule lumen, or reduced GFR. True resistant hypertension is unusual, and a methodologic approach should be taken to help facilitate BP control in these patients because a lack of control puts these patients at a greater risk of cardiovascular complications.

“Refractory hypertension” refers to patients whose BP cannot be controlled with maximal medical therapy (more than four drugs with complementary mechanisms, given at maximal tolerable doses) under the care of a specialist.⁶⁸⁹ It is an important clinical phenotype that is commonly associated with those of African heritage, patients with diabetes, and/or those with albuminuria.⁶⁸⁹ Recent large observational studies have noted that 0.5% of patients receiving antihypertensive treatment and 3.6% of individuals with resistant hypertension have refractory hypertension, and that these individuals have much higher 10-year Framingham Coronary Heart Disease and Stroke risk scores. Strategies for improving treatment in these types of patients are unknown but may require higher doses of MRAs.

Many hypertensive patients are not controlled because of nonadherence or lack of tolerance to available antihypertensive therapy. In one study, in as many as 35% of these individuals, who may be prescribed as many as three to five antihypertensive medications, blood and urine samples revealed no trace of medication.^{690,691} Thus there is a need for novel drug therapies and devices to lower BP in individuals with refractory hypertension. Fortunately, there are many new drugs in preclinical and clinical trials.⁶⁹² A recent large,

blinded clinical trial evaluating catheter-based renal denervation did not show a significant reduction of SBP 6 months after the procedure compared with that of sham controls.⁶⁹³ However, it has become clear that results are difficult to interpret because of the procedural and technical shortcomings of the renal denervation procedure itself as only a few participants (9 of 253) received a complete four-quadrant ablation (covering 360 degrees of the renal artery), and SBP was significantly higher in participants who underwent incomplete renal denervation compared with those with complete renal denervation.^{692,694}

Hopefully, one may be optimistic that these shortcomings will be resolved in the near-future to allow renal denervation procedures to reach clinical applicability.⁶⁹² Drugs in development and innovative novel interventional approaches should provide physicians with the tools to provide better care for their patients with hypertension that is not yet controlled. Novel antihypertensive drug and device treatments are the subject of a recent review⁶⁹² (Table 49.37).

DRUG TREATMENT OF HYPERTENSIVE URGENCIES AND EMERGENCIES

It is important to distinguish between hypertensive urgency and hypertensive emergency. These terms are used loosely in clinical practice, with a great deal of overlap. The distinction between the two is important because the management approach is substantially different.

A hypertensive emergency is a clinical syndrome in which marked elevation in BP results in ongoing target-organ damage in the body (Table 49.38). The syndrome can be manifested by encephalopathy, retinal hemorrhage, papilledema, acute MI, stroke, or AKI. Any delay in the control of BP may lead to irreversible sequelae, including death. This syndrome is unusual but requires immediate hospitalization in an intensive care unit (ICU), with judicious use of intravenous vasodilators to lower SBPs and DBPs to approximately 140/90 mm Hg.

Hypertensive urgency is a clinical situation in which a patient may have a marked elevation in BP (e.g., >200/130 mm Hg) but no evidence of ongoing target-organ damage. Such a patient can be treated with rapid-onset drugs, such as captopril or clonidine, and be observed cautiously, with the administration of long-acting medications on an outpatient basis as progressive restoration of a more appropriate BP level is attained.

Thus the history and physical examination findings are the critical factors in delineating the difference between these two syndromes. The decision about whether to hospitalize the patient in an ICU and use intravenous medication or to observe the patient carefully and use oral medications to facilitate better BP control depends in large part on the presence or absence of ongoing target-organ injury.

PARENTERAL DIRECT-ACTING VASODILATORS

Hydralazine is a direct-acting vasodilator and may be given intramuscularly or as a rapid intravenous bolus injection (Table 49.39). It acts rapidly, and the BP effect persists for up to 6 hours.^{695,696} Hydralazine is less potent than diazoxide, and the BP response is less predictable. It may also

Table 49.37 New Drugs for Hypertension

Drug	Mechanism of Action	Status
BAY-94-8862 (Finerenone)	Mineralocorticoid receptor antagonist	Phase IIb
LC1699	Aldosterone synthase inhibitor	Phase II trials ^a
C21	AT ₂ receptor agonist	Preclinical
XNT	ACE2 activator	Preclinical ^a
DIZE	ACE2 activator	Phase 1
rhACE2	ACE2 activator	Phase 1
HP-β-CD/Ang 1-7	Ang 1-7 analogue	Preclinical
AVE0991	Nonpeptide agonist of Mas receptor	Preclinical
CGEN-856S	Peptide agonist of Mas receptor	Preclinical
Alamandine/HPβCD	Mas-related-G-protein coupled receptor, member D agonist	Preclinical
PC18	Aminopeptidase N inhibitor	Preclinical
RB150 (OGC001)	Aminopeptidase N inhibitor	Phase 1
LCZ696	Dual-acting angiotensin receptor-neprilysin inhibitor	Phase III
SLV-306 (Daglutril)	Dual-acting endothelin-converting enzymes-neprilysin inhibitor	Phase II
PL-3994	Natriuretic peptide A agonist	Phase II
C-ANP ₄₋₂₃	ANP analog, selective for natriuretic peptide clearance receptor (NPR-C)	Preclinical
AR9281	Soluble epoxide hydrolase inhibitors	Phase II ^a
Vasomera (PB1046)	Vasoactive intestinal peptide receptor 2 (VPAC2) agonist	Phase II
AZD1722 (Tenapanor)	Intestinal Na ⁺ /H ⁺ exchanger 3 inhibitor	Phase 1
Etamicastat	Dopamine β-hydroxylase inhibitor	Phase 1
Vaccines		
CYT006-Ang0β	Vaccine against angiotensin II	Phase II
Angil-KLH	Vaccine against angiotensin II	Preclinical
pHAV-4Anglis	Vaccine against angiotensin II	Preclinical
ATROβ-001	Vaccine against angiotensin II	Preclinical
ATR12181	Vaccine against angiotensin II type 1 receptor	Preclinical
Preeclampsia Drugs		
DIF	Antidigoxin antibody fragment	Phase II expedited
ATryn	Recombinant antithrombin	Phase III

^aStopped.

ANP, Atrial natriuretic peptide; ATR, angiotensin II type 1 receptors; DIF, digoxin-immune Fab; KLH, keyhole limpet hemocyanin; rhACE2, recombinant human ACE2.

From Oparil S, Schmieder RE. New approaches in the treatment of hypertension. *Circ Res*. 2015;116:1074–1095.

cause a reflex increase in heart rate and sodium and water retention.

Sodium nitroprusside is the most potent of the parenteral vasodilators.^{695,696} Nitroprusside acts on the excitation-contraction coupling of vascular smooth muscle by interfering with the intracellular activation of calcium. Unlike diazoxide and hydralazine, nitroprusside dilates arteriolar resistance and venous capacitance vessels. It has the advantages of being immediately effective when given as an infusion and of having an extremely short duration of action, which permits minute-to-minute adjustments in BP control (Table 49.39). Disadvantages of nitroprusside therapy include the following: (1) need for intraarterial BP monitoring; (2) need for the drug to be prepared fresh every 4 hours; (3) need to protect the solution from light during infusion; and (4) potential for toxic effects from metabolic side products. Nitroprusside is not excreted intact; it is rapidly metabolized to cyanide and thiocyanate through a reaction with hemoglobin, which yields methemoglobin and an unstable intermediate that dissociates to release cyanide. The major elimination pathway of cyanide is conversion in the liver and kidney to thiocyanate. Back conversion of thiocyanate to cyanide may occur. Thiocyanate is largely excreted in the urine; it has a plasma half-life of

1 week in normal individuals and accumulates in patients with CKD.

Toxic concentrations of cyanide or thiocyanate may occur if nitroprusside infusions are given for longer than 48 hours or at infusion rates higher than 2 mg/kg/min; the drug should not be administered at the maximal dose of 10 mg/kg/min for longer than 10 minutes.⁶⁹⁷ Toxic manifestations include air hunger, hyperreflexia, confusion, and seizures. Lactic acidosis and venous hyperoxemia are laboratory indicators of cyanide intoxication. The appearance of drug unresponsiveness may reflect an increase in the concentration of free cyanide. In such cases, the drug should be promptly discontinued and the levels of cyanide measured. Nitroprusside is hemodialyzable.

Intravenous nitroglycerin produces dilation of arterial and venous beds in a dose-related manner. At lower dosages, the primary effect is on preload; at higher infusion rates, afterload is reduced. Nitroglycerin may also dilate epicardial coronary vessels and their collaterals, increasing the blood supply to ischemic regions. Effective coronary perfusion is maintained, provided that the BP does not fall excessively or that the heart rate does not increase significantly. Nitroglycerin has an immediate onset of action but is rapidly metabolized to dinitrates and mononitrates (Table 49.39). Because

nitroglycerin is absorbed by many plastics, dilution should be performed only in glass parenteral solution bottles. Nitroglycerin is also absorbed by polyvinyl chloride (PVC) tubing; non-PVC intravenous administration sets should be used.

Patients with normal or low left ventricular filling pressure or pulmonary wedge pressure may be hypersensitive to the

effects of nitroglycerin. Therefore continuous monitoring of BP, heart rate, and pulmonary capillary wedge pressure must be performed to assess the correct dose. Intravenous nitroglycerin may be the drug of choice in the treatment of patients with moderate hypertension associated with coronary ischemia because it provides collateral coronary vasodilation, a property that is not seen with the other direct-acting arteriolar vasodilators. The principal AEs are headache, nausea, and vomiting. Tolerance may develop with prolonged use.

Table 49.38 Hypertensive Emergencies

Hypertensive encephalopathy ^a
Acute aortic dissection ^a
Central nervous system bleeding ^a
Intracranial hemorrhage
Thrombotic cerebrovascular accident
Subarachnoid hemorrhage
Acute left ventricular failure refractory to conventional medical therapy ^a
Myocardial ischemia or infarction associated with persistent chest pain ^a
Accelerated or malignant hypertension ^b
Toxemia of pregnancy – eclampsia ^a
Renal failure or insufficiency ^b
Hypertension associated with hyperadrenergic states ^a
Pheochromocytoma
Interaction between monoamine oxidase inhibitors and tyramine-containing foods
Interaction between an α -adrenergic agonist and nonselective β -adrenergic antagonist
After abrupt withdrawal of clonidine or guanabenz
After severe body burns
Neurogenic hypertension
Hypertension in the surgical patient ^b
Associated with postoperative bleeding
After open heart or vascular surgery
Preceding emergency surgery
After kidney transplantation
Hypertension in a diabetic patient with retinal hemorrhage ^a

^aConsidered by some authors to be a true hypertensive emergency.

^bConsidered by some authors to be a hypertensive urgency.

β_1 -SELECTIVE ADRENERGIC ANTAGONIST

Esmolol hydrochloride is a short-acting, β_1 -selective adrenergic antagonist. Esmolol hydrochloride concentrate for injection must be diluted to a final concentration of 10 mg/mL.^{695,696} Extravasation of esmolol hydrochloride may cause serious local irritation and skin necrosis. Esmolol shares all the toxic potential of the β_1 -adrenergic antagonists, as previously discussed.

After intravenous injection of a loading dose of 250 to 500 mg/kg and then infusion of a maintenance dose ranging from 50 to 100 mg/kg/min, steady-state blood concentrations are achieved within 5 minutes (Table 49.39). Efficacy should be assessed after the 1-minute loading dose and 4 minutes of maintenance infusion. If an adequate therapeutic effect is observed, as assessed by BP and heart rate response, then the maintenance infusion should be maintained. If an adequate therapeutic effect is not observed, the same loading dose can be repeated for 1 minute, followed by maintenance infusion at an increased rate.

Esmolol has pharmacologic actions similar to those of other β_1 -selective adrenergic antagonists; it produces negative chronotropic and inotropic activity. It has been used to prevent or treat hemodynamic changes induced by surgical events, including increases in SBPs and DBPs and doubling of the product of heart rate multiplied by SBP. Esmolol may be particularly useful for the treatment of postoperative hypertension and hypertension associated with coronary insufficiency.^{695,696} Esmolol is hydrolyzed rapidly in the blood, and negligible concentrations are present 30 minutes after discontinuance. Because the kidneys eliminate the de-esterified metabolite of esmolol, the drug should be used cautiously

Table 49.39 Parenteral Drugs Used in the Treatment of Hypertensive Emergencies

Drug	Dosage	Onset of Action	Peak Effect	Duration of Action
Hydralazine	0.5–1.0-mg/min infusion or 10–50 mg intramuscularly	1–5 min	10–80 min	3–6 h
Nitroglycerine	5–100- μ g/min infusion	1–2 min	2–5 min	3–5 min
Nitroprusside	0.25–10- μ g/kg/min infusion	Immediate	1–2 min	2–5 min
Esmolol	250–500 μ g/kg/min $\times$ 1 (loading dose), then 50–100 μ g/kg/min $\times$ 4 (maintenance); maintenance dosage may be increased to maximum of 300 μ g/kg/min	1–2 min	5 min	0–30 min
Labetalol	20 mg bolus then 0.5 to 2 mg/minute infusion	5 min	10 min	3–6 h
Methyldopa	250–500-mg bolus every 6 h (2 g maximum)	2–3 h	3–5 h	6–12 h
Trimethaphan	0.5–10-mg/min infusion bolus over 2 min (300 mg maximum)	Immediate	1–2 min	5–10 min
Enalaprilat	0.625–5.0-mg bolus over 5 min q6h	5–15 min	1–4 h	6 h
Phentolamine	0.5–1.0-mg/min infusion or 2.5–5.0-mg bolus q5–15 minutes	Immediate	3–5 min	10–15 min
Nicardipine	5–15 mg/h	5–10 min	45 min	50 h
Clevidipine	1–2 mg/hour (16 mg/hour maximum)	2–4 min	5–6 min	15 min
Fenoldopam	0.01–1.6- μ g/min constant infusion	5–15 min	30 min	5–10 min

in patients with CKD. Esmolol is the preferred agent for treatment of acute aortic dissection.

α_1 - AND β -ADRENERGIC ANTAGONISTS

The α_1 - and β -adrenergic antagonist labetalol may be given by repeated intravenous injection or slow continuous infusion^{695,696} (Table 49.39). The maximal BP-lowering effect occurs within 5 minutes of the first injection. The drug should be administered to patients in the supine position to avoid symptomatic postural hypotension. The AEs of labetalol have been previously discussed. This drug has been proven safe and useful in hypertensive urgencies and emergencies in pregnant women.

CENTRAL α_2 -ADRENERGIC AGONIST

Methyldopate hydrochloride is a central α_2 -adrenergic agonist that may be administered intravenously as a bolus infusion^{695,696} (Table 49.39). It has a delayed onset of action and peak effect, and its effect on BP is unpredictable. The AEs of methyldopa are addressed earlier.

ANGIOTENSIN-CONVERTING ENZYME INHIBITOR

Enalaprilat, the active metabolite of the oral ACE inhibitor enalapril, can be administered as a slow intravenous infusion for 5 minutes (Table 49.39) in an intravenous dose that is approximately 25% of the oral dose. The onset of action occurs within 15 minutes, and the maximal effect is observed within 1 to 4 hours.⁶⁹⁸ The duration of action is approximately 6 hours. In patients with CKD, the initial dose should be no more than 0.625 mg.

α -ADRENERGIC ANTAGONIST

Phentolamine mesylate is a nonselective, α -adrenergic antagonist used primarily in the treatment of hypertension associated with pheochromocytoma.^{695,696} It has a rapid onset of action when administered intravenously as a bolus or continuous infusion (Table 49.39). The duration of action is 10 to 15 minutes. The drug has a plasma half-life of 19 minutes. Approximately 13% of a single dose appears in the urine as unchanged drug. AEs include those associated with nonselective α -adrenergic blockade, as previously discussed.

CALCIUM CHANNEL BLOCKERS

Nicardipine hydrochloride, a dihydropyridine CCB, is administered by slow continuous infusion at a concentration of 0.1 mg/mL; each 1-mL ampule (25 mg) should be diluted with 240 mL of a compatible intravenous fluid (not including sodium bicarbonate or lactated Ringer's solution) to produce 250 mL of solution at a concentration of 0.1 mg/mL.⁶⁹⁹ There is a dose-dependent decrease in BP. The onset of action is within minutes; 50% of the ultimate decrease in BP occurs within 45 minutes, but a final steady state is not reached for approximately 50 hours (Table 49.39). The discontinuation of infusion is followed by a 50% offset of action within 30 minutes, but gradually decreasing antihypertensive effects exist for approximately 50 hours. This drug has been shown to be safe and effective in the treatment of pediatric hypertensive emergencies.^{699,700}

Clevidipine is a dihydropyridine CCB that is available in a lipid emulsion for intravenous infusion for the treatment of hypertensive emergencies.⁷⁰¹ Steady-state concentrations

of clevidipine in arterial or venous blood are attained within 2 and 10 minutes in healthy volunteers receiving 0.91 and 3.2 $\mu\text{g/kg/min}$, respectively. The relationship between the intravenous infusion dose and steady-state blood concentrations is linear in patients with mild-to-moderate hypertension and in healthy volunteers. Clevidipine is highly protein bound and rapidly distributed. This drug is rapidly hydrolyzed by esterases in the blood and extravascular tissues. Blood concentrations decrease rapidly after termination of the infusion. The initial phase is rapid (half-life of ≈ 1 minute) and accounts for 85% to 90% of elimination. The terminal elimination half-life is 15 minutes. Clevidipine treatment results in a prompt reduction in BP ($\geq 15\%$ from baseline) in less than 6 minutes.⁷⁰² Clevidipine has a safety profile comparable to that of nitroglycerin, sodium nitroprusside, and nicardipine. The most common adverse events are sinus tachycardia, headache, nausea, and chest discomfort.⁷⁰³

DOPAMINE D_1 -LIKE RECEPTOR AGONIST

Fenoldopam mesylate, a dopamine D_1 -like receptor agonist, is formulated as a solution to be diluted for intravenous infusion for the treatment of acute hypertension.^{701,704} It is a rapid-acting agent that produces vasodilation by functioning as an agonist for dopamine D_1 -like receptors and has moderate affinity for α_2 -adrenoreceptors.

Fenoldopam is a racemic mixture in which the R isomers are responsible for its biologic activity. It has vasodilatory effects on coronary, renal, mesenteric, and peripheral arteries in experimental studies; however, not all vascular beds respond uniformly. In humans, the drug increases renal blood flow in hypertensive and normotensive patients.

Fenoldopam comes in 1-mL ampules that contain 10 mg of fenoldopam and is diluted for administration as a constant infusion at a rate of 0.01 to 1.6 mg/kg/min (Table 49.39). It produces steady-state plasma concentrations in proportion to its infusion rate, its elimination half-life is 5 minutes, and steady-state concentrations are reached within 20 minutes.

Clearance of the active compound is not altered by ESKD or hepatic disease. Approximately 90% of infused fenoldopam is eliminated in urine and 10% in feces. Elimination occurs largely by conjugation that does not involve CYP enzymes. There are no data on drug-drug interactions.

AEs include reflex increase in heart rate, increase in intraocular pressure, headache, flushing, nausea, and hypotension.

RAPID-ACTING ORAL DRUGS

A more gradual, progressive reduction in systemic BP may be achieved after the oral administration of drugs with rapid absorption.⁶²² These drugs include the following: (1) α_1 - and β -adrenergic antagonist labetalol; (2) central α_2 -adrenergic agonist clonidine; (3) CCBs diltiazem and verapamil; (4) ACE inhibitors captopril and enalapril; (5) postsynaptic α_1 -adrenergic antagonist prazosin; and (6) a combination of oral therapies. The dosages and pharmacodynamic effects of rapid-acting oral drugs that are commonly used in the treatment of hypertensive emergencies are given in Table 49.40. Note that rapid-acting oral dihydropyridine CCBs, such as sublingual nifedipine, are no longer recommended because they may cause large and unpredictable reductions in BP, with resultant ischemic events.⁷⁰⁵

Table 49.40 Rapid-Acting Oral Drugs Used in the Treatment of Hypertensive Emergencies

Drug	Dosage	Onset of Action	Peak Effect	Duration of Action
α_1- and β-Adrenergic Antagonist				
Labetalol	100–400 mg q12h (2400 mg maximum)	1–2 h	2–4 h	8–12 h
α_1-Adrenergic Antagonist				
Prazosin	1–5 mg q2h (20 mg maximum)	<60 min	2–4 h	6–12 h
Central α_2-Agonist				
Clonidine	0.2 mg initially, then 0.1 mg/h (0.8 mg maximum)	30–60 min	2–4 h	6–8 h
Calcium Channel Blockers				
Diltiazem	30–120 mg q8h (480 mg maximum)	<15 min	2–3 h	8 h
Verapamil	80–120 mg q8h (480 mg maximum)	< 60 min	2–3 h	8 h
Angiotensin-Converting Enzyme Inhibitors				
Captopril	12.5–25 mg q6h (150 mg maximum)	<15 min	1 h	6–12 h
Enalapril	2.5–10 mg q6h (40 mg maximum)	<60 min	4–8 h	12–24 h

CLINICAL CONSIDERATIONS IN THE RAPID REDUCTION OF BLOOD PRESSURE

The rapid reduction of BP carries the risk of impairing blood supply to vital structures, such as the brain and heart. The 2017 ACC/AHA guidelines⁵⁵ recommend an initial reduction in SBP by no more than 25%, except in certain clinical scenarios such as acute aortic dissection or preeclampsia/eclampsia, when a larger and more rapid reduction in BP may be preferred. The risk of overreduction of BP in a rapid fashion is linked to the use of sublingual nifedipine capsules with stroke and heart attack in hypertensive patients.⁷⁰⁵ Because this approach is variable and rapid, clinicians are unable to set a lower limit of BP that is achieved with therapy.

Cerebral blood flow is normally carefully autoregulated so that perfusion is maintained at a sufficient level when BP is low but is diminished during states of chronic hypertension to avoid cerebral edema. With chronic hypertension, the short-term, rapid reduction of BP may decrease cerebral blood flow sufficiently to precipitate ischemia and infarction. This decrease may be particularly important in patients with atherosclerotic disease of the cerebral blood vessels in whom there may be areas of uneven cerebral perfusion. Although drugs that do penetrate the blood–brain barrier, such as hydralazine, sodium nitroprusside, and nicardipine, dilate cerebral vessels, which may lessen the likelihood of ischemia, intrinsic vascular disease may render some areas more ischemic than others with BP reduction. In addition, potent cerebral vasodilators can conceivably cause an increase in intracranial pressure, creating the potential for cerebral edema and possible herniation.

Sudden drops in BP can also interfere with coronary perfusion during diastole and result in myocardial ischemia, infarction, or arrhythmia. In addition, rapid reduction of BP may result in a reflex increase in heart rate, which would also interfere with coronary perfusion during diastole. For these reasons, a cautious and controlled reduction in BP is

necessary for these patients. For most hypertensive emergencies, a parenteral drug, such as sodium nitroprusside, is ideal. However, if the patient has coronary disease, the use of intravenous nitroglycerin, esmolol, or both is a useful approach because these drugs can induce coronary dilation and slow heart rate, respectively. Intravenous nicardipine can also be used because it facilitates coronary vasodilation. Patients with acute aortic dissection are best treated with a β -adrenergic antagonist first, followed by a vasodilating agent. Patients with hypertensive encephalopathy or CNS hemorrhage are best treated with drugs that do not cause cerebral vasodilation, such as hydralazine, nitroprusside, nicardipine, or fenoldopam.

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BOARD REVIEW QUESTIONS

1. A 43-year-old (G1P0) previously normotensive woman at 37 weeks gestation presents to clinic with complaints of increasing lower extremity edema. Blood pressure is 188/99 mm Hg. Urinalysis shows 4+ proteinuria and protein-to-creatinine ratio is 4 g/g. Serum creatinine 1.5 mg/dL (was 0.6 mg/dL at baseline). What is the most appropriate next step in management?
 - a. Begin oral labetalol 200 mg tid and follow-up in 1 week.
 - b. Admit to the hospital administer and start IV enalaprilat.
 - c. Admit to the hospital administer and start IV labetalol.
 - d. Begin oral magnesium sulfate and follow-up in 1 week.

Answer: c

Rationale: The American College of Obstetricians and Gynecologists recommend intravenous labetalol, hydralazine or immediate release oral nifedipine as first-line agents for treatment of acute-onset, severe hypertension in pregnant women.

2. A 65-year-old man comes for blood pressure management follow-up visit. He feels completely well. His mean office BP (2 readings) is 167/78 mm Hg. Two weeks ago his mean office BP was 173/69 mm Hg. He currently takes losartan 100 mg daily, chlorthalidone 25 mg daily and amlodipine 10 mg daily. He also has a history of hyperlipidemia for which he takes atorvastatin 20 mg daily. Which of the following is the next best step for his blood pressure management?
 - a. Congratulate him that his blood pressure is at goal and make no changes to his medications.
 - b. Start doxazosin 8 mg once daily.
 - c. Start bisoprolol 5 mg once daily.
 - d. Start spironolactone 25 mg once daily and check serum electrolytes in 2 weeks.

Answer: d

Rationale: The PATHWAY-2 trial (Williams et al, Lancet 2015) was a randomized, placebo-controlled cross-over trial that administered spironolactone, bisoprolol, doxazosin or placebo to patients for 12 weeks each. This study found that spironolactone was superior to all other active treatments and to placebo on home systolic blood pressure lowering. Serum electrolytes should be checked for hyperkalemia and renal dysfunction after initiating the mineralocorticoid receptor antagonist.

3. A 45-year-old man with a history of hypertension and hyperlipidemia presents to the emergency department with a 1-hour history of sudden onset knife-like chest pain radiating to the back. On physical exam, blood pressure on presentation is 220/130 mm Hg. The patient is in some discomfort, but exam is otherwise unremarkable. Neurological exam is completely intact. Computed tomography confirms descending aortic dissection. After placing an arterial line, which of the following options is the most appropriate for this patient?
 - a. Administer metoprolol 100 mg PO $\times$ 1 and continue to monitor blood pressure.
 - b. Begin esmolol infusion, titrating dose upwards until systolic blood pressure <120 mm Hg.
 - c. Begin clevidipine infusion, titrating dose to reduce systolic blood pressure by no more than 25% over the first hour.
 - d. Proceed directly to operating room for repair without administering any antihypertensive medications.

Answer: b

Rationale: The use of beta adrenergic receptor blockers like esmolol is preferred in aortic dissection, to decrease aortic shear stress and the tendency for the dissection to propagate.